

Time Series

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Chapter 1

ARMA models

1.1 Starters

1.1.1 Moments

Denote a **sample** of size T of some random variable Y_t by

$$\{y_1, y_2, \dots, y_T\}$$

The **expectation** of the t -th observation is given by

$$\mathbb{E}(Y_t) = \mu_t = \int_{-\infty}^{\infty} y_t f_{Y_t}(y_t) dy_t$$

The **variance** of Y_t is given by

$$\gamma_{0t} \equiv \mathbb{E}(Y_t - \mu_t)^2 = \int_{-\infty}^{\infty} (y_t - \mu_t)^2 f_{Y_t}(y_t) dy_t$$

The j -th **autocovariance** of Y_t is given by

$$\begin{aligned} \gamma_{jt} &= E((Y_t - \mu_t)(Y_{t-j} - \mu_{t-j})) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (y_t - \mu_t)(y_{t-j} - \mu_{t-j}) \times f_{Y_t, Y_{t-1}, \dots, Y_{t-j}}(y_t, y_{t-1}, \dots, y_{t-j}) dy_t dy_{t-1} \dots dy_{t-j} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y_t - \mu_t)(y_{t-j} - \mu_{t-j}) f_{Y_t, Y_{t-j}}(y_t, y_{t-j}) dy_t dy_{t-j}. \end{aligned}$$

The j -th **autocorrelation** is defined by

$$\text{Corr}(Y_t, Y_{t-j}) = \frac{\text{Cov}(Y_t, Y_{t-j})}{\sqrt{\text{Var}(Y_t) \text{Var}(Y_{t-j})}} = \frac{\gamma_j}{\sqrt{\gamma_0 \gamma_0}} = \frac{\gamma_j}{\gamma_0} = \rho_j.$$

The **skewness** of Y_t and its moment-based sample estimator are given by

$$A(Y_t) = \mathbb{E} \left[\left(\frac{Y_t - \mu}{\sigma} \right)^3 \right], \quad \hat{A} = \frac{1}{T \hat{\sigma}^3} \sum_{t=1}^T (Y_t - \bar{Y})^3.$$

where $\mu = \mathbb{E}(Y_t)$, $\sigma^2 = \text{Var}(Y_t)$, $\bar{Y} = \frac{1}{T} \sum_{t=1}^T Y_t$, and $\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (Y_t - \bar{Y})^2$. The **kurtosis** of Y and its moment-based sample estimator are given by

$$K(Y_t) = \mathbb{E} \left(\frac{(Y_t - \mu)^4}{\sigma^4} \right), \quad \hat{K} = \frac{1}{T \hat{\sigma}^4} \sum_{t=1}^T (Y_t - \bar{Y})^4.$$

If Y_t is a Gaussian i.i.d. process and T is large, then we have the approximations

$$\hat{A} \sim \mathcal{N}\left(0, \frac{6}{T}\right), \quad \hat{K} \sim \mathcal{N}\left(3, \frac{24}{T}\right).$$

Moreover, a Jarque-Bera test uses the above distributions to test for normality in i.i.d. samples, with test statistic

$$S = \left(\frac{T}{6}\right) \hat{A}^2 + \left(\frac{T}{24}\right) (\hat{K} - 3)^2.$$

where under $H_0 : Y_t \stackrel{i.i.d.}{\sim} \mathcal{N}$, we have $S \sim \chi_2^2$.

1.1.2 Stationarity

The process Y_t is **covariance-stationary** or weakly stationary if neither μ_t nor γ_{jt} depend on the date t and if the second moment is finite:

$$\mathbb{E}(Y_t) = \mu \quad \text{for all } t \quad (1.1)$$

$$\mathbb{E}((Y_t - \mu)(Y_{t-j} - \mu)) = \gamma_j \quad \text{for all } t \text{ and any } j \quad (1.2)$$

$$\mathbb{E}(Y_t^2) < \infty \quad (1.3)$$

and **strictly stationary** if for all $\tau \in \mathbb{Z}$:

$$F(Y_{t_1}, Y_{t_2}, \dots, Y_{t_k}) \stackrel{d}{=} F(Y_{t_1+\tau}, Y_{t_2+\tau}, \dots, Y_{t_k+\tau}) \quad (1.4)$$

where the joint distribution only depends on the intervals between dates. If a process is strictly stationary with finite second order moments (1.3), then it must be covariance-stationary. Note that higher moments, such as $E(Y_t)^3$, could depend on time given the covariance-stationary definition.

A process $\{Y_t\}$ is said to be **Gaussian** if the joint density

$$f_{Y_t, Y_{t+j_1}, \dots, Y_{t+j_n}}(y_t, y_{t+j_1}, \dots, y_{t+j_n})$$

is Gaussian for any $j_1, j_2, \dots, j_n$. Moreover, a covariance-stationary Gaussian process is strictly stationary.

The following properties hold for the autocovariance in the presence of stationarity ¹:

1. $\gamma_0 > 0$.
2. $\gamma_{-j} = \gamma_j$.
3. $|\gamma_j| \leq \gamma_0$.
4. γ_τ is **non-negative definite** in the sense that, for all a_j, a_k , $\sum_{j=1}^n a_j a_k \gamma_{j-k} \geq 0$.

1.1.3 Ergodicity

A covariance-stationary process is said to be **ergodic for the mean** if $\bar{Y}$ converges in probability to $E(Y_t)$ as $T \rightarrow \infty$:

$$\frac{1}{T} \sum_{t=1}^T Y_t \xrightarrow{p} E(Y_t), \quad \text{as } T \rightarrow \infty \quad (1.5)$$

¹Proof in section 1.12.1.

A process will be ergodic for the mean provided that the autocovariance γ_j goes to zero sufficiently quickly as j becomes large. If the autocovariances of a covariance-stationary process satisfy

$$\sum_{j=0}^{\infty} |\gamma_j| < \infty \quad (1.6)$$

then $\{Y_t\}$ is ergodic for the mean. Similarly, a covariance-stationary process is said to be **ergodic for second moments** if

$$\left[\frac{1}{T-j} \sum_{t=j+1}^T (Y_t - \mu)(Y_{t-j} - \mu) \right] \xrightarrow{p} \gamma_j \quad \text{for all } j. \quad (1.7)$$

Proof of Ergodicity for the Mean

Let $Z_t \equiv Y_t - \mu$ and $\bar{Z}_T = \frac{1}{T} \sum_{t=1}^T Z_t$. We will show that $\bar{Z}_T$ converges to 0 in mean square (and therefore converges in probability to 0). We have

$$\begin{aligned} \mathbb{E}(\bar{Z}_T^2) &= \mathbb{E} \left[\left(\frac{1}{T} \sum_{s=1}^T Z_s \right) \left(\frac{1}{T} \sum_{t=1}^T Z_t \right) \right] = \frac{1}{T^2} \sum_{s=1}^T \sum_{t=1}^T \mathbb{E}(Z_s Z_t) \\ &= \frac{1}{T^2} \sum_{s=1}^T \sum_{t=1}^T \gamma_{s-t} = \frac{1}{T} \sum_{j=-T+1}^{T-1} \left(\frac{T-|j|}{T} \right) \gamma_j \\ &\leq \frac{1}{T} \sum_{j=-T+1}^{T-1} |\gamma_j| = \frac{1}{T} \left(|\gamma_0| + 2 \sum_{j=1}^{T-1} |\gamma_j| \right) \\ &\leq \frac{1}{T} \left(|\gamma_0| + 2 \sum_{j=1}^{\infty} |\gamma_j| \right) \xrightarrow{T \rightarrow \infty} 0, \end{aligned}$$

where we used the symmetry $\gamma_{-j} = \gamma_j$ and assumed that $\sum_{j=0}^{\infty} |\gamma_j| < \infty$. Therefore, $\mathbb{E}(\bar{Z}_T^2) \rightarrow 0$, so $\bar{Z}_T \rightarrow 0$ in mean square. Since $\bar{Z}_T = \bar{Y}_T - \mu$, it follows that

$$\bar{Y}_T \xrightarrow{L^2} \mu,$$

and hence $\bar{Y}_T \xrightarrow{p} \mu$.

Ergodicity for the mean implies that the average over time within a single realization (a single sample path) equals the average over all possible realizations (the ensemble average). For an ergodic process, a long observation of one sample path is assumed to contain all the statistical information of the entire process.

Ergodicity for the first and second moments does not generally imply stationarity. Stationarity concerns invariance of the probabilistic structure over time, whereas ergodicity concerns the convergence of time averages to ensemble averages. Thus, a process can be stationary but not ergodic, or ergodic but not stationary. However, a process that is both stationary and ergodic is powerful because a single, long observation provides all the statistical insights into the entire system

1.2 White Noise processes

A sequence $\{\varepsilon_t\}$ is a **white noise process** if its elements have mean zero, variance σ^2 , and are uncorrelated over time. That is,

$$\mathbb{E}(\varepsilon_t) = 0 \quad (1.8)$$

$$\mathbb{E}(\varepsilon_t^2) = \sigma^2 \quad (1.9)$$

$$\mathbb{E}(\varepsilon_t \varepsilon_{t+\tau}) = 0 \quad \text{for } t \neq t + \tau. \quad (1.10)$$

We may use a stronger condition that gives an **independent white noise process**:

$$\varepsilon_t, \varepsilon_{t+\tau} \text{ independent for } t \neq t + \tau. \quad (1.11)$$

Finally, if (1.8)-(1.11) hold along with

$$\varepsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (1.12)$$

then we have the **Gaussian white noise process**².

1.3 Moving Average processes

1.3.1 The MA(1) process

Consider the **MA(1)** process where $\{\varepsilon_t\} \sim w.n.(0, \sigma^2)$:

$$Y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad (1.13)$$

The expectation of Y_t is

$$\mathbb{E}(Y_t) = \mathbb{E}(\mu + \varepsilon_t + \theta \varepsilon_{t-1}) = \mu + \mathbb{E}(\varepsilon_t) + \theta \mathbb{E}(\varepsilon_{t-1}) = \mu. \quad (1.14)$$

The variance of Y_t is

$$\begin{aligned} \mathbb{E}[(Y_t - \mu)^2] &= \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})^2] = \mathbb{E}[(\varepsilon_t^2 + 2\theta \varepsilon_t \varepsilon_{t-1} + \theta^2 \varepsilon_{t-1}^2)] \\ &= \sigma^2 + 0 + \theta^2 \sigma^2 \\ &= (1 + \theta^2) \sigma^2. \end{aligned} \quad (1.15)$$

The first autocovariance is

$$\begin{aligned} \mathbb{E}[(Y_t - \mu)(Y_{t-1} - \mu)] &= \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t-1} + \theta \varepsilon_{t-2})] \\ &= \mathbb{E}(\varepsilon_t \varepsilon_{t-1} + \theta \varepsilon_t \varepsilon_{t-2} + \theta \varepsilon_{t-1}^2 + \theta^2 \varepsilon_{t-1} \varepsilon_{t-2}) \\ &= 0 + 0 + \theta \sigma^2 + 0 \\ &= \theta \sigma^2, \end{aligned} \quad (1.16)$$

and higher autocovariances are all zero:

$$\mathbb{E}[(Y_t - \mu)(Y_{t-j} - \mu)] = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t-j} + \theta \varepsilon_{t-j-1})] = 0 \quad \text{for } j > 1. \quad (1.17)$$

The MA(1) process is **covariance-stationary** regardless of the value of θ as the mean and autocovariances are not functions of time. Furthermore, (1.6) is always satisfied:

$$\sum_{j=0}^{\infty} |\gamma_j| = (1 + \theta^2) \sigma^2 + |\theta \sigma^2| < \infty \quad (1.18)$$

²For Gaussian random variables, independence and non-correlation are equivalent concepts. Thus a Gaussian white noise process is an independent white noise process.

which means that the $MA(1)$ process is **ergodic** for the mean. The first autocorrelation for an $MA(1)$ process is given by

$$\rho_1 = \frac{\theta\sigma^2}{(1+\theta^2)\sigma^2} = \frac{\theta}{1+\theta^2}, \quad (1.19)$$

and higher autocorrelations are all zero. Notice that the value of $\theta/(1+\theta^2)$ is unchanged if θ is replaced by $1/\theta$:

$$\rho_1 = \frac{(1/\theta)}{1+(1/\theta)^2} = \frac{\theta^2 \cdot (1/\theta)}{\theta^2[1+(1/\theta)^2]} = \frac{\theta}{\theta^2+1}.$$

For example, the processes

$$\begin{aligned} Y_t &= \varepsilon_t + 0.5\varepsilon_{t-1}, \\ Y_t &= \varepsilon_t + 2\varepsilon_{t-1} \end{aligned}$$

would have the same autocorrelation function: $\rho_1 = \frac{2}{1+2^2} = \frac{0.5}{1+0.5^2} = 0.4$. However, you should choose the process that is **invertible**, because some algorithms for estimation³ and forecasting⁴ are valid only if the invertible representation is used.⁵

1.3.2 The $MA(q)$ process

The $MA(q)$ is written as

$$Y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_q\varepsilon_{t-q}, \quad (1.20)$$

where $\{\varepsilon_t\} \sim w.n.(0, \sigma^2)$ and $(\theta_1, \theta_2, \dots, \theta_q) \in \mathbb{R}^q$. The mean of (1.20) is:

$$\mathbb{E}(Y_t) = \mu + \mathbb{E}(\varepsilon_t) + \theta_1\mathbb{E}(\varepsilon_{t-1}) + \theta_2\mathbb{E}(\varepsilon_{t-2}) + \cdots + \theta_q\mathbb{E}(\varepsilon_{t-q}) = \mu. \quad (1.21)$$

The variance of an $MA(q)$ process is⁶

$$\begin{aligned} \gamma_0 &= \mathbb{E}[(Y_t - \mu)^2] = \mathbb{E}[(\varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_q\varepsilon_{t-q})^2] \\ &= \sigma^2 + \theta_1^2\sigma^2 + \theta_2^2\sigma^2 + \cdots + \theta_q^2\sigma^2 = (1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2)\sigma^2. \end{aligned} \quad (1.22)$$

The autocovariance for $j = 1, 2, \dots, q$

$$\begin{aligned} \gamma_j &= \mathbb{E}[(\varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q})(\varepsilon_{t-j} + \theta_1\varepsilon_{t-j-1} + \cdots + \theta_q\varepsilon_{t-j-q})] \\ &= \mathbb{E}[\theta_j\varepsilon_{t-j}^2 + \theta_1\theta_{j+1}\varepsilon_{t-j-1}^2 + \cdots + \theta_q\theta_{q-j}\varepsilon_{t-q}^2] \\ &= (\theta_j + \theta_1\theta_{j+1} + \theta_2\theta_{j+2} + \cdots + \theta_q\theta_{q-j})\sigma^2. \end{aligned} \quad (1.23)$$

³**Estimation:** the invertible representation uniquely identifies the MA parameters among observationally equivalent likelihoods. Furthermore, the effect of initial errors eventually die out, which is desirable in conditional MLE.

⁴**Forecasting:** invertibility allows expressing the process as $AR(\infty)$ through a convergent filter. Truncating the $AR(\infty)$, one can obtain forecasts from this representation.

⁵A noninvertible $MA(q)$ process admits an observationally equivalent invertible representation, obtained by reflecting the roots of the MA polynomial that lie inside the unit circle and appropriately adjusting the innovation variance. Both representations have the same autocovariance function and, under Gaussianity, the same distribution, since a Gaussian process is completely characterized by its mean and autocovariance function. Without Gaussianity, the two processes have the same second-order properties. This follows only if there are no roots **in** the unit circle.

⁶ ε_t uncorrelated.

For $j > q$, the autocovariance is zero. Thus

$$\gamma_j = \begin{cases} (\theta_j + \theta_1\theta_{j+1} + \theta_2\theta_{j+2} + \cdots + \theta_q\theta_{q-j})\sigma^2, & \text{for } j = 1, 2, \dots, q, \\ 0, & \text{for } j > q. \end{cases} \quad (1.24)$$

For example, for an MA(2) process:

$$\begin{aligned} \gamma_0 &= [1 + \theta_1^2 + \theta_2^2]\sigma^2, \\ \gamma_1 &= [\theta_1 + \theta_1\theta_2]\sigma^2, \\ \gamma_2 &= [\theta_2]\sigma^2, \\ \gamma_3 &= \gamma_4 = \cdots = 0. \end{aligned} \quad (1.25)$$

For any values of $(\theta_1, \theta_2, \dots, \theta_q)$, the MA(q) process is **covariance-stationary**. Condition (1.6) is satisfied for any $(\theta_1, \theta_2, \dots, \theta_q) \in \mathbb{R}^q$, so for white noise ε_t , the MA(q) process is also **ergodic** for the mean.

1.3.3 The MA(∞) process

The MA(q) process can be written $Y_t = \mu + \sum_{j=0}^q \theta_j \varepsilon_{t-j}$ with $\theta_0 = 1$. As $q \rightarrow \infty$, we have the MA(∞) process⁷:

$$Y_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} = \mu + \psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \cdots \quad (1.26)$$

The infinite sequence in (1.26) generates a **covariance-stationary** process provided that the sequence $\{\psi_j\}$ is **square summable** (Appendix 1.12.2), that is:

$$\sum_{j=0}^{\infty} \psi_j^2 < \infty, \quad (1.27)$$

but it is convenient to work with a stronger condition of **absolutely summability**:

$$\sum_{j=0}^{\infty} |\psi_j| < \infty. \quad (1.28)$$

Absolute summability implies square-summability, but the converse does not hold.

The mean and autocovariances of an MA(∞) process with absolutely summable coefficients can be calculated from a simple extrapolation of the results for an MA(q) process:

$$\mathbb{E}(Y_t) = \lim_{T \rightarrow \infty} \mathbb{E}(\mu + \psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \cdots + \psi_T \varepsilon_{t-T}) = \mu, \quad (1.29)$$

$$\begin{aligned} \gamma_0 &= \mathbb{E}[(Y_t - \mu)^2] = \lim_{T \rightarrow \infty} \mathbb{E}[(\psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \cdots + \psi_T \varepsilon_{t-T})^2] \\ &= \lim_{T \rightarrow \infty} (\psi_0^2 + \psi_1^2 + \psi_2^2 + \cdots + \psi_T^2) \sigma^2 \\ &= \sigma^2 \sum_{i=0}^{\infty} (\psi_i^2), \end{aligned} \quad (1.30)$$

$$\begin{aligned} \gamma_j &= \mathbb{E}[(Y_t - \mu)(Y_{t-j} - \mu)] = \sigma^2 (\psi_j \psi_0 + \psi_{j+1} \psi_1 + \psi_{j+2} \psi_2 + \psi_{j+3} \psi_3 + \cdots) \\ &= \sigma^2 \sum_{i=0}^{\infty} (\psi_{j+i} \psi_i) \end{aligned} \quad (1.31)$$

⁷We use ψ 's for the coefficients of an infinite-order moving average process and θ 's for the coefficients of a finite-order moving average process.

An MA(∞) process with the absolute summability property (1.28) has absolutely summable autocovariances (see proof on 1.75):

$$\sum_{j=0}^{\infty} |\gamma_j| < \infty, \quad (1.32)$$

being **ergodic** for the mean⁸. If the ε 's are Gaussian, then the process is ergodic for all moments. Square summability alone does not guarantee ergodicity, only absolute summability.

1.4 Autorregressive Processes

1.4.1 The First-order autorregressive process

An AR(1) process is given by

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t \quad (1.33)$$

If $|\phi| \geq 1$, then the effect of shocks ε_t accumulates over time, and there does not exist a covariance-stationary process for Y_t with finite variance. When $|\phi| < 1$, there is a covariance-stationary process Y_t satisfying the equation above, given by

$$\begin{aligned} Y_t &= (c + \varepsilon_t) + \phi(c + \varepsilon_{t-1}) + \phi^2(c + \varepsilon_{t-2}) + \phi^3(c + \varepsilon_{t-3}) + \cdots \\ &= \left[\frac{c}{1-\phi} \right] + \varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \phi^3\varepsilon_{t-3} + \cdots, \end{aligned} \quad (1.34)$$

Notice that this is an MA(∞) process with $\mu = \frac{c}{1-\phi}$ and $\psi_j = \phi^j$. When $|\phi| < 1$, the absolute summability condition is satisfied, since

$$\sum_{j=0}^{\infty} |\psi_j| = \sum_{j=0}^{\infty} |\phi^j| = \frac{1}{1-|\phi|} < \infty, \quad (1.35)$$

which ensures that the MA(∞) representation exists and the AR(1) process is ergodic for the mean (1.28) and stationary. The mean is:

$$\mathbb{E}(Y_t) = \left[\frac{c}{1-\phi} \right] + 0 + 0 + \cdots = \frac{c}{1-\phi} \quad (1.36)$$

The variance is

$$\gamma_0 = \mathbb{E}[(Y_t - \mu)^2] = \mathbb{E}[(\varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \cdots)^2] = (1 + \phi^2 + \phi^4 + \phi^6 + \cdots)\sigma^2 = \frac{\sigma^2}{1-\phi^2} \quad (1.37)$$

The j th autocovariance is

$$\begin{aligned} \gamma_j &= \mathbb{E}[(Y_t - \mu)(Y_{t-j} - \mu)] \\ &= \mathbb{E}[(\varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \cdots)(\varepsilon_{t-j} + \phi\varepsilon_{t-j-1} + \phi^2\varepsilon_{t-j-2} + \cdots)] \\ &= (\phi^j + \phi^{j+2} + \phi^{j+4} + \cdots)\sigma^2 = \phi^j \cdot \frac{\sigma^2}{1-\phi^2} \end{aligned} \quad (1.38)$$

The autocorrelation function is

$$\rho_j = \gamma_j / \gamma_0 = \phi^j, \quad (1.39)$$

⁸Ergodicity not guaranteed if square summable, i.e. $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

There are two main ways to derive the moments of a stationary AR(1) process. The first is to represent the process as an MA(∞), as above. The second is to assume covariance stationarity and derive the moments directly from the difference equation. From $\mathbb{E}(Y_t) = \mathbb{E}(Y_{t-1}) = \mu$, the mean is given by

$$\mu = c + \phi\mu + 0 \quad \Rightarrow \quad \mu = \frac{c}{1 - \phi}, \quad (1.40)$$

Notice that the formula $\mu = c/(1 - \phi)$ is not valid when $|\phi| \geq 1$, since stationarity was assumed in deriving it. To find the second moments of Y_t in a similar way, use

$$\begin{aligned} Y_t &= \mu(1 - \phi) + \phi Y_{t-1} + \varepsilon_t \\ Y_t - \mu &= \phi(Y_{t-1} - \mu) + \varepsilon_t. \end{aligned}$$

Now square both sides and take expectations, using $\mathbb{E}[(Y_{t-1} - \mu)\varepsilon_t] = 0$:

$$\begin{aligned} \mathbb{E}[(Y_t - \mu)^2] &= \phi^2 \mathbb{E}[(Y_{t-1} - \mu)^2] + 2\phi \mathbb{E}[(Y_{t-1} - \mu)\varepsilon_t] + \mathbb{E}(\varepsilon_t^2) \\ \gamma_0 &= \phi^2 \gamma_0 + 0 + \sigma^2 \quad \Rightarrow \quad \gamma_0 = \frac{\sigma^2}{1 - \phi^2} \end{aligned} \quad (1.41)$$

Similarly, we can multiply $(Y_t - \mu)$ by $(Y_{t-j} - \mu)$ for $j > 0$ and take expectations:

$$\begin{aligned} \mathbb{E}[(Y_t - \mu)(Y_{t-j} - \mu)] &= \phi \mathbb{E}[(Y_{t-1} - \mu)(Y_{t-j} - \mu)] + \mathbb{E}[\varepsilon_t(Y_{t-j} - \mu)] \\ \gamma_j &= \phi \gamma_{j-1}. \end{aligned}$$

where $(Y_{t-j} - \mu)$ is a linear function of $\varepsilon_{t-j}, \varepsilon_{t-j-1}, \dots$. Thus, for $j > 0$, $(Y_{t-j} - \mu)$ is uncorrelated with ε_t , implying $\mathbb{E}[\varepsilon_t(Y_{t-j} - \mu)] = 0$. Substituting for γ_{j-1} recursively yields:

$$\gamma_j = \phi^j \gamma_0, \quad (1.42)$$

which agrees with our earlier result that:

$$\rho_j = \gamma_j / \gamma_0 = \phi^j.$$

1.4.2 The Second-order autorregressive process

An AR(2) satisfies

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t, \quad (1.43)$$

being equivalently expressed as

$$\begin{aligned} (1 - \phi_1 L - \phi_2 L^2)Y_t &= c + \varepsilon_t \\ \Phi(L)Y_t &= c + \varepsilon_t \\ Y_t &= \Psi(L)c + \Psi(L)\varepsilon_t \end{aligned}$$

where the MA(∞) representation can be seen from

$$\Psi(L) = \Phi(L)^{-1} = (1 - \phi_1 L - \phi_2 L^2)^{-1} = \psi_0 + \psi_1 L + \psi_2 L^2 + \psi_3 L^3 + \dots$$

The AR(2) is stationary provided that the roots of $\Phi(L) = (1 - \phi_1 L - \phi_2 L^2)$ lie outside the unit circle. Its **mean** is:

$$\mu = \Psi(L)c = \frac{c}{1 - \phi_1 - \phi_2} \quad (1.44)$$

or we can assume the process is stationary and take expectations of the original equation:

$$\mu = c + \phi_1\mu + \phi_2\mu \quad \Rightarrow \quad \mu = \frac{c}{1 - \phi_1 - \phi_2}$$

To compute the **autocovariances**, rewrite the AR(2) as:

$$\begin{aligned} Y_t &= \mu(1 - \phi_1 - \phi_2) + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t \\ Y_t - \mu &= \phi_1(Y_{t-1} - \mu) + \phi_2(Y_{t-2} - \mu) + \varepsilon_t \end{aligned}$$

Multiplying both sides by $(Y_{t-j} - \mu)$ and taking expectations:

$$\gamma_j = \phi_1\gamma_{j-1} + \phi_2\gamma_{j-2} \quad \text{for } j = 1, 2, \dots \quad (1.45)$$

The autocorrelations are $(/\gamma_0)$:

$$\rho_j = \phi_1\rho_{j-1} + \phi_2\rho_{j-2} \quad \text{for } j = 1, 2, \dots \quad (1.46)$$

In particular, setting $j = 1$:

$$\rho_1 = \phi_1 + \phi_2\rho_1 \quad \Rightarrow \quad \rho_1 = \frac{\phi_1}{1 - \phi_2}$$

For $j = 2$:

$$\rho_2 = \phi_1\rho_1 + \phi_2$$

The **variance** comes from:

$$\begin{aligned} \mathbb{E}[(Y_t - \mu)^2] &= \phi_1\mathbb{E}[(Y_{t-1} - \mu)(Y_t - \mu)] + \phi_2\mathbb{E}[(Y_{t-2} - \mu)(Y_t - \mu)] + \mathbb{E}[\varepsilon_t(Y_t - \mu)] \\ \gamma_0 &= \phi_1\gamma_1 + \phi_2\gamma_2 + \sigma^2 \end{aligned} \quad (1.47)$$

which can be written as:

$$\begin{aligned} \gamma_0 &= \phi_1\rho_1\gamma_0 + \phi_2\rho_2\gamma_0 + \sigma^2 \\ \gamma_0 &= \phi_1 \left(\frac{\phi_1}{1 - \phi_2} \right) \gamma_0 + \phi_2 \left(\frac{\phi_1^2}{1 - \phi_2} + \phi_2 \right) \gamma_0 + \sigma^2 \\ &= \frac{(1 - \phi_2)\sigma^2}{(1 + \phi_2)(1 - \phi_2)^2 - \phi_1^2} \end{aligned}$$

1.4.3 The p th-order autoregressive process

A p th-order autoregression, denoted AR(p), satisfies

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t \quad (1.48)$$

$$\Phi(L)Y_t = c + \varepsilon_t. \quad (1.49)$$

Provided that the roots of

$$\Phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p = 0 \quad (1.50)$$

all lie outside the unit circle, we can write (1.48) in the MA(∞) covariance-stationary representation:

$$Y_t = \mu + \Psi(L)\varepsilon_t, \quad (1.51)$$

where

$$\Psi(L) = (1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p)^{-1}, \quad \sum_{j=0}^{\infty} |\psi_j| < \infty. \quad (1.52)$$

Assuming stationarity, the mean is found by taking expectations:

$$\begin{aligned} \mu &= c + \phi_1 \mu + \phi_2 \mu + \cdots + \phi_p \mu \\ \mu &= \frac{c}{1 - \phi_1 - \phi_2 - \cdots - \phi_p} \end{aligned} \quad (1.53)$$

Using this result, the process can be written as

$$Y_t - \mu = \phi_1(Y_{t-1} - \mu) + \phi_2(Y_{t-2} - \mu) + \cdots + \phi_p(Y_{t-p} - \mu) + \varepsilon_t$$

Autocovariances are obtained by multiplying both sides by $(Y_{t-j} - \mu)$ and taking expectations:

$$\gamma_j = \phi_1 \gamma_{j-1} + \phi_2 \gamma_{j-2} + \cdots + \phi_p \gamma_{j-p}, \quad j = 1, 2, \dots \quad (1.54)$$

$$\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \cdots + \phi_p \gamma_p + \sigma^2 \quad (1.55)$$

Yule-Walker equations

Dividing the general equation (1.54) by γ_0 gives the Yule-Walker equations:

$$\rho_j = \phi_1 \rho_{j-1} + \phi_2 \rho_{j-2} + \cdots + \phi_p \rho_{j-p}, \quad j = 1, 2, \dots \quad (1.56)$$

In matrix form, they are given by:

$$\begin{bmatrix} 1 & \rho_1 & \cdots & \rho_{p-1} \\ \rho_1 & 1 & \cdots & \rho_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{p-1} & \rho_{p-2} & \cdots & 1 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_p \end{bmatrix}.$$

The **Yule-Walker estimator** for the $\text{AR}(p)$ estimates $\phi_1, \dots, \phi_p$ by substituting ρ_j by their sample counterparts $\hat{\rho}_j$.

1.5 ARMA(p,q) processes

An $\text{ARMA}(p, q)$ model is represented by

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}; \quad (1.57)$$

or in lag operator form:

$$\begin{aligned} (1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p) Y_t &= c + (1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q) \varepsilon_t \\ \Phi(L) Y_t &= c + \Theta(L) \varepsilon_t \end{aligned} \quad (1.58)$$

Provided that the roots of $\Phi(L)$ lie outside the unit circle, the ARMA process admits the $\text{MA}(\infty)$ representation:

$$Y_t = \mu + \Psi(L) \varepsilon_t \quad (1.59)$$

where

$$\Psi(L) = \frac{1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q}{1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p}, \quad \sum_{j=0}^{\infty} |\psi_j| < \infty \quad (1.60)$$

and

$$\mu = \frac{c}{1 - \phi_1 - \phi_2 - \cdots - \phi_p} \quad (1.61)$$

It follows that the ARMA(p, q) can be written in terms of deviations from the mean:

$$Y_t - \mu = \phi_1(Y_{t-1} - \mu) + \phi_2(Y_{t-2} - \mu) + \cdots + \phi_p(Y_{t-p} - \mu) + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_q\varepsilon_{t-q} \quad (1.62)$$

Autocovariances are obtained by multiplying both sides by $(Y_{t-j} - \mu)$ and taking expectations. For $j > q$, the resulting equations take the form:

$$\gamma_j = \phi_1\gamma_{j-1} + \phi_2\gamma_{j-2} + \cdots + \phi_p\gamma_{j-p}, \quad j = q+1, q+2, \dots \quad (1.63)$$

This takes the same form as the autocovariances for an $AR(p)$ process, though the initial values $\gamma_0, \gamma_1, \dots, \gamma_q$ differ, as $\theta_j\varepsilon_{t-j}$ and Y_{t-j} are correlated for $j \leq q$.

There is a potential for redundant parameterizations in ARMA processes. For example, in an ARMA(p, q) model. Consider factoring the lag polynomial operators:

$$(1 - \lambda_1 L)(1 - \lambda_2 L) \cdots (1 - \lambda_p L)(Y_t - \mu) = (1 - \eta_1 L)(1 - \eta_2 L) \cdots (1 - \eta_q L)\varepsilon_t$$

Assume $|\lambda_i| < 1$ for all i , so that the process is covariance-stationary. If $(1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p)$ and $(1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q)$ have any roots in common, $\lambda_i = \eta_j$ for some i, j , then both sides can be divided by $(1 - \lambda_i L)$:

$$(1 - \phi_1^* L - \phi_2^* L^2 - \cdots - \phi_{p-1}^* L^{p-1})(Y_t - \mu) = (1 + \theta_1^* L + \theta_2^* L^2 + \cdots + \theta_{q-1}^* L^{q-1})\varepsilon_t \quad (1.64)$$

The stationary ARMA(p, q) process satisfying the previous expression is identical to an ARMA($p-1, q-1$) process.

1.6 Wold's Decomposition

Wold's Decomposition: Any covariance-stationary process Y_t can be decomposed into a deterministic component and a purely nondeterministic component. If Y_t is purely nondeterministic, it admits the MA(∞) representation

$$Y_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad (1.65)$$

where $\psi_0 = 1$, $\sum_{j=0}^{\infty} \psi_j^2 < \infty$, and ε_t is the innovation process, which is white noise with variance σ^2 . The moments of Y_t are

$$E(Y_t) = \mu, \quad \gamma_0 = \sigma^2 \sum_{j=0}^{\infty} \psi_j^2, \quad \gamma_k = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+k}.$$

In practice, the infinite-order Wold representation can be approximated parametrically by specifying an ARMA(p, q) process. Its corresponding approximated MA(∞) representation is obtained from

$$\Psi(L) = \sum_{j=0}^{\infty} \psi_j L^j = \frac{\Theta(L)}{\Phi(L)} = \frac{1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q}{1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p}.$$

1.7 Stationarity and Invertibility

Consider the ARMA(p, q):

$$\Phi(L)(Y_t - \mu) = \Theta(L)\varepsilon_t, \quad (1.66)$$

where $\Phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$, $\Theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$ and $\varepsilon_t \sim \text{wn}(0, \sigma^2)$.

Theorem 1.1. *The ARMA(p, q) process is **covariance-stationary** if all roots of $\Phi(L) = 0$ lie outside the unit circle, and **invertible** if all roots of $\Theta(L) = 0$ lie outside the unit circle.*

To see this, let λ_i^{-1} , $i = 1, \dots, p$, denote the roots of $\Phi(L) = 0$. Then

$$\begin{aligned} \Phi(L) &= \prod_{i=1}^p (1 - \lambda_i L), \\ \Phi(L)^{-1} &= \frac{1}{\prod_{i=1}^p (1 - \lambda_i L)} = \sum_{i=1}^p \frac{A_i}{1 - \lambda_i L} = \sum_{i=1}^p A_i \sum_{j=0}^{\infty} (\lambda_i L)^j = \sum_{j=0}^{\infty} \underbrace{\left(\sum_{i=1}^p A_i \lambda_i^j \right)}_{\psi_j} L^j \end{aligned}$$

where we have used partial fraction decomposition in the second equality. Assume at first a pure AR process (i.e. $\Theta(L) = 1$):

$$Y_t - \mu = \Phi(L)^{-1} \varepsilon_t = \sum_{j=0}^{\infty} \psi_j L^j \varepsilon_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}.$$

Thus, Y_t admits an MA(∞) representation. If all roots λ_i^{-1} lie outside the unit circle, then $|\lambda_i| < 1 \forall i$, and by the triangle inequality

$$\sum_{j=0}^{\infty} |\psi_j| = \sum_{j=0}^{\infty} \left| \sum_{i=1}^p A_i \lambda_i^j \right| \leq \sum_{i=1}^p |A_i| \sum_{j=0}^{\infty} |\lambda_i|^j = \sum_{i=1}^p \frac{|A_i|}{1 - |\lambda_i|} < \infty.$$

Hence, the MA(∞) coefficients are absolutely summable, which implies the process is **stationary**. For $\Theta(L) = \sum_{k=0}^q \theta_k L^k$, the result continues to hold since $\Theta(L)$ is a finite-order polynomial.

Similarly, if all roots of $\Theta(L) = 0$ lie outside the unit circle, the coefficients of $\Theta(L)^{-1}$ are absolutely summable. Hence,

$$\varepsilon_t = \Theta(L)^{-1} \Phi(L)(Y_t - \mu) = \sum_{j=0}^{\infty} \pi_j (Y_{t-j} - \mu),$$

with $\sum_{j=0}^{\infty} |\pi_j| < \infty$ and $\pi_0 = 1$, and the process is **invertible**. Thus, **stationarity** means that the ARMA(p, q) process admits an MA(∞) representation, while **invertibility** means that it admits an AR(∞) representation.

1.8 Box and Jenkins Modelling Philosophy

The Box and Jenkins method can be broken down into four steps:

1. Perform unit root tests and transform the data accordingly for stationarity (ex: diff). Also consider transformations for variance stability (ex: ln or $\sqrt{\cdot}$).
- (a) ACF, PACF, Q -statistics and t -statistics assume the data is stationary⁹

⁹The usual asymptotic distributions of the ACF, PACF, Q -statistics, and t -statistics fail.

- (b) Box Jenkins also assumes invertibility for estimation and forecasting.
- 2. Identification: select different candidate $\text{ARMA}(p, q)$ models.
 - (a) Plot the time series. Try to identify seasonality, outliers, missing values and structural breaks and make the appropriate adjustments.
 - (b) Plot ACF and the PACF. They may suggest several candidate models.
 - (c) Consider the parsimony principle (e.g. $\text{AR}(1)$ for $\text{MA}(\infty)$, common factor problem).
- 3. Estimate the parameters from the $\text{ARMA}(p, q)$ candidate models.
 - (a) Check for significance.
 - (b) Check for stationarity and invertibility.
 - (c) Estimation with the MA component is usually performed using a nonlinear search procedure. Consider that if it fails to converge rapidly, the parameters may be unstable, in which a small perturbation of the model alters its parameters. In this case try a more parsimonious representation, check for stationarity and invertibility, try different initializations.
- 4. Perform diagnostic analysis to confirm that the model is adequate
 - (a) Check for residual autocorrelation using ACF plot and Q -tests.
 - (b) Plot the residuals, and look for outliers and structural breaks, in which the data does not fit model well. In this case add dummy variables, consider a markov switching approach or consider intervention/transfer function analysis.
 - (c) Test for normality. Statistical tests, Q -plot.
 - (d) Check the variance of the residuals. In the presence of heteroskedasticity consider a GARCH approach. Furthermore, if the variance is increasing over time, consider remodelling with a $\ln$ transformation (e.g. GDP).
 - (e) You can also verify the out-of-sample forecasts and compare it with baseline alternatives.
 - (f) Check information criteria of candidate models.

After these four steps, you select the most satisfactory model among the candidate ones. We highlight that model selection is not always clear. There is a sentence in Enders [1995] that captures this model choosing issue: "These ambiguities may lead two equally skilled econometricians to estimate and forecast the same series using very different ARMA processes"

1.9 Box-Jenkins: Identification

1. ARIMA model identification starts from plotting the data and identifying seasonality, missings, outliers and structural breaks and eventual transformations. In this step, consider controlling for such eventualities.
2. Also test for unit roots and make the appropriate transformations for stationarity. Notice that the ACF of an undifferenced series is very persistent.
3. Then, plot the ACF and PACF to select candidate values for p and q . You may consider different portions of the data to check the robustness of the ACF and PACF. Considerable changes must be approached carefully, since they may arise from parameter instability and/or structural breaks.
4. Take M candidate models to perform estimation, diagnostics and forecasting.

1.9.1 ACF

An estimate of the population autocorrelation ρ_j is provided by the sample moments:

$$\hat{\rho}_j = \hat{\gamma}_j / \hat{\gamma}_0 \quad (1.67)$$

where

$$\begin{aligned} \hat{\gamma}_j &= \frac{1}{T} \sum_{t=j+1}^T (Y_t - \bar{Y})(Y_{t-j} - \bar{Y}) \quad \text{for } j = 0, 1, 2, \dots, T-1 \\ \bar{Y} &= \frac{1}{T} \sum_{t=1}^T Y_t \end{aligned}$$

The denominator in $\hat{\gamma}_j$ is T rather than $T-j$. Thus, for large j , the estimate $\hat{\gamma}_j$ is shrunk towards zero, consistent with the fact that the population autocovariances γ_j go to zero as $j \rightarrow \infty$.

Recall that for an MA(q) process, $\rho_j = 0$ for $j > q$ and for an AR(p), ρ_j will gradually decay toward zero as a mixture of exponentials or damped sinusoids¹⁰ To test whether $\rho_j = 0$ for $j = q+1, q+2, \dots$ we use the following. An approximate expression for the variance of $\hat{\rho}_j$ in a gaussian process is given by:

$$\text{Var}(\hat{\rho}_j) \simeq \frac{1}{T} \sum_{i=-\infty}^{\infty} [\rho_i^2 + \rho_{i+j}\rho_{i-j} - 4\rho_j\rho_i\rho_{i-j} + 2\rho_i^2\rho_j^2].$$

For an MA(q), we have $\rho_j = 0$ when $j > q$. Thus, for $j > q$, all terms vanish except the first, yielding

$$\text{Var}(\hat{\rho}_j) \approx \frac{1}{T} \left(1 + 2 \sum_{i=1}^q \rho_i^2 \right), \quad \text{for } j > q \quad (1.68)$$

For sufficiently large T and under the hypothesis that $\rho_j = 0$ for $j > q$, the distribution of $\hat{\rho}_j$ is approximately normal, with mean equal to zero and variance given by (1.68) From the TCL, an approximate confidence interval for ρ_j can be constructed as

$$\hat{\rho}_j \pm t_\gamma \sqrt{\text{Var}(\hat{\rho}_j)},$$

If the data is a **Gaussian white noise** ($q = 0$), the estimate $\hat{\rho}_j$ for any $j \neq 0$ should lie between $\pm 2/\sqrt{T}$ about 95% of the time. This is because $\rho_j = 0$ for $j > 0$ in this case, implying $\sqrt{\text{Var}(\hat{\rho}_j)} = 1/\sqrt{T}$. We can verify if the data is Gaussian white noise by plotting the autocorrelations $\hat{\rho}_j$ and checking if they lie inside the limits $\pm 2/\sqrt{T}$.

Summarizing:

1. AR(p): The ACF tails off toward zero as a mixture of exponential decays and/or damped sinusoids.
2. MA(q): The ACF cuts off after lag q , i.e., $\rho_j = 0$ for $j > q$.
3. ARMA(p, q): The ACF tails off toward zero as a mixture of exponential decays and/or damped sinusoids for $j > q$.

¹⁰See Section 3.2 of Shumway and Stoffer [2006] for an introduction to difference equations and why real roots lead to exponential decay, while complex conjugate roots lead to damped oscillations.

1.9.2 PACF

The j -th PACF is the correlation between Y_t and Y_{t-j} after removing the linear effects of the intermediate variables $Y_{t-1}, \dots, Y_{t-j+1}$. Accordingly, the j -th population partial autocorrelation $\alpha_j^{(j)}$ is the last coefficient in the linear projection

$$Y_t - \mu = \alpha_1^{(j)}(Y_{t-1} - \mu) + \alpha_2^{(j)}(Y_{t-2} - \mu) + \dots + \alpha_j^{(j)}(Y_{t-j} - \mu) + e_t^{(j)}. \quad (1.69)$$

The vector $\boldsymbol{\alpha}^{(j)} = (\alpha_1^{(j)}, \dots, \alpha_j^{(j)})'$ can be obtained from the Yule–Walker equations¹¹:

$$\begin{bmatrix} \alpha_1^{(j)} \\ \alpha_2^{(j)} \\ \vdots \\ \alpha_j^{(j)} \end{bmatrix} = \begin{bmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{j-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{j-2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{j-1} & \gamma_{j-2} & \cdots & \gamma_0 \end{bmatrix}^{-1} \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_j \end{bmatrix} \quad (1.70)$$

If the data were really generated by an $AR(p)$ process, then $\alpha_j^{(j)} = 0$ for $j = p+1, p+2, \dots$. By contrast, if the data were generated by an $MA(q)$, then $\alpha_j^{(j)}$ asymptotically approaches zero instead of cutting off abruptly, since the $MA(q)$ is represented by an $AR(\infty)$.

Alternatively to (1.70), a natural estimator of the j -th partial autocorrelation is the last coefficient in an OLS regression of Y_t on a constant and its j most recent values:

$$Y_{t+1} = \hat{c} + \hat{\alpha}_1^{(j)}Y_t + \hat{\alpha}_2^{(j)}Y_{t-1} + \dots + \hat{\alpha}_j^{(j)}Y_{t-j+1} + \hat{\varepsilon}_t \quad (1.71)$$

If the data were really generated by an $AR(p)$ process, then the sample estimate $\hat{\alpha}_j^{(j)}$ for $j = p+1, p+2, \dots$ would be asymptotically distributed as:

$$\hat{\alpha}_j^{(j)} \stackrel{a}{\sim} \mathcal{N}\left(0, \frac{1}{T}\right), \quad j = p+1, p+2, \dots$$

If T is large, $\hat{\alpha}_j^{(j)}$ is approximately Gaussian and we can build the limits $\pm 2/\sqrt{T}$ on a 95% Gaussian confidence interval. Moreover, if the data were really generated by an $AR(p)$ process, then $\hat{\alpha}_i^{(i)}$ and $\hat{\alpha}_l^{(l)}$ would be asymptotically independent for $i, l > p$. Summarizing:

1. $AR(p)$: PACF finite, 0 for $j > p$.
2. $MA(q)$: PACF infinite, gradually decays toward zero as a mixture of exponentials or damped sinusoids.
3. $ARMA(p, q)$: PACF infinite, behaves like the PACF of a MA process.

1.9.3 Information Criterion

The **Akaike Information Criterion (AIC)** for an $ARMA(p, q)$ model is given by

$$AIC(p, q) = \ln \hat{\sigma}_{p,q}^2 + \frac{2(p+q)}{T}, \quad (1.72)$$

The **Bayesian Information Criterion (BIC)** for an $ARMA(p, q)$ model is given by

$$BIC(p, q) = \ln \hat{\sigma}_{p,q}^2 + \frac{(p+q) \ln T}{T}. \quad (1.73)$$

¹¹The linear projection error $e_t^{(j)} = (Y_t - \mu) - \sum_{i=1}^j \alpha_i^{(j)}(Y_{t-i} - \mu)$ is orthogonal to each regressor, so $\mathbb{E}[e_t^{(j)}(Y_{t-k} - \mu)] = 0$ for $k = 1, \dots, j$. Under covariance stationarity, this gives $\gamma_k = \sum_{i=1}^j \alpha_i^{(j)} \gamma_{|i-k|}$, $k = 1, \dots, j$.

While the BIC is asymptotically consistent, the AIC is biased toward selecting an overparameterized model. However, in small samples, Monte Carlo studies show that the AIC can perform better than the BIC. Furthermore, notice that as $\ln(T) > 2$, the BIC will always select a more parsimonious model than the AIC, since the marginal cost of adding regressors is greater in the BIC than in the AIC.

If the AIC and the BIC select different models, you should be careful. Since the BIC selects a more parsimonious model, you should check if the residuals behave as white noise. And as the AIC selects a more parameterized model, you should check if the t -statistics of the coefficients are significantly different from 0.

- When comparing models with different lags, all must be estimated with the same sample size T .
- Model selection may proceed purely from AIC/BIC criteria. Setting limits P and Q for p and q , and compute AIC/BIC for all $0 \leq p \leq P$ and $0 \leq q \leq Q$. Then one chooses the model with smallest AIC/BIC. The problem with such approach is that it may select a nonparsimonious model or even a misspecified model that does not pass diagnostics.
- Thus, taking many candidate models and using AIC/BIC as auxiliary for model choosing seems to be the best practice.

1.10 Box-jenkins: Estimation

Chapter 2 is devoted entirely to estimation for ARMA models.

- Common approaches include the Method of Moments (MM), Ordinary Least Squares (OLS), and Maximum Likelihood Estimation (MLE). In practice, MM estimators usually do not possess good statistical properties and are often used only as initial values for iterative estimation procedures. Thereby, we will focus on OLS and MLE.
- After identifying p and q for M candidate models, one estimates the ARMA parameters and check their significance.
- Be advised that removing nonsignificant coefficients with nonorthogonal regressors alters the estimates and significance of the remaining coefficients. A more robust approach is to use coefficient regularization, which account for multicollinearity and perform feature selection.

1.11 Box-Jenkins: Diagnostics

After fitting an ARIMA model, residual diagnostics are used to assess **model adequacy**. These include:

- Checking the residuals for remaining serial correlation using the ACF plots and Q-tests.
- Performing tests of conditional heteroskedasticity.
- Test for departures from the assumed innovation distribution (e.g., the Jarque–Bera test). Another useful diagnostic is the Q–Q plot, which compares the empirical distribution with the assumed distribution of the residuals.
- Test for structural breaks and check for outliers.

If any of these assumptions are violated, respecify the model accordingly and repeat the diagnostic checks. We focus here on residual autocorrelation, using the residual ACF and the Ljung-Box test to **assess whether the residuals behave as white noise**.

1.11.1 ACF diagnostics on the residuals

Given the estimates $\hat{\Phi}$ and $\hat{\Theta}$ of an ARMA(p, q) model, the quantities

$$\hat{\varepsilon}_t = \hat{\Theta}^{-1}(B) \hat{\Phi}(B) Y_t$$

are the **residuals**. If the model is adequate, $\hat{\varepsilon}_t$ should be white noise and hence uncorrelated. Thus, under the H_0 of an adequate model we have $\hat{\rho}_k = \text{Corr}(\hat{\varepsilon}_t \hat{\varepsilon}_{t-k}) = 0$ with distribution

$$\hat{\rho}_k \sim N\left(0, \frac{1}{T}\right).$$

The sample autocorrelations are computed as

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T \hat{\varepsilon}_t \hat{\varepsilon}_{t-k}}{\sum_{t=1}^T \hat{\varepsilon}_t^2}$$

A practical rule is to compare $\hat{\rho}_k$ at lag k with the bounds $\pm 2/\sqrt{T}$ to detect departures from white noise, although these bounds may be inaccurate at small lags k ¹².

1.11.2 Ljung-Box Test

To further corroborate your conclusions, you can perform a **Ljung-Box test**. In addition to the residual ACF, we can use the **Q-statistic** to test if a group of residual autocorrelations is significantly different from 0. The original statistic of Box-Pierce is given by

$$Q(K) = \sum_{k=1}^K \hat{\rho}_k^2,$$

where under $H_0 : \rho_k = 0$ for $k = 1, \dots, K$. Notice that Q has an asymptotic χ_{K-p-q}^2 distribution. If the calculated $Q(K)$ exceeds the appropriate value in a χ_{K-p-q}^2 table, we can reject H_0 in favor of H_1 : at least one ρ_k is not zero.

A problem with the above Q -statistic is that it performs poorly in small samples. A superior performance is obtained with the **modified Q-statistic**:

$$Q(K) = n(n+2) \sum_{k=1}^K \frac{\hat{\rho}_k^2}{n-k},$$

which also has an asymptotic χ_{K-p-q}^2 distribution¹³. Again, $\hat{\rho}_k$ represents the sample autocorrelations of the residuals. Therefore, to perform the Ljung-Box test, first compute the residuals from the estimated ARMA(p, q) model and then calculate the Q -statistic. Finally, compare Q_s with the critical value of a χ_{K-p-q}^2 distribution. The null hypothesis of white noise in the residuals is rejected for large values of $Q(K)$. In practice, the first 8,12,16,20 autocorrelations $\hat{\rho}_k$ are usually used in the calculation.

1.11.3 Plot checking

Another diagnostic check involves **plotting**

$$\{\varepsilon_t/\sigma\}_{t=1}^T$$

For a simple **normality** diagnostics, if $\varepsilon_t \sim N(0, \sigma)$, less than 5% of the $\{\varepsilon_t/\sigma\}_{t=1}^T$ must be within the $[-2, 2]$ band. One can even check visually for **structural breaks and outliers** in $\{\varepsilon_t/\sigma\}_{t=1}^T$.

¹²Autocorrelations at smaller lags incorporate parameter-estimation uncertainty. At higher lags, this influence tends to vanish for stationary and invertible models.

¹³If a constant is included the degrees of freedom are reduced to $K - p - q - 1$.

1.12 Appendix - Chapter 1

1.12.1 Properties of the Autocovariance Function

Let $\gamma_j = \text{Cov}(Y_t, Y_{t-j})$ denote the autocovariance function of a stationary process $\{Y_t\}$.

Proof of (1): $\gamma_0 > 0$. By definition, $\gamma_0 = \text{Cov}(Y_t, Y_t) = \text{Var}(Y_t)$. Since the variance of a non-degenerate random variable is positive, we have $\gamma_0 > 0$.

Proof of (2) : $\gamma_{-j} = \gamma_j$. This follows from symmetry:

$$\begin{aligned}\gamma_{-j} &= \text{Cov}(Y_t, Y_{t+j}) \\ &= \text{Cov}(Y_{t+j}, Y_t) = \gamma_j.\end{aligned}$$

Notice that this does not hold in general for vector processes, for which $\mathbf{\Gamma}_{-j} = \mathbf{\Gamma}_j^\top$.

Proof of (3) : $|\gamma_j| \leq \gamma_0$. By the Cauchy-Schwarz inequality,

$$|\gamma_j| = |\text{Cov}(Y_t, Y_{t-j})| \leq \sqrt{\text{Var}(Y_t) \text{Var}(Y_{t-j})}.$$

Take $\text{Var}(Y_t) = \text{Var}(Y_{t-j}) = \gamma_0$. Thus $|\gamma_j| \leq \gamma_0$.

Proof of (4) : γ_τ is non-negative definite. Consider the random variable $Z = \sum_{j=1}^n a_j Y_j$. Since variances are non-negative,

$$\begin{aligned}\text{Var}(Z) &= \text{Var}\left(\sum_{j=1}^n a_j Y_j\right) = \sum_{j=1}^n \sum_{k=1}^n a_j a_k \text{Cov}(Y_j, Y_k) \\ &= \sum_{j=1}^n \sum_{k=1}^n a_j a_k \gamma_{j-k} \geq 0.\end{aligned}$$

Therefore, the autocovariance function is non-negative definite.

1.12.2 Convergence Results for MA(∞)

The proofs presented below are based on the Appendix to Chapter 3 of Hamilton [1994].

(1) Absolute summability $\sum_{j=0}^{\infty} |\psi_j| < \infty$ implies square-summability $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

Suppose that $\{\psi_j\}_{j=0}^{\infty}$ is absolutely summable. Then there exists an $N < \infty$ such that $|\psi_j| < 1$ for all $j \geq N$, implying $\psi_j^2 < |\psi_j|$ for all $j \geq N$. Then

$$\sum_{j=0}^{\infty} \psi_j^2 = \sum_{j=0}^{N-1} \psi_j^2 + \sum_{j=N}^{\infty} \psi_j^2 < \sum_{j=0}^{N-1} \psi_j^2 + \sum_{j=N}^{\infty} |\psi_j|.$$

But $\sum_{j=0}^{N-1} \psi_j^2$ is finite, since N is finite, and $\sum_{j=N}^{\infty} |\psi_j|$ is finite, since $\{\psi_j\}$ is absolutely summable. Hence $\sum_{j=0}^{\infty} \psi_j^2 < \infty$, establishing that $\sum_{j=0}^{\infty} |\psi_j| < \infty$ implies $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

(2) Square-summability does *not* imply absolute summability

For an example, consider $\psi_j = 1/j$ for $j = 1, 2, \dots$. Notice that $1/j > 1/x$ for all $x > j$. Thus, integrating both sides over the interval $(j, j+1]$ yields

$$\int_j^{j+1} \frac{1}{j} dx = \frac{1}{j}(j+1-j) = \frac{1}{j} > \int_j^{j+1} \frac{1}{x} dx,$$

and so

$$\sum_{j=1}^N \frac{1}{j} > \sum_{j=1}^N \int_j^{j+1} \frac{1}{x} dx = \int_1^{N+1} \frac{1}{x} dx = \ln(N+1) - \ln(1) = \ln(N+1),$$

which diverges to ∞ as $N \rightarrow \infty$. Hence $\{\psi_j\}_{j=1}^\infty$ for $\psi_j = 1/j$ is not absolutely summable. However, it is square-summable, since $\frac{1}{j^2} \leq \frac{1}{x^2}$ for $x \in [j-1, j]$, meaning that

$$1/j^2 < \int_{j-1}^j \frac{1}{x^2} dx,$$

Using $\sum_{j=2}^N \frac{1}{j^2} < \sum_{j=2}^N \int_{j-1}^j \frac{1}{x^2} dx = \int_1^N \frac{1}{x^2} dx$ and adding 1 to both sides:

$$\sum_{j=1}^N \frac{1}{j^2} < 1 + \int_1^N \frac{1}{x^2} dx = 1 + \left(-\frac{1}{x}\right) \Big|_1^N = 2 - \frac{1}{N},$$

which converges to 2 as $N \rightarrow \infty$. Hence $\{\psi_j\}_{j=1}^\infty$ for $\psi_j = 1/j$ is square-summable.

(3) Square-summability of the coefficients ψ_j implies that the $MA(\infty)$ representation generates a mean square convergent random variable.

The Cauchy criterion states that $\sum_{j=0}^\infty a_j$ converges to a finite number if and only if, for any $\varepsilon > 0$, there exists a suitably large integer N such that, for any integer $M > N$,

$$\left| \sum_{j=0}^M a_j - \sum_{j=0}^N a_j \right| < \varepsilon.$$

In words, once we have summed N terms, calculating the sum out to a larger number M does not change the total by any more than an arbitrarily small number ε .

For the $MA(\infty)$ stochastic process, the question is whether $\sum_{j=0}^\infty \psi_j \varepsilon_{t-j}$ converges in mean square to some finite random variable $Y_t - \mu$ as $T \rightarrow \infty$. In this case, the Cauchy criterion states that $\sum_{j=0}^\infty \psi_j \varepsilon_{t-j}$ converges if and only if, for any $\varepsilon > 0$, there exists a suitably large integer N such that for any integer $M > N$,

$$\mathbb{E} \left(\sum_{j=0}^M \psi_j \varepsilon_{t-j} - \sum_{j=0}^N \psi_j \varepsilon_{t-j} \right)^2 < \varepsilon.$$

In words, once N terms have been summed, the difference between that sum and the one obtained from summing to M is a random variable whose mean and variance are both arbitrarily close to zero. Taking $\varepsilon_t \sim w.n.(0, \sigma^2)$, the left-side becomes

$$\begin{aligned} \mathbb{E}[\psi_M \varepsilon_{t-M} + \psi_{M-1} \varepsilon_{t-M+1} + \cdots + \psi_{N+1} \varepsilon_{t-N-1}]^2 &= (\psi_M^2 + \psi_{M-1}^2 + \cdots + \psi_{N+1}^2) \sigma^2 \\ &= \left[\sum_{j=0}^M \psi_j^2 - \sum_{j=0}^N \psi_j^2 \right] \sigma^2. \end{aligned} \quad (1.74)$$

But if $\sum_{j=0}^\infty \psi_j^2$ converges (square summability), then by the Cauchy criterion the right side of (1.74) may be made as small as desired by choice of a suitably large N . Thus the infinite series $\sum_{j=0}^\infty \psi_j \varepsilon_{t-j}$ in the $MA(\infty)$ converges in mean square to $Y_t - \mu$ provided that $\sum_{j=0}^\infty \psi_j^2 < \infty$ is satisfied; and also $\sigma^2 < \infty$ (1.3).

(4) $\sum_{j=0}^\infty |\psi_j| < \infty$ implies $\sum_{j=0}^\infty |\gamma_j| < \infty$ (that is, the process is ergodic for the mean)

Write the autocovariance as

$$\gamma_j = \sigma^2 \sum_{k=0}^{\infty} \psi_{j+k} \psi_k. \implies |\gamma_j| = \sigma^2 \left| \sum_{k=0}^{\infty} \psi_{j+k} \psi_k \right|.$$

A key property of the absolute value operator is the triangle inequality, in which $|a + b + c| \leq |a| + |b| + |c|$, implying

$$\sum_{j=0}^{\infty} |\gamma_j| \leq \sigma^2 \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} |\psi_{j+k} \psi_k| = \sigma^2 \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} |\psi_{j+k}| \cdot |\psi_k| = \sigma^2 \sum_{k=0}^{\infty} |\psi_k| \sum_{j=0}^{\infty} |\psi_{j+k}|.$$

But there exists an $M < \infty$ such that $\sum_{j=0}^{\infty} |\psi_j| < M$, and therefore $\sum_{j=0}^{\infty} |\psi_{j+k}| < M$ for $k = 0, 1, 2, \dots$, meaning that

$$\sum_{j=0}^{\infty} |\gamma_j| < \sigma^2 \sum_{k=0}^{\infty} |\psi_k| \cdot M < \sigma^2 M^2 < \infty. \quad (1.75)$$

Hence absolute summability implies that the process is ergodic for the mean. Additionally, finite second moments condition must be satisfied (1.3) in proofs (3) and (4) for stationarity and consequently ergodicity. Note that square summability alone does not guarantee ergodicity, only absolute summability.

Chapter 2

ARMA - Estimation

2.1 Introduction

Denote the ARMA(p, q) model by

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j}, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

Let $\theta \equiv (c, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q, \sigma^2)'$ represent the vector of population parameters and $(y_1, y_2, \dots, y_T)$ be the observed sample. Write the joint probability density as

$$f_{Y_T, Y_{T-1}, \dots, Y_1}(y_T, y_{T-1}, \dots, y_1; \theta). \quad (2.1)$$

The MLE of θ is the value for which (2.1) is maximized. Given the distribution of the residuals, e.g. $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$, we can calculate the likelihood function and find $\hat{\theta}_{MLE}$ that maximizes this function.

2.2 ARMA(p,q)

2.2.1 AR(1)

Assume stationarity ($|\phi| < 1$) in the AR(1) model:

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2).$$

Further assume from stationarity that Y_1 follows the unconditional distribution $Y_1 \sim N\left(\frac{c}{1-\phi}, \frac{\sigma^2}{1-\phi^2}\right)$:

$$f_{Y_1}(y_1; \theta) = \frac{1}{\sqrt{2\pi \sigma^2 / (1-\phi^2)}} \exp \left[-\frac{\left(y_1 - \frac{c}{1-\phi}\right)^2}{2\sigma^2 / (1-\phi^2)} \right]. \quad (2.2)$$

In general, given Y_1 and the conditional distribution of $Y_t | Y_{t-1}$

$$Y_t | Y_{t-1} \sim N(c + \phi Y_{t-1}, \sigma^2), \quad t \geq 2. \quad (2.3)$$

with conditional density

$$\begin{aligned} f_{Y_t | Y_{t-1}, \dots, Y_1}(y_t | y_{t-1}, \dots, y_1; \theta) &= f_{Y_t | Y_{t-1}}(y_t | y_{t-1}; \theta) \\ &= \frac{1}{\sqrt{2\pi \sigma^2}} \exp \left[-\frac{(y_t - c - \phi y_{t-1})^2}{2\sigma^2} \right]. \end{aligned} \quad (2.4)$$

The joint density of the first t observations then is

$$f_{Y_t, Y_{t-1}, \dots, Y_1}(y_t, y_{t-1}, \dots, y_1; \theta) = f_{Y_t|Y_{t-1}}(y_t | y_{t-1}; \theta) f_{Y_{t-1}, \dots, Y_1}(y_{t-1}, \dots, y_1; \theta). \quad (2.5)$$

The likelihood of the complete sample can thus be calculated as

$$L(\theta) = f_{Y_1}(y_1; \theta) \prod_{t=2}^T f_{Y_t|Y_{t-1}}(y_t | y_{t-1}; \theta). \quad (2.6)$$

The log-likelihood function is

$$\mathcal{L}(\theta) = \log f_{Y_1}(y_1; \theta) + \sum_{t=2}^T \log f_{Y_t|Y_{t-1}}(y_t | y_{t-1}; \theta). \quad (2.7)$$

Substituting (2.2) and (2.4) into the log-likelihood, we get

$$\begin{aligned} \mathcal{L}(\theta) = & -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log\left(\frac{\sigma^2}{1-\phi^2}\right) - \frac{\left[y_1 - \frac{c}{1-\phi}\right]^2}{2\sigma^2/(1-\phi^2)} \\ & - \frac{T-1}{2} \log(2\pi) - \frac{T-1}{2} \log(\sigma^2) - \sum_{t=2}^T \frac{(y_t - c - \phi y_{t-1})^2}{2\sigma^2}. \end{aligned} \quad (2.8)$$

to be maximized w.r.t. $\theta = \{c, \phi, \sigma^2\}$. Alternatively, in **matrix form**, write $\mathbf{y}_{(T \times 1)} = (y_1, y_2, \dots, y_T)'$ where

$$\begin{aligned} E(\mathbf{Y}) &= \boldsymbol{\mu} \\ E[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})'] &= \boldsymbol{\Omega}. \end{aligned}$$

Given that the autocovariance function of the AR(1) process is $E[(Y_t - \mu)(Y_{t-j} - \mu)] = \frac{\sigma^2 \phi^{|j|}}{1-\phi^2}$, the variance-covariance matrix $\boldsymbol{\Omega}$ can be written as

$$\boldsymbol{\Omega} = \sigma^2 \mathbf{V}, \quad (2.9)$$

where $\mathbf{V}$ is the correlation matrix with entries

$$V_{ij} = \frac{\phi^{|i-j|}}{1-\phi^2}. \quad (2.10)$$

Viewing y as a single draw from a $\mathcal{N}(\mu, \Omega)$ distribution, the sample likelihood is:

$$f_Y(y; \theta) = (2\pi)^{-T/2} |\Omega^{-1}|^{1/2} \exp \left[-\frac{1}{2} (y - \mu)' \Omega^{-1} (y - \mu) \right], \quad (2.11)$$

with log likelihood

$$\mathcal{L}(\theta) = (-T/2) \log(2\pi) + \frac{1}{2} \log |\Omega^{-1}| - \frac{1}{2} (y - \mu)' \Omega^{-1} (y - \mu). \quad (2.12)$$

Note that (2.8) and (2.12) are the same, but (2.8) is computationally preferable for not requiring the inversion of a $T \times T$ matrix.

MLE for the exact likelihood

Finding the MLE $\hat{\theta}$ from the **exact likelihood** (2.8) results is a system of nonlinear equations in θ and $(y_1, y_2, \dots, y_T)$, for which there is no analytical solution for θ in terms of $(y_1, y_2, \dots, y_T)$. Maximization of (2.8) thus requires numerical iterative procedures.

MLE for the conditional likelihood

The **conditional likelihood** (on y_1) is given by

$$f_{Y_T, Y_{T-1}, \dots, Y_2 | Y_1}(y_T, y_{T-1}, \dots, y_2 | y_1; \theta) = \prod_{t=2}^T f_{Y_t | Y_{t-1}}(y_t | y_{t-1}; \theta), \quad (2.13)$$

with the objective then being to maximize

$$\begin{aligned} & \log f_{Y_T, Y_{T-1}, \dots, Y_2 | Y_1}(y_T, y_{T-1}, \dots, y_2 | y_1; \theta) \\ &= - \left[\frac{(T-1)}{2} \right] \log(2\pi) - \left[\frac{(T-1)}{2} \right] \log(\sigma^2) - \sum_{t=2}^T \frac{(y_t - c - \phi y_{t-1})^2}{2\sigma^2} \end{aligned} \quad (2.14)$$

Maximization of (2.14) with respect to c and ϕ is equivalent to minimization of

$$\sum_{t=2}^T (y_t - c - \phi y_{t-1})^2,$$

which is achieved by OLS regression of y_t on a constant and y_{t-1} . The conditional maximum likelihood estimates of c and ϕ are therefore given by

$$\begin{bmatrix} \hat{c} \\ \hat{\phi} \end{bmatrix} = \begin{bmatrix} T-1 & \sum y_{t-1} \\ \sum y_{t-1} & \sum y_{t-1}^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y_t \\ \sum y_t y_{t-1} \end{bmatrix}, \quad (2.15)$$

where we derived *wrt* c and ϕ and equated to zero, forming a systems of equations, finally inverting the matrix. The derivative for σ^2 on the conditional MLE is:

$$-\frac{(T-1)}{2\sigma^2} + \sum_{t=2}^T \frac{(y_t - \hat{c} - \hat{\phi} y_{t-1})^2}{2\sigma^4} = 0 \implies \hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=2}^T (y_t - \hat{c} - \hat{\phi} y_{t-1})^2. \quad (2.16)$$

which is the average squared residual from the OLS regression. The **conditional maximum likelihood estimates** are trivial to compute, in contrast to the **exact maximum likelihood estimates**. The exact MLE and conditional MLE turn out to have the same large-sample distribution, provided that $|\phi| < 1$. And when $|\phi| > 1$, the conditional MLE continues to provide consistent estimates, whereas maximization of (2.8) does not. This is because (2.8) is derived from the AR(1) equation, which does not accurately describe the density of Y_1 when $|\phi| > 1$. For these reasons, in most applications the parameters of an autoregression are estimated by **OLS**¹ rather than **exact MLE**.

2.2.2 AR(p)

For the Gaussian AR(p) process,

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim \text{i.i.d. } N(0, \sigma^2)$$

we have $\theta = (c, \phi_1, \phi_2, \dots, \phi_p, \sigma^2)$. The first p observations in the sample $(y_1, y_2, \dots, y_p)$ are collected in a $(p \times 1)$ vector y_p , with mean

$$\mu = \frac{c}{1 - \phi_1 - \phi_2 - \dots - \phi_p}$$

¹Conditional MLE.

and variance :

$$\sigma^2 V_p = \begin{bmatrix} \gamma_0 & \gamma_1 & \gamma_2 & \cdots & \gamma_{p-1} \\ \gamma_1 & \gamma_0 & \gamma_1 & \cdots & \gamma_{p-2} \\ \gamma_2 & \gamma_1 & \gamma_0 & \cdots & \gamma_{p-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma_{p-1} & \gamma_{p-2} & \gamma_{p-3} & \cdots & \gamma_0 \end{bmatrix}$$

The density of the first p observations is that of a $\mathcal{N}(\mu_p, \sigma^2 V_p)$ variable:

$$\begin{aligned} f_{Y_p, Y_{p-1}, \dots, Y_1}(y_p, y_{p-1}, \dots, y_1; \theta) &= (2\pi)^{-p/2} |\sigma^2 V_p|^{-1/2} \exp \left[-\frac{1}{2\sigma^2} (y_p - \mu_p)' V_p^{-1} (y_p - \mu_p) \right] \\ &= (2\pi)^{-p/2} (\sigma^2)^{-p/2} |V_p|^{-1/2} \exp \left[-\frac{1}{2\sigma^2} (y_p - \mu_p)' V_p^{-1} (y_p - \mu_p) \right] \end{aligned}$$

For $t \geq p$, the observation $y_t | y_{1:t-1}$ is Gaussian with mean

$$c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p}$$

and variance σ^2 . Therefore the conditional likelihood is:

$$\begin{aligned} f_{Y_t | Y_{t-1}, \dots, Y_1}(y_t | y_{t-1}, \dots, y_1; \theta) &= f_{Y_t | Y_{t-1}, \dots, Y_{t-p}}(y_t | y_{t-1}, \dots, y_{t-p}; \theta) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_t - c - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \cdots - \phi_p y_{t-p})^2}{2\sigma^2} \right] \end{aligned}$$

The **exact likelihood function for the complete sample** is then

$$\begin{aligned} f_{Y_T, Y_{T-1}, \dots, Y_1}(y_T, y_{T-1}, \dots, y_1; \theta) &= f_{Y_p, Y_{p-1}, \dots, Y_1}(y_p, y_{p-1}, \dots, y_1; \theta) \\ &\quad \times \prod_{t=p+1}^T f_{Y_t | Y_{t-1}, \dots, Y_{t-p}}(y_t | y_{t-1}, \dots, y_{t-p}; \theta) \end{aligned}$$

with log likelihood:

$$\begin{aligned} \mathcal{L}(\theta) &= \log f_{Y_T, Y_{T-1}, \dots, Y_1}(y_T, y_{T-1}, \dots, y_1; \theta) \\ &= -\frac{p}{2} \log(2\pi) - \frac{p}{2} \log(\sigma^2) + \frac{1}{2} \log |V_p| - \frac{1}{2\sigma^2} (y_p - \mu_p)' V_p^{-1} (y_p - \mu_p) - \frac{T-p}{2} \log(2\pi) \\ &\quad - \frac{T-p}{2} \log(\sigma^2) - \sum_{t=p+1}^T \frac{(y_t - c - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \cdots - \phi_p y_{t-p})^2}{2\sigma^2} \\ &= -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma^2) + \frac{1}{2} \log |V_p| - \frac{1}{2\sigma^2} (y_p - \mu_p)' V_p^{-1} (y_p - \mu_p) \\ &\quad - \sum_{t=p+1}^T \frac{(y_t - c - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \cdots - \phi_p y_{t-p})^2}{2\sigma^2} \end{aligned} \tag{2.17}$$

Maximization of the exact likelihood function is numerical. In contrast, the **conditional log likelihood function** (on y_p) is:

$$\begin{aligned} \log f_{Y_T, Y_{T-1}, \dots, Y_{p+1} | Y_p, \dots, Y_1}(y_T, y_{T-1}, \dots, y_{p+1} | y_p, \dots, y_1; \theta) &= \\ &= -\frac{T-p}{2} \log(2\pi) - \frac{T-p}{2} \log(\sigma^2) - \sum_{t=p+1}^T \frac{(y_t - c - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \cdots - \phi_p y_{t-p})^2}{2\sigma^2} \end{aligned} \tag{2.18}$$

The values of $c, \phi_1, \phi_2, \dots, \phi_p$ that maximize the conditional log-likelihood (2.18) are the same as those that minimize

$$\sum_{t=p+1}^T (y_t - c - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \dots - \phi_p y_{t-p})^2$$

which is equivalent to OLS regression of y_t on a constant and p of its own lagged values. The conditional MLE of σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{T-p} \sum_{t=p+1}^T \left(y_t - \hat{c} - \hat{\phi}_1 y_{t-1} - \hat{\phi}_2 y_{t-2} - \dots - \hat{\phi}_p y_{t-p} \right)^2$$

The exact maximum likelihood estimates and the conditional maximum likelihood estimates again have the same large-sample distribution².

2.2.3 MA(1)

Consider the Gaussian MA(1) process

$$Y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim \text{i.i.d. } N(0, \sigma^2)$$

Let $\theta = (\mu, \theta, \sigma^2)'$. If the value of ε_{t-1} was known, then $Y_t \mid \varepsilon_{t-1} \sim N(\mu + \theta \varepsilon_{t-1}, \sigma^2)$ and

$$f_{Y_t \mid \varepsilon_{t-1}}(y_t \mid \varepsilon_{t-1}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_t - \mu - \theta \varepsilon_{t-1})^2}{2\sigma^2} \right]$$

Suppose that we knew for certain that $\varepsilon_0 = 0$. Then $Y_1 \mid \varepsilon_0 = 0 \sim N(\mu, \sigma^2)$. Moreover, given observation of y_1 , the value of $\varepsilon_1 = y_1 - \mu$ is known with certainty. Then, for $Y_2 \mid Y_1, \varepsilon_0$:

$$f_{Y_2 \mid Y_1, \varepsilon_0=0}(y_2 \mid y_1, \varepsilon_0 = 0; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_2 - \mu - \theta \varepsilon_1)^2}{2\sigma^2} \right]$$

Since ε_1 is known, ε_2 can be calculated from $\varepsilon_2 = y_2 - \mu - \theta \varepsilon_1$. Proceeding in this fashion, given $\varepsilon_0 = 0$, the full sequence $\{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_T\}$ can be calculated from $\{y_1, y_2, \dots, y_T\}$ by iterating on

$$\varepsilon_t = y_t - \mu - \theta \varepsilon_{t-1} \quad (2.19)$$

for $t = 1, 2, \dots, T$, starting from $\varepsilon_0 = 0$. The **conditional density** of the t th observation can then be calculated as

$$f_{Y_t \mid Y_{t-1}, Y_{t-2}, \dots, Y_1, \varepsilon_0=0}(y_t \mid y_{t-1}, y_{t-2}, \dots, y_1, \varepsilon_0 = 0; \theta) = f_{Y_t \mid \varepsilon_{t-1}}(y_t \mid \varepsilon_{t-1}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{\varepsilon_t^2}{2\sigma^2} \right] \quad (2.20)$$

The **conditional sample likelihood** would then be the product of these individual densities:

$$\begin{aligned} & f_{Y_T, Y_{T-1}, \dots, Y_1 \mid \varepsilon_0=0}(y_T, y_{T-1}, \dots, y_1 \mid \varepsilon_0 = 0; \theta) \\ &= f_{Y_1 \mid \varepsilon_0=0}(y_1 \mid \varepsilon_0 = 0; \theta) \prod_{t=2}^T f_{Y_t \mid Y_{t-1}, \dots, Y_1, \varepsilon_0=0}(y_t \mid y_{t-1}, \dots, y_1, \varepsilon_0 = 0; \theta) \end{aligned} \quad (2.21)$$

²Besides least squares estimators, Yule-Walker estimators can be obtained from the sample Yule-Walker equations, with good properties and computable via the Durbin-Levinson algorithm.

The corresponding **conditional log likelihood** is then

$$\begin{aligned}\mathcal{L}(\theta) &= \log f_{Y_T, Y_{T-1}, \dots, Y_1 | \varepsilon_0=0}(y_T, y_{T-1}, \dots, y_1 | \varepsilon_0 = 0; \theta) \\ &= -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma^2) - \sum_{t=1}^T \frac{\varepsilon_t^2}{2\sigma^2}\end{aligned}\quad (2.22)$$

Maximizing (2.22) results in a nonlinear function of $\theta = (\mu, \theta, \sigma^2)'$, requiring numerical optimization. Iteration given ε_0 will result in

$$\begin{aligned}\varepsilon_t &= (y_t - \mu) - \theta(y_{t-1} - \mu) + \theta^2(y_{t-2} - \mu) - \dots \\ &\quad + (-1)^{t-1} \theta^{t-1} (y_1 - \mu) + (-1)^t \theta^t \varepsilon_0\end{aligned}$$

If $|\theta| < 1$, imposing $\varepsilon_0 = 0$ will die out and the conditional likelihood will give a good approximation to the unconditional likelihood for a reasonably large sample size. By contrast, if $|\theta| > 1$, the consequences of imposing $\varepsilon_0 = 0$ accumulate over time and the conditional approach is not reasonable in such a case. This is one of the reasons why **invertibility** is a desirable property for estimation, as it allows representing $\{\varepsilon_t\}$ through the $\{y_t\}$ series.

If numerical optimization of the conditional log likelihood results in a value of θ that exceeds 1 in absolute value, the results must be discarded. The numerical optimization could be attempted again with the reciprocal of $\hat{\theta}$ used as a starting value for the numerical search procedure.

Exact Likelihood

For MA processes, the exact likelihood treats the initial residuals as additional parameters to be estimated. In AR processes, it simply augments the likelihood with the initial observed variables. There are two algorithms for calculating the exact likelihood of an MA process. The first is the triangular factorization of the variance-covariance matrix (Hamilton 5.4) and the second is the **Kalman Filter** (Hamilton 13).

2.2.4 MA(q)

For the MA(q) process,

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$$

a simple approach is to condition on the assumption that the first q values for ε were all zero:

$$\varepsilon_0 = \varepsilon_{-1} = \dots = \varepsilon_{-q+1} = 0$$

From these starting values we can iterate on

$$\varepsilon_t = y_t - \mu - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q} \quad (2.23)$$

for $t = 1, 2, \dots, T$. Let ε_0 denote the $(q \times 1)$ vector $(\varepsilon_0, \varepsilon_{-1}, \dots, \varepsilon_{-q+1})'$. The **conditional log likelihood** is then

$$\begin{aligned}\mathcal{L}(\theta) &= \log f_{Y_T, Y_{T-1}, \dots, Y_1 | \varepsilon_0=0}(y_T, y_{T-1}, \dots, y_1 | \varepsilon_0 = 0; \theta) \\ &= -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma^2) - \sum_{t=1}^T \frac{\varepsilon_t^2}{2\sigma^2}\end{aligned}\quad (2.24)$$

where $\theta = (\mu, \theta_1, \theta_2, \dots, \theta_q, \sigma^2)'$. The expression above is useful only if all values of z for which

$$1 + \theta_1 z + \theta_2 z^2 + \dots + \theta_q z^q = 0$$

lie outside the unit circle, since errors may accumulate when iterating on (2.23) and the representation of $\{\varepsilon_t\}$ through $\{y_t\}$ becomes meaningless. Again we have a nonlinear equation in the parameters for the conditional likelihood case, which requires numerical optimization. If a non-invertible MA(q) representation is obtained, the equivalent invertible representation, obtained by reflecting any roots inside the unit circle³, should be used as the initial value for the maximization.

The **exact likelihood function** considers ε_0 as a parameter to be estimated. It is given by

$$f_Y(y; \theta) = (2\pi)^{-T/2} |\Omega|^{-1/2} \exp \left[-\frac{1}{2} (y - \mu)' \Omega^{-1} (y - \mu) \right] \quad (2.25)$$

Again, the exact likelihood function can be evaluated using either the Kalman filter or the triangular factorization.

2.2.5 ARMA(p,q)

A Gaussian ARMA(p, q) process takes the form

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$$

where $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. The goal is to estimate the vector of population parameters $\theta = (c, \phi_1, \phi_2, \dots, \phi_p, \theta_1, \theta_2, \dots, \theta_q, \sigma^2)'$. Taking initial values for $y_0 = (y_0, y_{-1}, \dots, y_{-p+1})'$ and $\varepsilon_0 = (\varepsilon_0, \varepsilon_{-1}, \dots, \varepsilon_{-q+1})'$ as given, the sequence $\{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_T\}$ can be calculated from $\{y_1, y_2, \dots, y_T\}$ by iterating on

$$\varepsilon_t = y_t - c - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \cdots - \phi_p y_{t-p} - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \cdots - \theta_q \varepsilon_{t-q}$$

for $t = 1, 2, \dots, T$. We can set initial y 's and ε 's equal to their expected values; $y_{-j} = \frac{c}{1 - \phi_1 - \phi_2 - \cdots - \phi_p}$ and $\varepsilon_j = 0$ and proceed with iteration. The **conditional log likelihood** is then:

$$\mathcal{L}(\theta) = \log f_{Y_T, Y_{T-1}, \dots, Y_1 | Y_0, \varepsilon_0}(y_T, y_{T-1}, \dots, y_1 | y_0, \varepsilon_0; \theta) = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma^2) - \sum_{t=1}^T \frac{\varepsilon_t^2}{2\sigma^2} \quad (2.26)$$

Alternatively, one can set ε 's to zero but use the true y 's. The **conditional likelihood** in this last case is

$$\begin{aligned} & \log f(y_T, y_{T-1}, \dots, y_{p+1} | y_p, \dots, y_1, \varepsilon_0 = 0, \dots, \varepsilon_{-q+1} = 0) \\ &= -\frac{T-p}{2} \log(2\pi) - \frac{T-p}{2} \log(\sigma^2) - \sum_{t=p+1}^T \frac{\varepsilon_t^2}{2\sigma^2} \end{aligned}$$

As in the case for the moving average processes, these approximations should be used only if all values of z satisfying $1 + \theta_1 z + \theta_2 z^2 + \cdots + \theta_q z^q = 0$ lie outside the unit circle, thus proceeding with numerical optimization for the **conditional likelihood** in an invertible process, in which the effect of the initial $\varepsilon_0, \dots, \varepsilon_{q-1} = \mathbf{0}$ die out and not accumulate over time. For the calculating the **exact likelihood**, use the Kalman Filter, which naturally estimates the initial residuals as additional model parameters.

³Variance can be adjusted accordingly in such cases.

2.3 Numerical Optimization

2.3.1 Steepest ascent

1st order approximation

Consider the first-order Taylor expansion of $\mathcal{L}(\theta^{(m+1)})$ around $\theta^{(m)}$:

$$\mathcal{L}(\theta^{(m+1)}) \approx \mathcal{L}(\theta^{(m)}) + \mathbf{g}(\theta^{(m)})'(\theta^{(m+1)} - \theta^{(m)}). \quad (2.27)$$

For a given step length, the gradient gives the direction of maximal first-order increase in $\mathcal{L}(\theta)$ by the Cauchy–Schwarz inequality. Thus, for a sufficiently small step size $s > 0$, choosing

$$\theta^{(m+1)} = \theta^{(m)} + s \mathbf{g}(\theta^{(m)}) \quad (2.28)$$

yields the first-order increase

$$\mathcal{L}(\theta^{(m+1)}) - \mathcal{L}(\theta^{(m)}) \approx s \|\mathbf{g}(\theta^{(m)})\|^2 > 0, \quad (2.29)$$

provided that $\mathbf{g}(\theta^{(m)}) \neq \mathbf{0}$.

The **steepest ascent method** allows to numerically maximize a continuously differentiable function with a large number of parameters. Let $\mathbf{g}(\theta)$ denote the gradient vector of the log likelihood function:

$$\mathbf{g}(\theta) \equiv \frac{\partial \mathcal{L}(\theta)}{\partial \theta}$$

where $\mathbf{g}(\theta)$ is a $(k \times 1)$ vector. Given an initial $\theta^{(0)}$, the gradient

$$\mathbf{g}(\theta^{(0)}) = \left. \frac{\partial \mathcal{L}(\theta)}{\partial \theta} \right|_{\theta=\theta^{(0)}}$$

gives the direction in which the log likelihood function $\mathcal{L}(\theta)$ increases most steeply from $\theta^{(0)}$. If it is too difficult to differentiate $\mathcal{L}(\theta)$ *analytically*, get a *numerical approximation* to the gradient by seeing how $\mathcal{L}(\theta)$ changes for a small change in each element of θ . The i th element of $\mathbf{g}(\theta^{(0)})$ might be approximated by

$$g_i(\theta^{(0)}) \approx \frac{1}{\Delta} \left\{ \mathcal{L}(\theta_1^{(0)}, \theta_2^{(0)}, \dots, \theta_{i-1}^{(0)}, \theta_i^{(0)} + \Delta, \theta_{i+1}^{(0)}, \dots, \theta_k^{(0)}) - \mathcal{L}(\theta_1^{(0)}, \theta_2^{(0)}, \dots, \theta_{i-1}^{(0)}, \theta_i^{(0)}, \theta_{i+1}^{(0)}, \dots, \theta_k^{(0)}) \right\}$$

where Δ is some arbitrarily small scalar such as $\Delta = 10^{-6}$. Proceeding analogously, an estimate of the full vector $\mathbf{g}(\theta^{(0)})$ can be uncovered. From (2.28), for a general step m we choose:

$$\theta^{(m+1)} = \theta^{(m)} + s \cdot \mathbf{g}(\theta^{(m)})$$

until some convergence criterion is satisfied. One should start from different starting values $\theta^{(0)}$ to avoid local maximums.

2.3.2 Newton Rhapson

The **Newton-Raphson method** often converges more quickly than the **steepest ascent method** provided that (1) second derivatives of the log likelihood function $\mathcal{L}(\theta)$ exist and (2) the function $\mathcal{L}(\theta)$ is concave.

Let $\mathbf{g}(\theta^{(0)})$ denote the gradient vector of the log likelihood function at $\theta^{(0)}$:

$$\mathbf{g}(\theta^{(0)}) = \left. \frac{\partial \mathcal{L}(\theta)}{\partial \theta} \right|_{\theta=\theta^{(0)}}$$

and let $\mathbf{H}(\theta^{(0)})$ denote minus the Hessian matrix of second derivatives of $\mathcal{L}(\theta)$:

$$\mathbf{H}(\theta^{(0)}) = - \left. \frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\theta^{(0)}}$$

Approximate $\mathcal{L}(\theta)$ with a second-order Taylor series around $\theta^{(0)}$:

$$\mathcal{L}(\theta) \approx \mathcal{L}(\theta^{(0)}) + \mathbf{g}(\theta^{(0)})'(\theta - \theta^{(0)}) - \frac{1}{2}(\theta - \theta^{(0)})'\mathbf{H}(\theta^{(0)})(\theta - \theta^{(0)}) \quad (2.30)$$

Then maximize it with respect to θ to get:

$$\mathbf{g}(\theta^{(0)}) - \mathbf{H}(\theta^{(0)})(\theta - \theta^{(0)}) = 0$$

Therefore the first Newton-Raphson update is:

$$\theta^{(1)} = \theta^{(0)} + [\mathbf{H}(\theta^{(0)})]^{-1}\mathbf{g}(\theta^{(0)})$$

The m th step in the iteration proceeds analogously:

$$\theta^{(m+1)} = \theta^{(m)} + [\mathbf{H}(\theta^{(m)})]^{-1}\mathbf{g}(\theta^{(m)}) \quad (2.31)$$

If $\mathcal{L}(\theta)$ is exactly quadratic, then (2.30) holds exactly and this update yields the MLE in a single step:

$$\theta^{(1)} = \hat{\theta}_{MLE}$$

When $\mathcal{L}(\theta)$ is not perfectly quadratic, Newton-Raphson in (2.31) often includes a step size s :

$$\theta^{(m+1)} = \theta^{(m)} + s[\mathbf{H}(\theta^{(m)})]^{-1}\mathbf{g}(\theta^{(m)})$$

2.4 Maximum Likelihood Estimation

For MLE, we assume that the observed data is strictly stationary (same distribution) and that neither $\hat{\theta}$ nor θ_0 fall on a boundary of the allowable parameter space. The distribution of the MLE $\hat{\theta}$ is asymptotically approximated by:

$$\hat{\theta} \approx N(\theta_0, T^{-1}I_f^{-1}), \quad (2.32)$$

where θ_0 denotes the true parameter vector and $I_f = E[\mathbf{H}(\theta)]$ is the information matrix. Matrix I_f can be estimated by two ways:

- The second-derivative estimate:

$$\hat{I}_{2D} = -T^{-1} \left. \frac{\partial^2 L(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\hat{\theta}}. \quad (2.33)$$

where $L(\theta) = \sum_{t=1}^T \log f_{Y_t|Y_{t-1}, \dots, Y_1}(y_t|y_{t-1}, \dots, y_1; \theta)$ is the log likelihood.

- The outer-product estimate:

$$\hat{I}_{OP} = T^{-1} \sum_{t=1}^T [g(\hat{\theta}, y_t)] \cdot [g(\hat{\theta}, y_t)]'. \quad (2.34)$$

where $g(\theta, y_t) = \frac{\partial}{\partial \theta} \log f_{Y_t|Y_{t-1}, \dots, Y_1}(y_t|y_{t-1}, \dots, y_1; \theta)$ is the individual score vector.

By substituting on (2.32), the covariance matrix of $\hat{\theta}$ can be either approximated by:

$$E[(\hat{\theta} - \theta_0)(\hat{\theta} - \theta_0)'] \approx \left[-\frac{\partial^2 L(\theta)}{\partial \theta \partial \theta'} \Big|_{\theta=\hat{\theta}} \right]^{-1} \approx \sum_{t=1}^T [g(\hat{\theta}, y_t)] \cdot [g(\hat{\theta}, y_t)]'$$

where the T 's cancel.

2.4.1 Likelihood Ratio test

Suppose H_0 implies m restrictions on θ of size $(k \times 1)$. To perform an **LR test**, first obtain the unrestricted MLE $\hat{\theta}$ and then obtain the restricted MLE $\tilde{\theta}$. Estimation of $\tilde{\theta}$ is obtained by defining a $[(k - m) \times 1]$ vector λ containing the free parameters under H_0 , such that $\theta = \theta(\lambda)$ satisfies the m restrictions. For example, if the last m elements of θ are restricted to zero, then

$$\lambda = (\theta_1, \dots, \theta_{k-m})', \quad \theta(\lambda) = (\lambda_1, \dots, \lambda_{k-m}, 0, \dots, 0)'.$$

In general, the restricted MLE is obtained by maximizing the log-likelihood over the $k - m$ free parameters:

$$\hat{\lambda} = \arg \max_{\lambda} \mathcal{L}(\theta(\lambda)), \quad \tilde{\theta} = \theta(\hat{\lambda}).$$

The LR test statistic is then given by

$$LR = 2 \left[\mathcal{L}(\hat{\theta}) - \mathcal{L}(\tilde{\theta}) \right] \xrightarrow{D} \chi^2(m). \quad (2.35)$$

2.4.2 Lagrange Multiplier Test

Use the LM test when it is easier to estimate the restricted model under the null hypothesis than to obtain the unrestricted MLE $\hat{\theta}$. Let $\tilde{\theta}$ denote the restricted MLE and define the individual score contribution evaluated under H_0 as

$$\mathbf{g}(\tilde{\theta}, y_t) = \frac{\partial \log f(y_t | y_{t-1}, y_{t-2}, \dots; \theta)}{\partial \theta} \Big|_{\theta=\tilde{\theta}}.$$

The total score evaluated at the restricted estimate is therefore

$$\mathbf{g}(\tilde{\theta}) = \sum_{t=1}^T \mathbf{g}(\tilde{\theta}, y_t).$$

The LM statistic measures how far the score at the restricted estimate is from zero, weighted by the inverse Fisher information matrix. It is given by

$$LM = T^{-1} \mathbf{g}(\tilde{\theta})' \mathbf{I}_f^{-1}(\tilde{\theta}) \mathbf{g}(\tilde{\theta}) \xrightarrow{D} \chi^2(m), \quad (2.36)$$

under $H_0 : \tilde{\theta}$ is true, where $\mathbf{I}_f(\tilde{\theta})$ is the Fisher information matrix per observation evaluated at the restricted estimate. Thus, unlike the LR test, the LM test requires estimation only of the restricted model.

2.4.3 Quasi-Maximum Likelihood Standard Errors

If T is large and the assumed probability density is correctly specified, $\hat{\mathbf{I}}_{2D}$ and $\hat{\mathbf{I}}_{OP}$ should be close, since the information matrix equality implies $\mathbf{I}_{2D} = \mathbf{I}_{OP}$ asymptotically. However, maximum likelihood may still provide a useful estimator when the assumed density is misspecified, although the usual MLE covariance matrix based on either $\hat{\mathbf{I}}_{2D}^{-1}$ or $\hat{\mathbf{I}}_{OP}^{-1}$ is generally no longer

valid. In this case, the estimator is referred to as the **quasi-maximum likelihood estimator (QMLE)**. Under some regularity conditions, QMLE remains **consistent** for the pseudo-true parameter θ_0 , while its asymptotic covariance matrix is given by the form

$$E \left[(\hat{\theta} - \theta_0)(\hat{\theta} - \theta_0)' \right] \approx T^{-1} \{ \mathbf{I}_{2D} \mathbf{I}_{OP}^{-1} \mathbf{I}_{2D} \}^{-1} = T^{-1} \mathbf{I}_{2D}^{-1} \mathbf{I}_{OP} \mathbf{I}_{2D}^{-1}. \quad (2.37)$$

Thus, QMLE corresponds to maximizing a likelihood based on a potentially misspecified distribution: consistency is retained for the pseudo-true parameter under appropriate conditions, but the usual MLE variance must be replaced by the robust sandwich covariance matrix.

2.4.4 Inequality Constraints

One way to ensure that a numerical optimization always satisfies a set of parameter constraints is to reparameterize the likelihood in terms of an unconstrained $(k \times 1)$ vector λ , such that

$$\theta = g(\lambda),$$

where $g : \mathbb{R}^k \rightarrow \mathbb{R}^k$ incorporates the restrictions, mapping the unconstrained parameter space into the admissible parameter space. The likelihood is then maximized with respect to λ rather than θ .

Input	Procedure	Output
$y_1, \dots, y_T$ and λ	set $\theta = g(\lambda)$; calculate $\mathcal{L}(\theta)$	$\mathcal{L}(g(\lambda))$

Examples:

- To impose $-1 < \phi < 1$, one may use

$$\phi = g(\lambda) = \frac{\lambda}{1 + |\lambda|}.$$

- For a symmetric positive-definite covariance matrix Σ , a convenient parameterization is

$$\Sigma = PP',$$

where P is lower triangular with strictly positive diagonal elements. The positivity of the diagonal can itself be imposed using, for example, $P_{ii} = \exp(\lambda_i)$.

- For probabilities $p_1, \dots, p_K$, the restrictions $0 \leq p_i \leq 1$ and $\sum_{i=1}^K p_i = 1$ can be imposed by parameterizing

$$p_i = \frac{\lambda_i^2}{1 + \sum_{j=1}^{K-1} \lambda_j^2}, \quad i = 1, \dots, K-1, \quad p_K = \frac{1}{1 + \sum_{j=1}^{K-1} \lambda_j^2}.$$

or with the softmax function.

Furthermore, reparameterization changes the asymptotic covariance matrix. If $\hat{\lambda}$ is asymptotically normal and $\hat{\theta} = g(\hat{\lambda})$, the **delta method** gives

$$\text{AVar}(\hat{\theta}) = G(\lambda_0) \text{AVar}(\hat{\lambda}) G(\lambda_0)',$$

where $G(\lambda_0) = \left. \frac{\partial g(\lambda)}{\partial \lambda'} \right|_{\lambda=\lambda_0}$ is the Jacobian of the transformation. For more complicated inequality constraints that do not admit a convenient reparameterization, the likelihood can be maximized directly subject to the constraints using constrained optimization methods, such as interior-point or sequential quadratic programming methods.

2.5 Examples of Large Sample Distributions

For stationary and invertible ARMA processes, under suitable regularity conditions, the maximum likelihood, unconditional least squares, and conditional least squares estimators are consistent and asymptotically equivalent. In particular, they have the same asymptotic distribution as the maximum likelihood estimator:

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{D} \mathcal{N}\left(\mathbf{0}, \mathbf{I}_f^{-1}(\theta_0)\right),$$

or, equivalently, for large T ,

$$\hat{\theta} \stackrel{a}{\sim} \mathcal{N}\left(\theta_0, T^{-1}\mathbf{I}_f^{-1}(\theta_0)\right),$$

where $\mathbf{I}_f(\theta_0)$ denotes the Fisher information matrix per observation,

$$\mathbf{I}_f(\theta_0) = -E\left[\frac{\partial^2 \ell_t(\theta_0)}{\partial \theta \partial \theta'}\right] = E\left[\frac{\partial \ell_t(\theta_0)}{\partial \theta} \frac{\partial \ell_t(\theta_0)}{\partial \theta'}\right],$$

where $\ell_t(\theta) = \log f(y_t | y_{t-1}, y_{t-2}, \dots; \theta)$. The asymptotic covariance matrix of the estimator $\hat{\theta}$ is the inverse of the information matrix per observation times T^{-1} ⁴. In particular, for an ARMA(p, q) model with parameter vector $\theta = (\Phi', \Theta')'$, where $\Phi = (\phi_1, \dots, \phi_p)'$ and $\Theta = (\theta_1, \dots, \theta_q)'$, the inverse Fisher information matrix can be partitioned as

$$\mathbf{I}_f^{-1}(\theta_0) = \Sigma_{p,q} = \begin{pmatrix} \Sigma_{\Phi\Phi} & \Sigma_{\Phi\Theta} \\ \Sigma_{\Theta\Phi} & \Sigma_{\Theta\Theta} \end{pmatrix},$$

Example - Some Specific Asymptotic Distributions.

AR(1):

$$\hat{\phi} \sim AN(\phi, T^{-1}(1 - \phi^2)).$$

AR(2):

$$\begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} \sim AN\left(\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}, T^{-1} \begin{pmatrix} 1 - \phi_2^2 & -\phi_1(1 + \phi_2) \\ \text{sym} & 1 - \phi_1^2 \end{pmatrix}^{-1}\right).$$

MA(1):

$$\hat{\theta} \sim AN(\theta, T^{-1}(1 - \theta^2)).$$

MA(2):

$$\begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{pmatrix} \sim AN\left(\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, T^{-1} \begin{pmatrix} 1 - \theta_2^2 & \theta_1(1 + \theta_2) \\ \text{sym} & 1 - \theta_1^2 \end{pmatrix}^{-1}\right).$$

ARMA(1,1):

$$\begin{pmatrix} \hat{\phi} \\ \hat{\theta} \end{pmatrix} \sim AN\left(\begin{pmatrix} \phi \\ \theta \end{pmatrix}, T^{-1} \begin{pmatrix} (1 - \phi^2)^{-1} & (1 + \phi\theta)^{-1} \\ \text{sym} & (1 - \theta^2)^{-1} \end{pmatrix}\right).$$

⁴If the information matrix is defined for the full sample, the factor T^{-1} is already incorporated.

Chapter 3

ARMA - Forecasting

3.1 Introduction

Given an observed time series $Y_1, Y_2, \dots, Y_T$, the objective of forecasting is to predict the future values

$$Y_{T+1}, Y_{T+2}, \dots, Y_{T+h},$$

where h denotes the forecast horizon. Let $g(Y_1, Y_2, \dots, Y_T)$ be any predictor of Y_{T+h} built upon the information set $\mathcal{F}_T$. Its MSE is given by:

$$\text{MSE}[g(Y_1, Y_2, \dots, Y_T)] := \mathbb{E}[(Y_{T+h} - g(Y_1, Y_2, \dots, Y_T))^2].$$

The predictor g^* that minimizes the MSE is the conditional expectation:

$$\mathbb{E}(Y_{T+h} \mid Y_1, \dots, Y_T) = \arg \min_g \mathbb{E}[(Y_{T+h} - g(Y_1, Y_2, \dots, Y_T))^2].$$

Given $y = Y_{T+h}$, we write the MSE:

$$\mathbb{E}[(Y_{T+h} - g(Y_1, \dots, Y_T))^2] = \int (y - g(Y_1, \dots, Y_T))^2 f(y \mid Y_1, \dots, Y_T) dy.$$

Differentiating with respect to g and equating to zero:

$$\int -2(y - g)f(y \mid Y_1, \dots, Y_T) dy = -2(\mathbb{E}[Y_{T+h} \mid Y_1, \dots, Y_T] - g) = 0$$

$$\longleftrightarrow$$

$$g^*(Y_1, \dots, Y_T) = \mathbb{E}[Y_{T+h} \mid Y_1, \dots, Y_T].$$

In general, the functional form of $\mathbb{E}(Y_{T+h} \mid Y_T, \dots, Y_1)$ is unknown. However, we restrict our attention to linear forecasting models¹:

$$\pi(Y_{T+h} \mid Y_T, \dots, Y_1) = \alpha + \beta_1 Y_T + \beta_2 Y_{T-1} + \dots + \beta_T Y_1.$$

with the **estimated best linear forecast** as the one that minimizes the MSE:

$$Y_{T+h|T} = \hat{\pi}(Y_{T+h} \mid Y_T, \dots, Y_1) = \hat{\alpha} + \hat{\beta}_1 Y_T + \dots + \hat{\beta}_T Y_1.$$

¹Let $\mathbf{Y}_T = (Y_T, \dots, Y_1)'$. If $(Y_{T+h}, \mathbf{Y}_T)'$ is jointly Gaussian, then $\mathbb{E}(Y_{T+h} \mid \mathbf{Y}_T) = \mu_{T+h} + \boldsymbol{\Sigma}_{T+h,T} \boldsymbol{\Sigma}_{T,T}^{-1} (\mathbf{Y}_T - \boldsymbol{\mu}_T)$, which is affine in $\mathbf{Y}_T$. Hence, the best linear predictor coincides with the conditional expectation and therefore with the minimum MSE predictor. Gaussianity is not necessary: if $Y_{T+h} = \alpha + \boldsymbol{\beta}' \mathbf{Y}_T + \varepsilon_{T+h}$ with $\mathbb{E}(\varepsilon_{T+h} \mid \mathbf{Y}_T) = 0$, then $\mathbb{E}(Y_{T+h} \mid \mathbf{Y}_T) = \alpha + \boldsymbol{\beta}' \mathbf{Y}_T$. Thus, linear forecasting is exact whenever the conditional mean is affine in the observed history.

The h -step ahead **prediction error** is then given by:

$$u_{t+h} = Y_{T+h} - Y_{T+h|T} \quad (3.1)$$

To calculate linear forecasts, use the following:

$$\begin{aligned} \text{(a)} \quad E(Y_{T+h} \mid Y_T, Y_{T-1}, \dots) &= \begin{cases} Y_{T+h}, & h \leq 0 \\ Y_{T+h|T}, & h > 0 \end{cases} \\ \text{(b)} \quad E(\varepsilon_{T+h} \mid Y_T, Y_{T-1}, \dots) &= \begin{cases} \varepsilon_{T+h}, & h \leq 0 \\ 0, & h > 0 \end{cases} \end{aligned}$$

If $\{Y_t\}$ is covariance-stationary, it can be written as an $MA(\infty)$. Thus expressing Y_{T+h} of (3.1) as an $MA(\infty)$ form yields the prediction error:

$$\begin{aligned} u_{T+h} &= Y_{T+h} - Y_{T+h|T} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{T+h-j} - E \left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{T+h-j} \mid \mathcal{F}_T \right) \\ &= \varepsilon_{T+h} + \psi_1 \varepsilon_{T+h-1} + \dots + \psi_{h-1} \varepsilon_{T+1} \\ &= \sum_{j=0}^{h-1} \psi_j \varepsilon_{T+h-j}, \end{aligned} \quad (3.2)$$

with corresponding variance²:

$$\text{Var}(u_{T+h}) = \sigma^2 (1 + \psi_1^2 + \dots + \psi_{h-1}^2). \quad (3.3)$$

3.2 Forecasting ARMA processes

For linear forecasts, we use the law of total expectations. For example, see $Y_{T+2|T}$ for the AR(1).

3.2.1 AR(1)

Recall that an AR(1) is given by $Y_t = c + \phi Y_{t-1} + \varepsilon_t$. Since $\mathbb{E}(\varepsilon_{T+h} \mid Y_T) = 0 \forall h > 0$,

$$\begin{aligned} Y_{T+1|T} &= \pi(Y_{T+1} \mid Y_T, \dots, Y_1) = c + \phi Y_T \\ Y_{T+2|T} &= \pi(Y_{T+2} \mid Y_T, \dots, Y_1) = \pi(\pi(Y_{T+2} \mid Y_{T+1}, Y_T, \dots, Y_1) \mid Y_T, \dots, Y_1) \\ &= \pi(c + \phi Y_{T+1} \mid Y_T, \dots, Y_1) = \pi(c + \phi Y_{T+1} \mid Y_T) \\ &= c + \phi(c + \phi Y_T) = (1 + \phi)c + \phi^2 Y_T \\ &\vdots \\ Y_{T+h|T} &= (1 + \phi + \phi^2 + \dots + \phi^{h-1})c + \phi^h Y_T. \end{aligned}$$

The forecast errors are equivalent to (3.2), being given by:

$$\begin{aligned} u_{T+1|T} &= Y_{T+1} - Y_{T+1|T} = Y_{T+1} - c - \phi Y_T = \varepsilon_{T+1} \\ u_{T+2|T} &= Y_{T+2} - Y_{T+2|T} = c + \phi Y_{T+1} + \varepsilon_{T+2} - (c + \phi Y_{T+1|T}) \\ &= \phi(Y_{T+1} - Y_{T+1|T}) + \varepsilon_{T+2} \\ &= \phi \varepsilon_{T+1} + \varepsilon_{T+2} \\ &\vdots \\ u_{T+h|T} &= (\varepsilon_{T+h} + \phi \varepsilon_{T+h-1} + \dots + \phi^{h-1} \varepsilon_{T+1}) \end{aligned}$$

² $\psi_0 = 1$

The forecast and error variance tend to the unconditional mean and variance:

$$\begin{aligned}\lim_{h \rightarrow \infty} Y_{T+h|T} &= c \lim_{h \rightarrow \infty} \sum_{i=0}^{h-1} \phi^i + Y_T \lim_{h \rightarrow \infty} \phi^h = \frac{c}{1-\phi} =: \mu \\ \mathbb{V}(u_{T+h|T}) &= (1 + \phi^2 + \phi^4 + \dots + \phi^{2(h-1)})\sigma^2 \\ \lim_{h \rightarrow \infty} \mathbb{V}(u_{T+h|T}) &= \sigma^2 \lim_{h \rightarrow \infty} \sum_{i=0}^{h-1} \phi^{2i} = \frac{\sigma^2}{1-\phi^2} =: \gamma_0.\end{aligned}$$

3.2.2 AR(p)

For an AR(p):

$$\begin{aligned}Y_{T+1|T} &= \pi(Y_{T+1} | Y_T, \dots, Y_1) = c + \phi_1 Y_T + \dots + \phi_p Y_{T+1-p} \\ Y_{T+2|T} &= c + \phi_1 Y_{T+1|T} + \phi_2 Y_T + \phi_3 Y_{T-1} + \dots + \phi_p Y_{T+2-p}, \\ Y_{T+3|T} &= c + \phi_1 Y_{T+2|T} + \phi_2 Y_{T+1|T} + \phi_3 Y_T + \dots + \phi_p Y_{T+3-p}, \\ &\vdots \\ Y_{T+h|T} &= c + \phi_1 Y_{T+h-1|T} + \phi_2 Y_{T+h-2|T} + \dots + \phi_p Y_{T+h-p|T}.\end{aligned}$$

The forecast error is given by

$$\begin{aligned}u_{T+1|T} &= Y_{T+1} - Y_{T+1|T} = \varepsilon_{T+1} \\ u_{T+2|T} &= \phi_1 u_{T+1|T} + \varepsilon_{T+2}, \\ u_{T+3|T} &= \phi_1 u_{T+2|T} + \phi_2 u_{T+1|T} + \varepsilon_{T+3}, \\ &\vdots \\ u_{T+h|T} &= \phi_1 u_{T+h-1|T} + \dots + \phi_{p-1} u_{T+h-p+1|T} + \phi_p u_{T+h-p|T} + \varepsilon_{T+h},\end{aligned}$$

3.2.3 MA(1)

An MA(1) process is given by $Y_t = c + \theta \varepsilon_{t-1} + \varepsilon_t$. Given an initial value $\hat{\varepsilon}_0$, the innovation series can be reconstructed recursively as

$$\begin{aligned}\hat{\varepsilon}_1 &= Y_1 - c - \theta \hat{\varepsilon}_0, \\ \hat{\varepsilon}_2 &= Y_2 - c - \theta \hat{\varepsilon}_1, \\ &\vdots \\ \hat{\varepsilon}_T &= Y_T - c - \theta \hat{\varepsilon}_{T-1},\end{aligned}$$

where we may initialize $\hat{\varepsilon}_0 = 0$. If T is large and $|\theta| < 1$, the effect of this initialization decays geometrically and becomes negligible. Since future innovations have zero conditional mean,

$$E(\varepsilon_{T+h} | \mathcal{F}_T) = 0, \quad h \geq 1,$$

the forecasts are

$$\begin{aligned}Y_{T+1|T} &= c + \theta \hat{\varepsilon}_T \\ Y_{T+2|T} &= c + \theta \hat{\varepsilon}_{T+1} = c + \theta(Y_{T+1|T} - c - \theta \hat{\varepsilon}_T) = c. \\ &\vdots \\ Y_{T+h|T} &= c\end{aligned}$$

Thus, beyond one step ahead, the MA(1) process cannot be predicted beyond its unconditional mean. The forecast errors are

$$\begin{aligned} u_{T+1|T} &= Y_{T+1} - Y_{T+1|T} = \varepsilon_{T+1} \\ u_{T+2|T} &= Y_{T+2} - Y_{T+2|T} = \varepsilon_{T+2} + \theta\varepsilon_{T+1} \\ &\vdots \\ u_{T+h|T} &= Y_{T+h} - Y_{T+h|T} = \varepsilon_{T+h} + \theta\varepsilon_{T+h-1}, \quad \forall h > 1. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Var}(u_{T+1|T}) &= \sigma^2, \\ \text{Var}(u_{T+h|T}) &= (1 + \theta^2)\sigma^2 = \gamma_0, \quad h > 1. \end{aligned}$$

3.2.4 MA(q)

Set the first q innovations before the first observation to zero:

$$\hat{\varepsilon}_0 = \hat{\varepsilon}_{-1} = \cdots = \hat{\varepsilon}_{-q+1} = 0.$$

Then get innovation series $\{\hat{\varepsilon}_t\}$ iteratively :

$$\begin{aligned} \hat{\varepsilon}_1 &= Y_1 - c - \theta_1\hat{\varepsilon}_0 - \theta_2\hat{\varepsilon}_{-1} - \cdots - \theta_q\hat{\varepsilon}_{1-q}, \\ \hat{\varepsilon}_2 &= Y_2 - c - \theta_1\hat{\varepsilon}_1 - \theta_2\hat{\varepsilon}_0 - \cdots - \theta_q\hat{\varepsilon}_{2-q}, \\ &\vdots \\ \hat{\varepsilon}_T &= Y_T - c - \theta_1\hat{\varepsilon}_{T-1} - \theta_2\hat{\varepsilon}_{T-2} - \cdots - \theta_q\hat{\varepsilon}_{T-q}. \end{aligned}$$

and make the forecasts iteratively, using $E(\varepsilon_{T+h} | \mathcal{F}_T) = 0$, $h \geq 1$,

$$\begin{aligned} Y_{T+1|T} &= c + \theta_1\hat{\varepsilon}_T + \cdots + \theta_q\hat{\varepsilon}_{T+1-q}, \\ Y_{T+2|T} &= c + \theta_2\hat{\varepsilon}_T + \cdots + \theta_q\hat{\varepsilon}_{T+2-q}, \\ &\vdots \\ Y_{T+q|T} &= c + \theta_q\hat{\varepsilon}_T \\ Y_{T+h|T} &= c \quad \forall h > q \end{aligned}$$

where after q periods, the forecast is simply the unconditional mean c . The forecast errors are obtained from:

$$\begin{aligned} u_{T+1|T} &= Y_{T+1} - Y_{T+1|T} = \varepsilon_{T+1}, \\ u_{T+2|T} &= Y_{T+2} - Y_{T+2|T} = \varepsilon_{T+2} + \theta_1\varepsilon_{T+1}, \\ u_{T+3|T} &= Y_{T+3} - Y_{T+3|T} = \varepsilon_{T+3} + \theta_1\varepsilon_{T+2} + \theta_2\varepsilon_{T+1}, \\ &\vdots \\ u_{T+h|T} &= \varepsilon_{T+h} + \theta_1\varepsilon_{T+h-1} + \cdots + \theta_{h-1}\varepsilon_{T+1}, \quad 1 \leq h \leq q, \\ u_{T+h|T} &= \varepsilon_{T+h} + \theta_1\varepsilon_{T+h-1} + \cdots + \theta_q\varepsilon_{T+h-q}, \quad h > q. \end{aligned}$$

The forecast error variances are

$$\begin{aligned}
\text{Var}(u_{T+1|T}) &= \sigma^2, \\
\text{Var}(u_{T+2|T}) &= (1 + \theta_1^2)\sigma^2, \\
\text{Var}(u_{T+3|T}) &= (1 + \theta_1^2 + \theta_2^2)\sigma^2, \\
&\vdots \\
\text{Var}(u_{T+h|T}) &= \left(1 + \sum_{j=1}^{h-1} \theta_j^2\right) \sigma^2, \quad 1 \leq h \leq q, \\
\text{Var}(u_{T+h|T}) &= \left(1 + \sum_{j=1}^q \theta_j^2\right) \sigma^2 = \gamma_0, \quad h > q.
\end{aligned}$$

Thus, the forecast error variance increases with the forecast horizon until $h = q + 1$, after which it reaches the unconditional variance γ_0 and remains constant.

3.2.5 ARMA(p,q)

Combining the AR(p) and MA(q) components, the one-step-ahead forecast is given by

$$Y_{T+1|T} = c + \phi_1 Y_T + \cdots + \phi_p Y_{T+1-p} + \theta_1 \hat{\varepsilon}_T + \cdots + \theta_q \hat{\varepsilon}_{T+1-q},$$

where the innovations $\{\hat{\varepsilon}_t\}$ are reconstructed recursively given the initial observations $\{Y_0, \dots, Y_{1-p}\}$ and pre-sample innovations $\{\hat{\varepsilon}_0, \dots, \hat{\varepsilon}_{1-q}\}$, typically initialized at zero. In general,

$$\hat{\varepsilon}_t = Y_t - c - \sum_{i=1}^p \phi_i Y_{t-i} - \sum_{j=1}^q \theta_j \hat{\varepsilon}_{t-j}, \quad t = 1, \dots, T.$$

which allows us to reconstruct the innovation series $\{\hat{\varepsilon}_t\}_{t=1}^T$ and recursively obtain the forecasts $Y_{T+h|T}$. Alternatively, conditioning on the first p observations, the innovation recursion may be started at $t = p + 1$. From this point, forecasts can be obtained recursively for any horizon $h \geq 1$. For illustration, suppose $h > p > q$. Then

$$\begin{aligned}
Y_{T+1|T} &= c + \phi_1 Y_T + \cdots + \phi_p Y_{T+1-p} + \theta_1 \hat{\varepsilon}_T + \cdots + \theta_q \hat{\varepsilon}_{T+1-q}, \\
Y_{T+2|T} &= c + \phi_1 Y_{T+1|T} + \phi_2 Y_T + \cdots + \phi_p Y_{T+2-p} + \theta_2 \hat{\varepsilon}_T + \cdots + \theta_q \hat{\varepsilon}_{T+2-q}, \\
&\vdots \\
Y_{T+q|T} &= c + \sum_{i=1}^p \phi_i Y_{T+q-i|T} + \theta_q \hat{\varepsilon}_T, \\
Y_{T+q+1|T} &= c + \sum_{i=1}^p \phi_i Y_{T+q+1-i|T}, \\
&\vdots \\
Y_{T+h|T} &= c + \sum_{i=1}^p \phi_i Y_{T+h-i|T},
\end{aligned}$$

Equivalently, using the MA(∞) representation, the h -step-ahead forecast error and its variance are given by (3.2) and (3.3), respectively. From these formulas, it can be easily seen that, for a stationary ARMA(p, q) process, as $h \rightarrow \infty$, the forecast converges to the unconditional mean, while the forecast error variance converges to the unconditional variance of the process.

3.3 Confidence intervals for prediction

3.3.1 Normal Errors

Assume $\varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$. From (3.2), the h -step-ahead forecast error is distributed as

$$u_{T+h|T} \sim \mathcal{N} \left(0, \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \right),$$

And using $Y_{T+h} = Y_{T+h|T} + u_{T+h|T}$, the conditional distribution of Y_{T+h} is then

$$Y_{T+h} | \mathcal{F}_T \sim \mathcal{N} (Y_{T+h|T}, \mathbb{V}(u_{T+h|T})). \quad (3.4)$$

Therefore, assuming the model parameters are known, a 95% **prediction interval** for Y_{T+h} is

$$Y_{T+h|T} \pm 1.96 \sqrt{\mathbb{V}(u_{T+h|T})}.$$

In practice, the model parameters are unknown. A consistent estimator of the forecast error variance is obtained by substituting $\hat{\sigma}^2$ and $\{\hat{\psi}_j\}$ into (3.3):

$$\widehat{\mathbb{V}}(u_{T+h|T}) = \hat{\sigma}^2 \left(1 + \sum_{j=1}^{h-1} \hat{\psi}_j^2 \right).$$

Hence, an approximate 95% prediction interval for Y_{T+h} is

$$\hat{Y}_{T+h|T} - 1.96 \hat{\sigma} \left(1 + \sum_{j=1}^{h-1} \hat{\psi}_j^2 \right)^{1/2} \leq Y_{T+h} \leq \hat{Y}_{T+h|T} + 1.96 \hat{\sigma} \left(1 + \sum_{j=1}^{h-1} \hat{\psi}_j^2 \right)^{1/2}. \quad (3.5)$$

3.3.2 Non-Normal errors: Bootstrap

If the innovations ε_t are non-Gaussian, **bootstrap** methods provide an alternative approach for constructing prediction intervals. The central idea of the bootstrap is to assume that ε_t is i.i.d. with c.d.f. F . Suppose we have T residuals $\hat{\varepsilon}_t$. The empirical distribution is simply given by

$$\hat{F}_T(v) \equiv \frac{\#\{\hat{\varepsilon}_t \leq v\}}{T}, \quad v \in \mathbb{R}.$$

The Law of Large Numbers ensures that, as $T \rightarrow \infty$,

$$\hat{F}_T(v) \xrightarrow{\text{a.s.}} F(v),$$

that is, the empirical distribution of the residuals becomes arbitrarily close to the true distribution of the error ε_t as the sample grows.

ARMA(p, q)

For an ARMA(p, q) model, bootstrap prediction intervals can be constructed using the following steps:

1. Estimate the ARMA(p, q) model:³

$$Y_t = c + \sum_{j=1}^p \phi_j Y_{t-j} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t,$$

³Gaussian maximum likelihood may be used even when the innovations are non-Gaussian, since the Gaussian QMLE is consistent under suitable regularity conditions.

obtaining $\hat{c}$, $\{\hat{\phi}_j\}_{j=1}^p$, and $\{\hat{\theta}_j\}_{j=1}^q$. Then compute the residuals

$$\hat{\varepsilon}_t = Y_t - \hat{c} - \sum_{j=1}^p \hat{\phi}_j Y_{t-j} - \sum_{j=1}^q \hat{\theta}_j \varepsilon_{t-j}, \quad (3.6)$$

and obtain the sequence $\{\tilde{\varepsilon}_t\}_{t=1}^T$.

2. **Resample and simulate:** Draw, with replacement, a sample of size $T + m$ from $\{\tilde{\varepsilon}_t\}_{t=1}^T$, denoted by

$$\{\varepsilon_1^*, \varepsilon_2^*, \dots, \varepsilon_{T+m}^*\}.$$

Generate the bootstrap series $\{Y_t^*\}_{t=1}^{T+m}$ as

$$Y_t^* = \begin{cases} Y_t, & 1 \leq t \leq \max(p, q), \\ \hat{c} + \sum_{j=1}^p \hat{\phi}_j Y_{t-j}^* + \sum_{j=1}^q \hat{\theta}_j \varepsilon_{t-j}^* + \varepsilon_t^*, & \max(p, q) < t \leq T + m. \end{cases}$$

Discard the first m observations and keep the bootstrap sample of size T :

$$\{Y_{m+1}^*, Y_{m+2}^*, \dots, Y_{m+T}^*\}.$$

3. **Forecasting:** Using the bootstrap sample $\{Y_t^*\}_{t=m}^{T+m}$, compute forecasts for horizon $h > 0$ based on the real-sample estimates $\hat{c}$, $\{\hat{\phi}_j\}_{j=1}^p$, and $\{\hat{\theta}_j\}_{j=1}^q$. This yields a forecast vector

$$(\hat{Y}_{T+1}^*, \hat{Y}_{T+2}^*, \dots, \hat{Y}_{T+h}^*).$$

Repeat Steps 2–3 S times (e.g. $S = 10,000$), stacking results into an $S \times h$ matrix.

4. **Empirical distribution and confidence intervals:** The empirical distribution of the S forecasts approximates the predictive distribution. For significance level α , the confidence interval is:

- Lower bound: empirical quantile $\alpha/2$ of the bootstrap distribution.
- Upper bound: empirical quantile $1 - \alpha/2$ of the bootstrap distribution.

We can also use the bootstrap samples $\{Y_t^*\}_{t=m}^{T+m}$ to approximate the sampling distributions of the parameter estimators $\hat{c}$, $\{\hat{\phi}_j\}_{j=1}^p$, and $\{\hat{\theta}_j\}_{j=1}^q$ for the observed sample, rather than relying on their asymptotic distributions. Notice that, under suitable regularity conditions, asymptotic normality applies to the parameter estimators as $T \rightarrow \infty$, but not applies to the predictive distribution of Y_{T+h} . In particular, increasing T reduces parameter-estimation uncertainty but does not make future innovations Gaussian; hence, if the innovations are non-Gaussian, the predictive distribution does not become Gaussian as $T \rightarrow \infty$. Indeed, the forecast error in (3.2) depends on the forecast horizon h , rather than the sample size T , so increasing T does not make the predictive distribution Gaussian.

3.3.3 Parameter Uncertainty

In practice, estimated parameters are used to compute point and interval forecasts. For these conditional forecasts⁴, the prediction variances does not consider the uncertainty in the parameter estimates. In theory, one can consider parameter uncertainty in forecasting **analytically** or by **bootstrapping**. A natural way to consider parameter and model uncertainty in forecasting is **Bayesian forecasting** with Markov chain Monte Carlo (MCMC) methods.

When the sample size used in estimation is sufficiently large, then the conditional forecast is close to the unconditional one, since the variances of the parameters tend to be negligible.

⁴Conditional on the parameters

3.4 Forecast Evaluation

Consider the mean squared prediction error (**MSPE**) for a sequence of one-step-ahead forecasts:

$$\text{MSPE} = \frac{1}{P} \sum_{t=1}^P u_{t|t-1}^2,$$

where P denotes the number of forecast observations and $u_{t|t-1} = Y_t - Y_{t|t-1}$ is the one-step-ahead forecast error. To compare the forecast accuracy of two models, consider the ratio of their MSPEs:

$$F = \frac{\text{MSPE}_1}{\text{MSPE}_2} = \frac{\sum_{t=1}^P u_{1,t|t-1}^2}{\sum_{t=1}^P u_{2,t|t-1}^2}.$$

Under H_0 of equal forecast accuracy, F follows an $\mathcal{F}_{P,P}$ distribution provided that:

1. The forecast errors have zero mean and are normally distributed;
2. The forecast errors are serially uncorrelated;
3. The forecast errors from the two models are contemporaneously uncorrelated.

If any of these assumptions is violated, the ratio of the MSPEs does not generally follow an F -distribution. The Diebold–Mariano test provides a more general comparison of predictive accuracy, allowing **asymptotically** for non-normal, serially correlated, and contemporaneously correlated forecast errors, as well as loss functions other than squared error.

3.4.1 Diebold Mariano test

The **Diebold–Mariano (DM) test** compares the predictive accuracy of two forecasting models without requiring the restrictive assumptions of the F -test. Let $L(\cdot)$ denote a loss function and define the **loss differential for the one-step-ahead forecast error** as

$$d_t = L(u_{1,t+1|t}) - L(u_{2,t+1|t}), \quad t = 1, \dots, P.$$

For squared-error loss, $d_t = u_{1,t+1|t}^2 - u_{2,t+1|t}^2$, so that the mean differential loss is

$$\bar{d} = \frac{1}{P} \sum_{t=1}^P d_t = \text{MSPE}_1 - \text{MSPE}_2.$$

The null hypothesis and alternative hypothesis of equal predictive accuracy are

$$\begin{aligned} H_0 &: E(d_t) = 0, \\ H_1 &: E(d_t) \neq 0 \end{aligned}$$

Under suitable regularity conditions, the central limit theorem implies that, under H_0 ⁵,

$$\sqrt{P} \bar{d} \xrightarrow{D} \mathcal{N}(0, \Omega_d),$$

where Ω_d is given by:

$$\Omega_d = \sum_{k=-\infty}^{\infty} \text{Cov}(d_t, d_{t-k}) = \gamma_0 + 2 \sum_{k=1}^{\infty} \gamma_k.$$

⁵Therefore, the normality assumption on the forecast errors is not required as in the last section.

Notice that $\Omega_d = \lim_{P \rightarrow \infty} P \text{Var}(\bar{d})$, in which

$$\begin{aligned} \text{Var}(\bar{d}) &= \text{Var}\left(\frac{1}{P} \sum_{t=1}^P d_t\right) \\ &= \frac{1}{P^2} \left[\sum_{t=1}^P \text{Var}(d_t) + \sum_{t=1}^P \sum_{s < t}^P \text{Cov}(d_t, d_s) + \sum_{t=1}^P \sum_{s > t}^P \text{Cov}(d_t, d_s) \right] \\ &= \frac{1}{P^2} \left[P\gamma_0 + 2 \sum_{k=1}^{P-1} (P-k)\gamma_k \right] \\ &= \frac{1}{P} \left[\gamma_0 + 2 \sum_{k=1}^{P-1} \left(1 - \frac{k}{P}\right) \gamma_k \right], \end{aligned}$$

where, from stationarity, $\gamma_k = \text{Cov}(d_t, d_{t-k})$ and $\gamma_{-k} = \gamma_k$. If $\text{Var}(\bar{d})$ were known, we could construct the ratio

$$\frac{\bar{d}}{\sqrt{\text{Var}(\bar{d})}} \xrightarrow{D} \mathcal{N}(0, 1),$$

and test the null hypothesis of equal forecast accuracy using a standard normal distribution. However, $\text{Var}(\bar{d})$ is generally unknown and must be estimated from the sample. If $\{d_t\}_{t=1}^P$ is **serially uncorrelated** with a sample variance of γ_0 , an **unbiased** estimate of $\text{Var}(\bar{d})$ is $\hat{\gamma}_0/(P-1)$, yielding the statistic:

$$\frac{\bar{d}}{\sqrt{\hat{\gamma}_0/(P-1)}} \sim t_{P-1}.$$

In the presence of **serial correlation** up to lag q , $\text{Var}(\bar{d})$ can be approximated for large P by the **unbiased** estimate

$$\text{Var}(\bar{d}) \approx \frac{1}{P-1} \left(\hat{\gamma}_0 + 2 \sum_{k=1}^q \hat{\gamma}_k \right).$$

Accordingly, one obtains the t -statistic

$$DM = \frac{\bar{d}}{\sqrt{(\hat{\gamma}_0 + 2 \sum_{k=1}^q \hat{\gamma}_k)/(P-1)}},$$

which can be implemented by regressing d_t on a constant and testing whether the intercept is zero using robust standard errors and a t_{P-1} distribution. Furthermore, the construction of DM can be sensitive to the choice of q and can even result in negative variance as the covariances may be negative. Therefore, it is preferable to use robust standard errors, e.g. Newey and West.

There is a small-sample-adjusted Diebold–Mariano statistic, given by

$$DM = \frac{\bar{d}}{\sqrt{(\gamma_0 + 2\gamma_1 + \dots + 2\gamma_q)/[P+1-2h+P^{-1}h(h-1)]}},$$

where $P+1-2h+P^{-1}h(h-1)$ replaces $P-1$ in the denominator.

OBS: the standard DM test is appropriate when the two competing models are non-nested. See Enders [1995] section 2.9 for an explanation and for a treatment in such cases.

Chapter 4

Integrated Processes

4.1 Random Walk

A **Random Walk** is an AR(1) with $\phi = 1$, being nonstationary:

$$Y_t = Y_{t-1} + \epsilon_t, \quad \epsilon_t \sim (0, \sigma^2),$$

$$Y_0 = c.$$

A **Random Walk with drift** is given by:

$$Y_t = \delta + Y_{t-1} + \epsilon_t, \quad \epsilon_t \sim (0, \sigma^2),$$

$$= c + \delta t + \sum_{i=1}^t \epsilon_i,$$

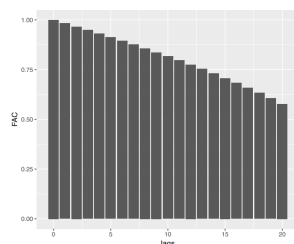
where the influence of past errors ϵ_i does not vanish over time, contrasting with stationary ARMA processes. Its moments are given by:

$$\mu_t := \mathbb{E}(Y_t) = c + \delta t$$

$$\gamma_{j,t} := \mathbb{C}(Y_t, Y_{t-j}) = (t-j)\sigma^2, \quad j \geq 0$$

$$\rho_{j,t} := \frac{\gamma_{j,t}}{\sqrt{\gamma_{0,t} \gamma_{0,t-j}}} = \frac{t-j}{\sqrt{t(t-j)}} = \sqrt{\frac{t-j}{t}} = \sqrt{1 - \frac{j}{t}}$$

Thus, as t grows, $\rho_{j,t}$ approaches 1 for fixed j . This means that the ACF of a random walk decays slowly over time



Note that δ and σ^2 are easily estimated for the differenced series, which behaves as a white noise with mean δ and variance σ^2 .

4.2 Unit root processes

The **Random Walk** is a specific case of the **Unit Root processes**:

$$Y_t = c + \delta t + u_t,$$

where $c, \delta \in \mathbb{R}$ and u_t has an $\text{ARIMA}(p, d, q)$ representation

$$(1 - L)^d \Phi_p(L) u_t = \Theta_q(L) \epsilon_t, \quad \epsilon_t \sim (0, \sigma^2),$$

where the roots of $\Phi_p(L)$ lie outside the unit circle. This implies

$$(1 - L)^d u_t = (1 - \lambda_2 L)^{-1} \cdots (1 - \lambda_p L)^{-1} \Theta_q(L) \epsilon_t =: \Psi(L) \epsilon_t$$

If $d = 1$, taking first-differences on the unit-root process Y_t :

$$\begin{aligned} (1 - L)Y_t &= (1 - L)c + (1 - L)\delta t + (1 - L)u_t = 0 + \delta + (1 - L)u_t = \delta + \Psi(L)\epsilon_t \\ \Delta Y_t &= \delta + \Psi(L)\epsilon_t \end{aligned}$$

Equivalently, for d unit roots, i.e. $Y_t \sim I(d) \rightarrow \Delta^d Y_t \sim I(0)$, we have

$$\Delta^d Y_t = (1 - L)^d Y_t = (1 - L)^d u_t \sim I(0)$$

4.3 ARIMA(p,d,q)

An $\text{ARIMA}(p, d, q)$ is a process that after d differences becomes a stationary $\text{ARMA}(p, q)$:

$$\Phi_p(L)(1 - L)^d Y_t = c + \Theta_q(L)\epsilon_t, \quad \epsilon_t \sim (0, \sigma^2), \quad (4.1)$$

where the roots of $\Phi_p(L)$ lie outside the unit circle. We can interpret $\text{ARIMA}(p, d, q)$ as a nonstationary $\text{ARMA}(p + d, q)$. Since $\Delta^d Y_t \sim \text{ARMA}(p, q)$, we can use the same properties for estimation and forecasting discussed in previous chapters. For example, consider a process with a single unit root. The h -step-ahead forecast is

$$Y_{T+h|T} = Y_T + \sum_{i=1}^h \Delta Y_{T+i|T}.$$

Defining $\tilde{\psi}_j = \sum_{k=0}^j \psi_k$, with $\tilde{\psi}_0 = 1$, we obtain

$$\begin{aligned} u_{T+h|T} &= Y_{T+h} - Y_{T+h|T} = \sum_{j=0}^{h-1} \tilde{\psi}_j \varepsilon_{T+h-j} \\ &= \varepsilon_{T+h} + (1 + \psi_1)\varepsilon_{T+h-1} + (1 + \psi_1 + \psi_2)\varepsilon_{T+h-2} + \cdots + (1 + \psi_1 + \cdots + \psi_{h-1})\varepsilon_{T+1}. \end{aligned}$$

Thus the forecast error variance is

$$\text{Var}(u_{T+h|T}) = \sigma^2 \sum_{j=0}^{h-1} \tilde{\psi}_j^2 = \sigma^2 \left(1 + \tilde{\psi}_1^2 + \cdots + \tilde{\psi}_{h-1}^2 \right).$$

Example: EWMA and ARIMA(0,1,1)

Write an $\text{ARIMA}(0,1,1)$ using $Y_t = X_t - X_{t-1}$ and assuming $X_t \sim I(1)$:

$$\begin{aligned} X_t &= X_{t-1} + w_t - \lambda w_{t-1} \\ Y_t &= w_t - \lambda w_{t-1} \end{aligned}$$

Consider $|\lambda| < 1$; then the process is invertible and Y_t has an $\text{AR}(\infty)$ representation:

$$Y_t = - \sum_{j=1}^{\infty} \lambda^j Y_{t-j} + w_t,$$

Writing $X_t = X_{t-1} + Y_t$ yields:

$$\begin{aligned} X_t &= X_{t-1} - \sum_{j=1}^{\infty} \lambda^j (X_{t-j} - X_{t-j-1}) + w_t \\ X_t &= X_{t-1} - \lambda X_{t-1} + \lambda X_{t-2} - \lambda^2 X_{t-2} + \lambda^2 X_{t-3} - \lambda^3 X_{t-3} + \lambda^3 X_{t-4} - \cdots + w_t \\ &= (1 - \lambda) X_{t-1} + \lambda(1 - \lambda) X_{t-2} + \lambda^2(1 - \lambda) X_{t-3} + \cdots + w_t \\ &= \sum_{j=1}^{\infty} \lambda^{j-1} (1 - \lambda) X_{t-j} + w_t. \end{aligned}$$

This EWMA method is often abused because some do not verify that the observations follow an IMA(1, 1) process, and often arbitrarily pick values of λ .

4.4 Regression with autocorrelated errors

4.4.1 General WLS

If the errors u_t are white noise, **OLS** estimation proceeds straightforwardly. However, if the errors u_t are correlated, estimation of the regression

$$Y_t = \beta X_t + u_t, \quad (4.2)$$

can instead be conducted using Generalized Least Squares (**GLS**) for efficiency. Consider that the autocovariances of u_t are stacked into a $T \times T$ matrix Ω . In matrix form:

$$Y = \beta X + u \quad u \sim (0, \Omega)$$

Then pre-multiplying by $\Omega^{-1/2}$ yields:

$$Y^* = \beta X^* + u^* \quad u^* \sim (0, I)$$

Therefore, the covariance matrix of the residuals becomes the identity and one can apply usual **OLS**. Therefore an estimate for $\hat{\beta}$ and $\text{Var}(\hat{\beta})$ are given by:

$$\begin{aligned} \hat{\beta} &= (X^{*\top} X^*)^{-1} X^{*\top} Y^* = (X^\top \Omega^{-1} X)^{-1} X^\top \Omega^{-1} Y \\ \text{Var}(\hat{\beta}) &= ((X^\top \Omega^{-1} X)^{-1}) \end{aligned}$$

4.4.2 ARMA WLS

Consider that the error u_t in (4.2) follows an ARMA process:

$$\Pi(L)u_t = \varepsilon_t \quad \text{or} \quad u_t = \Psi(L)\varepsilon_t$$

Thus, multiplying (4.2) with the transformation $\Pi(L)$ yields:

$$\begin{aligned} \Pi(L)Y_t &= \beta \Pi(L)X_t + \Pi(L)u_t, \\ Y_t^* &= \beta X_t^* + \varepsilon_t. \end{aligned} \quad (4.3)$$

Thus, Y_t , X_t and u_t are transformed, but β 's remain the same. We can solve (4.3) by minimizing the sum of squared residuals:

$$S(\Pi) = \sum_{t=1}^T \varepsilon_t^2 = \sum_{t=1}^t [\Pi(L)Y_t - \beta \Pi(L)X_t]^2. \quad (4.4)$$

Performing this optimization requires nonlinear numerical methods, with the order of $\Pi(L)$ ¹ being given. The problem is that we typically do not know the ARMA(p, q) behavior of u_t . A simple heuristic is the following:

¹ p and q .

1. First, run an OLS regression of Y_t on X_t , acting as if the errors are uncorrelated. Retain the residuals, $\hat{u}_t = Y_t - \hat{\beta}X_t$.
2. Check ACF and PACF of the residuals u_t for White Noise. If necessary, identify an ARMA(p, q) model for the residuals $\hat{u}_t$.
3. Estimate the regression model with autocorrelated errors using the criteria in (4.4).
4. Check if residuals $\hat{\varepsilon}_t$ are white noise.

The importance of accounting for autocorrelation in time series regressions is that, if serial correlation is present, $\hat{\beta}$ is not efficient. Furthermore, consistency requires that the regressors remain asymptotically uncorrelated with the error term u_t .

4.4.3 Consistent Covariance Matrix Estimation

Consider a situation in which the main goal is to provide inference for $\hat{\beta}$, in which u_t may have serial correlation and heteroskedasticity. If $\hat{\beta}$ is still consistent, i.e., if $\text{plim } T^{-1} \sum_{t=1}^T X_t u_t = 0$, it is possible to obtain a consistent estimate of the covariance matrix of $\hat{\beta}$ that accounts for residual autocorrelation and heteroskedasticity.

In traditional OLS regression (no serial correlation and no heteroskedasticity), we get $\hat{\beta} = (X^\top X)^{-1} X^\top Y$ and the associate covariance matrix $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$. However, $\text{Var}(\hat{\beta})$ is not efficient in the presence of serial correlations or heteroscedasticity, resulting in inflated t-ratios for $\hat{\beta}$.

White estimator

Considering $\text{Var}(\hat{\varepsilon}) = \hat{\Omega}^2$, the robust estimator of **White** is given by

$$\widehat{\text{Var}}_{\text{White}}(\hat{\beta}) = (X^\top X)^{-1} (X^\top \hat{\Omega} X) (X^\top X)^{-1}.$$

This estimator is **robust to heteroskedasticity, but not to serial correlation**.

Newey and West estimator

Newey and West proposed an estimator of $\text{Var}(\hat{\beta})$ that is robust to both **heteroskedasticity and serial correlation**. If the regression errors are suspected to exhibit heteroskedasticity or serial correlation, one may use

$$\widehat{\text{Var}}_{\text{NW}}(\hat{\beta}) = (X^\top X)^{-1} \hat{\Omega}_{\text{NW}} (X^\top X)^{-1},$$

where the Newey–West estimator of the middle matrix is

$$\hat{\Omega}_{\text{NW}} = \sum_{j=-m}^m \left(1 - \frac{|j|}{m+1}\right) \Gamma_j,$$

with

$$\Gamma_j := \sum_{t=j+1}^T X_t X_{t-j}^\top \hat{u}_t \hat{u}_{t-j}, \quad j \geq 0, \quad \Gamma_j = \Gamma_{-j}^\top \quad \text{for } j < 0,$$

where $\hat{u}_t = Y_t - X_t^\top \hat{\beta}$, and m denotes the maximum lag included in the estimator.

²Not $\sigma^2 I$.

- The choice of the maximum lag m is suggested by the authors to be of order $T^{1/4}$. The term $\sum_{j=-m}^m$ comes from the variance of an $MA(m)$ process.
- The $\left(1 - \frac{|j|}{m+1}\right)$ are Bartlett kernel weights that taper down higher-order autocovariances. It assigns zero weight to autocovariances beyond lag m .
- The Newey and West estimator $\widehat{\text{Var}}_{\text{NW}}(\hat{\beta})$ is a consistent estimate of $\text{Var}(\hat{\beta})$.

Summarizing, residual autocorrelation is a problem for inference on $\hat{\beta}$. There are two ways of fixing it:

1. Apply ARMA WLS to Y_t and X_t to guarantee residuals are white noise, thus addressing heteroskedasticity.
2. Use the Newey West estimator, which is robust to heteroskedasticity and serial correlation.

4.5 Summary: I(0) vs. I(1) processes

- $Y_t \sim I(0)$:
 1. $\text{Var}(Y_t) < \infty$, does not depend on t ;
 2. ε_t has a temporary effect on Y_t ;
 3. The process crosses the origin after a finite mean time;
 4. $\rho_k \rightarrow 0$, hence $\sum |\rho_k| < \infty$.
- $Y_t \sim I(1)$:
 1. $\text{Var}(Y_t) \rightarrow \infty$, as $t \rightarrow \infty$;
 2. ε_t has a permanent effect on Y_t (it is a sum of past innovations);
 3. The mean return time to the origin is infinite;
 4. $\rho_k \rightarrow 1$, $\forall k$, as $t \rightarrow \infty$.

Chapter 5

Seasonality

5.1 Introduction

A time series can display 2 types of seasonality: **deterministic seasonality** and **stochastic seasonality**. Suppose one starts by controlling the deterministic seasonality. After that, residual autocorrelation can still persist due to:

1. Low lags dependence - use an ARIMA model.
2. Seasonal lags of period s - use an SARIMA model.

5.2 Deterministic Seasonality

Consider a time series $\{Y_t\}$ with a deterministic seasonality behavior of period s :

$$Y_t = \mu_t + N_t \quad (5.1)$$

where N_t follows an ARMA process:

$$\Phi(L)N_t = \Theta(L)\varepsilon_t \quad (5.2)$$

and μ_t is a **deterministic periodic function**, satisfying $\mu_t - \mu_{t-s} = 0$ or $(1 - L^s)\mu_t = 0$. Its Fourier expansion is given by:

$$\mu_t = \mu + \sum_{j=1}^{s/2} \left[\alpha_j \cos\left(\frac{2\pi jt}{s}\right) + \beta_j \sin\left(\frac{2\pi jt}{s}\right) \right], \quad (5.3)$$

with $\mu, \alpha_j, \beta_j, j = 1, \dots, s$ are unknown constants. Applying $(1 - L^s)$ to Y_t results in:

$$(1 - L^s)Y_t = (1 - L^s)N_t \quad (5.4)$$

Substituting (5.4) into (5.2), we obtain

$$\Phi(B)(1 - L^s)Y_t = \Theta(B)(1 - L^s)\varepsilon_t, \quad (5.5)$$

5.2.1 Identification

Identification of models of the form (5.5) can be carried out in three steps:

1. Obtain preliminary estimates $\hat{\mu}, \hat{\alpha}_j, \hat{\beta}_j$ of $\mu, \alpha_j, \beta_j, j = 1, \dots, s$, in (5.3) by OLS.
2. Compute the residuals

$$\hat{N}_t = Y_t - \hat{\mu} - \sum_{j=1}^{s/2} \left[\hat{\alpha}_j \cos\left(\frac{2\pi jt}{s}\right) + \hat{\beta}_j \sin\left(\frac{2\pi jt}{s}\right) \right],$$

3. Examine the ACF and PACF to identify an ARMA(p, q) model for $\hat{N}_t$.

5.2.2 Estimation

Maximum likelihood estimation of the parameters $\mu, \alpha_i, \beta_i, \theta_j$ and ϕ_k , $i = 1, \dots, s/2$, $j = 1, \dots, q$ and $k = 1, \dots, p$ is similar to the one used for ARMA model estimation. One writes the likelihood and maximizes with respect to these parameters using an analytical procedure if available. Otherwise one may use a nonlinear optimization method.

5.2.3 Forecasting

The forecasts of future values Y_{t+h} , given $Y_1, \dots, Y_t$, are obtained by noting that

$$\begin{aligned}\hat{Y}_t(h) &= E[Y_{t+h} \mid Y_t, Y_{t-1}, \dots] \\ &= E[\mu_{t+h} + N_{t+h} \mid Y_t, Y_{t-1}, \dots] \\ &= \hat{\mu}_{t+h} + \hat{N}_{t+h|t} \\ &= \hat{\mu}_{t+h-s} + \hat{N}_{t+h|t}\end{aligned}\tag{5.6}$$

Example 1: Ozone series

The following example is extracted from Morettin and Toloi [2018] section 10.2.3. Its goal is to fit a model of the form $Y_t = \mu_t + N_t$ of period $s = 12$ on the Ozone series.

- They first fit the model

$$\mu_t = \mu + \sum_{j=1}^6 \left[\alpha_j \cos\left(\frac{2\pi jt}{12}\right) + \beta_j \sin\left(\frac{2\pi jt}{12}\right) \right] + \epsilon_t.$$

Then, they observe that $\frac{2\pi jt}{12}$, $\frac{2\pi 3t}{12}$, $\frac{2\pi 5t}{12}$ and $\frac{2\pi 6t}{12}$ are not significantly different from zero at the 5% level. Since the regressors are orthogonal to each other, t -statistics may be used for excluding these terms. Thus, a preliminary model for μ_t is:

$$\hat{\mu}_t = 5.08 - 2.0082 \cos\left(\frac{2\pi t}{12}\right) - 1.5502 \sin\left(\frac{2\pi t}{12}\right) - 0.2634 \cos\left(\frac{2\pi t}{6}\right) + 0.5253 \sin\left(\frac{2\pi t}{6}\right).$$

- Second, they fit an ARIMA for the residuals $\hat{N}_t = Y_t - \hat{\mu}_t$. Its ACF and PACF show that $\hat{\phi}_{11}$ and $\hat{\rho}_1$ are off the confidence bands of $[-1.96/\sqrt{T}, 1.96/\sqrt{T}]$. Thus, they start by proposing an AR(1) model, which results in a white noise ACF. The resulting model is

$$\hat{N}_t = (1 - 0.155L)^{-1} \epsilon_t$$

with residual variance $\sigma^2 = 0.981$.

- Finally, they aggregate both models in $\hat{Y}_t = \hat{\mu}_t + \hat{N}_t$, yielding

$$\begin{aligned}\hat{Y}_t &= -2.0082 \cos\left(\frac{2\pi t}{12}\right) - 1.5502 \sin\left(\frac{2\pi t}{12}\right) - 0.2634 \cos\left(\frac{2\pi t}{6}\right) \\ &\quad + 0.5253 \sin\left(\frac{2\pi t}{6}\right) + (1 - 0.155L)^{-1} \epsilon_t \quad \sigma^2 = 0.981\end{aligned}$$

- A more robust approach would now involve jointly fitting the coefficients for the above model by MLE and performing again model diagnostics.
- Notice that seasonal differencing $(1 - L^s)Y_t$ is unnecessary when seasonality is deterministic. Since N_t is already stationary, seasonal differencing introduces an artificial non-invertible seasonal MA factor $(1 - L^s)$ into the error process. Hence, it is preferable to model and remove the deterministic seasonal component directly through $\hat{\mu}_t$.

5.3 Stochastic Seasonality

Now consider μ_t is a stochastic process (not deterministic as in (5.3)) satisfying

$$(1 - L^s)\mu_t = X_t, \quad (5.7)$$

for X_t stationary. Applying $(1 - L^s)$ to equation (5.1) yields:

$$\begin{aligned} (1 - L^s)Y_t &= (1 - L^s)\mu_t + (1 - L^s)N_t \\ (1 - L^s)Y_t &= X_t + (1 - L^s)N_t. \end{aligned} \quad (5.8)$$

Since X_t and N_t are stationary process, Wold's decomposition allows us to approximate them in an ARMA form:

$$\begin{aligned} \Phi_X(L)X_t &= \Theta_X(L)v_t \\ \Phi_N(L)N_t &= \Theta_N(L)\varepsilon_t \end{aligned} \quad (5.9)$$

where v_t and ε_t are independent white noise processes. Substituting (5.9) into (5.8) results in

$$\begin{aligned} (1 - L^s)Y_t &= X_t + (1 - L^s)N_t \\ &= \Theta_X(L)\Phi_X^{-1}(L)v_t + (1 - L^s)\Theta_N(L)\Phi_N^{-1}(L)\varepsilon_t. \end{aligned} \quad (5.10)$$

Multiplying (5.10) by $\Phi_X(L)\Phi_N(L)$ yields

$$\Phi_X(L)\Phi_N(L)(1 - L^s)Y_t = \Theta_X(L)\Phi_N(L)v_t + (1 - L^s)\Theta_N(L)\Phi_X(L)\varepsilon_t. \quad (5.11)$$

Since v_t and ε_t are independent white noise processes, the right-hand side admits an equivalent finite-order MA representation $\Theta^*(L)\omega_t$. Hence,

$$\Phi^*(L)(1 - L^s)Y_t = \Theta^*(L)\omega_t, \quad (5.12)$$

where $\Phi^*(L) = \Phi_X(L)\Phi_N(L)$, up to possible common-factor cancellations. Thus, the seasonally differenced series $(1 - L^s)Y_t$ admits an ARMA representation.

More generally, allowing for both ordinary and seasonal integration, $(1 - L)^d(1 - L^s)^D Y_t$ is assumed to be stationary. Equation (5.10) may then be recast into a general SARIMA(p, d, q) $\times$ (P, D, Q) $_s$ model:

$$\Phi(L)\Phi_S(L^s)(1 - L^s)^D(1 - L)^d Y_t = \Theta(L)\Theta_S(L^s)\omega_t. \quad (5.13)$$

The ordinary AR and MA components are represented by $\Phi(L)$ and $\Theta(L)$, of orders p and q , respectively. The seasonal AR and MA components are represented by $\Phi_S(L^s)$ and $\Theta_S(L^s)$, of orders P and Q , respectively. The ordinary and seasonal difference components are given by $(1 - L)^d$ and $(1 - L^s)^D$, respectively.

Model (5.13) implies that $(1 - L^s)^D(1 - L)^d Y_t$ is stationary; and that there are different AR and MA dynamics for the seasonal and non-seasonal dynamics.

5.3.1 SMA(Q)

The SMA(Q) model with $s = 12$ takes the form:

$$Y_t = \varepsilon_t + \Theta_1\varepsilon_{t-12} + \cdots + \Theta_Q\varepsilon_{t-12Q} \quad (5.14)$$

with FAC

$$\begin{aligned} \rho_{12} &= \frac{\Theta_1 + \Theta_1\Theta_2 + \cdots + \Theta_{Q-1}\Theta_Q}{1 + \Theta_1^2 + \cdots + \Theta_Q^2} \\ &\vdots \\ \rho_{12Q} &= \frac{\Theta_Q}{1 + \Theta_1^2 + \cdots + \Theta_Q^2} \end{aligned}$$

For example, the model SMA(1) $Y_t = \varepsilon_t + \Theta\varepsilon_{t-12}$ will have non-zero autocorrelation only at lag 12, that is

$$\rho_{12} = \frac{\Theta}{1 + \Theta^2}, \quad \rho_j = 0, \quad j \neq 0, \pm 12.$$

5.3.2 SAR(P)

A SAR(P) model with $s = 12$ takes the form:

$$Y_t = \Phi_1 Y_{t-12} + \cdots + \Phi_P Y_{t-12P} + \varepsilon_t \quad (5.15)$$

The autocorrelation function will be non-zero only at lags that are multiples of 12. For example, the model SAR(1) $Y_t = \Phi Y_{t-12} + \varepsilon_t$ has autocorrelation given by

$$\begin{aligned} \rho_{12} &= \Phi \\ \rho_{24} &= \Phi^2 \\ \rho_{12j} &= \Phi^j, \quad j = 0, 1, \dots \end{aligned}$$

5.3.3 SARIMA(0,1,1) $\times$ (0,1,1)

A SARIMA(0,1,1) $\times$ (0,1,1) model with period $s = 12$ takes the form:

$$W_t = (1 - B)(1 - B^{12})Y_t = (1 + \Theta B + \Theta B^{12} + \theta \Theta B^{13})\varepsilon_t$$

The autocovariance function of W_t is

$$\begin{aligned} \gamma(0) &= \sigma^2(1 + \theta^2 + \Theta^2 + \theta^2\Theta^2) \\ \gamma(1) &= \sigma^2(\theta + \theta\Theta^2) \\ \gamma(2) &= \gamma(3) = \cdots = \gamma(10) = 0 \\ \gamma(11) &= \sigma^2(\Theta) \\ \gamma(12) &= \sigma^2(\theta + \theta\Theta^2) \\ \gamma(13) &= \sigma^2(\theta\Theta) \\ \gamma(j) &= 0, \quad j > 13 \end{aligned}$$

Thus, the autocorrelation function has nonzero values at lags 1, 11, 12, and 13, with $\rho(11) = \rho(13)$. A more general non-multiplicative model does not restrict $\theta_2 = \theta\Theta$:

$$W_t = (1 - \theta B - \theta_1 B^{12} - \theta_2 B^{13})\varepsilon_t$$

with

$$\begin{aligned} \gamma(0) &= \sigma^2(1 + \theta^2 + \theta_1^2 + \theta_2^2) \\ \gamma(1) &= \sigma^2(\theta + \theta_1\theta_2) \\ \gamma(11) &= \sigma^2(\theta\theta_1) \\ \gamma(12) &= \sigma^2(\theta + \theta\theta_2) \\ \gamma(13) &= \sigma^2\theta_2 \neq \gamma(11) \end{aligned}$$

5.3.4 Identification, Estimation, Diagnostics and Forecasting

There is no additional difficulty in the identification, estimation, and verification of seasonal models. The difference is that the series must be differenced with respect to Δ and Δ_{12} ($s = 12$) in order to produce stationarity. Furthermore, the interaction between the ordinary and seasonal polynomials may produce unconventional interactions in the ACF and PACF.

- **Identification:** Inspect the sample ACF and PACF of the differenced series. From the lags 1, 2, 3, ... we obtain values of p and q , and from the lags 12, 24, 36, ... we obtain values of P and Q . Also consider their interactions for lags around 12, 24, 36, ...
- **Estimation:** Use MLE.
- **Diagnostics:** Residual autocorrelation tests, the Box–Pierce statistic, and the cumulative periodogram (used to check if the residuals are not seasonal).
- **Forecasts:** calculated analogously to the $\text{ARIMA}(p, d, q)$ model

5.4 X-13ARIMA-SEATS

For seasonally adjusting a time series in practice, the X-13ARIMA-SEATS package, developed by the U.S. census bureau, is often employed. For references, check the **seasonal** package in R and Sax and Eddelbuettel [2018].

5.5 Enders: Seasonality

Section 2.10 of Enders [1995] is a must-read for a better practical understanding of seasonality. Here, I provide a summary of its main conclusions.

- Do not fall into the trap of ignoring seasonality when working with seasonally adjusted data (e.g. using X-13ARIMA-SEATS). Seasonal adjustment is estimated from the available data, and some residual seasonality may remain. Thus, always check seasonally adjusted data for residual seasonality.
- Furthermore, it is not advisable to take a seasonally adjusted series (e.g. seasonally adjusted GDP) and use only a portion of it for a real-time forecasting exercise. This may introduce look-ahead bias, since the seasonal adjustment of past observations may have been computed using data that were not available at the forecast origin. Ideally, seasonal adjustment should be re-estimated at each forecast origin using only the information available at that time. Alternatively, one may use data vintages.
- Accordingly, when working with only a portion of the data, it is preferable to use the raw series and perform the seasonal adjustment yourself using only the information available in that subsample.
- In practice, most of seasonal modelling is a two-step procedure. First, the seasonality is removed by seasonal adjusting, and second, the ARMA coefficients are estimated using Box–Jenkins techniques. However, surveys indicate that the seasonal and the ARMA coefficients are best identified and estimated jointly. Therefore, you may consider both cases when performing seasonal adjustments.
- Within the Box–Jenkins approach, the ACF and the PACF of the seasonal portion of the data appear at lags $s, 2s, 3s, \dots$ rather than at lags 1, 2, 3, ... Furthermore, identification

will be complicated by the fact that the **seasonal pattern** will interact with the **non-seasonal pattern** in the data, resulting in an ACF and PACF for the whole series that accounts for both elements.

- A SARIMA model is a multiplicative seasonality model that allows for the interaction of ARMA and seasonal effects. This multiplicative seasonality may be preferred, since a rich interaction pattern can be captured with a smaller number of coefficients, instead of a purely additive seasonal model. A recommended procedure is to estimate both specifications and compare their results.
- Enders exemplifies with the following multiplicative model:

$$(1 - a_1 L)Y_t = (1 + \beta_1 L)(1 + \beta_4 L^4)\varepsilon_t$$

$$Y_t = a_1 Y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \beta_4 \varepsilon_{t-4} + \beta_1 \beta_4 \varepsilon_{t-5}.$$

Its corresponding additive model is:

$$Y_t = a_1 Y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \beta_4 \varepsilon_{t-4} + \beta_5 \varepsilon_{t-5}.$$

The multiplicative model is more parsimonious by restricting $\beta_5 = \beta_1 \beta_4$, but has higher residual sum of squares. Experimentation and diagnostic checks are probably the best way to obtain the most appropriate model

- MINSA series
 - Take log to stabilize the variance.
 - Spikes at lags $s, 2s, 3s \dots$, do not exhibit rapid decay. May take a seasonal difference. A more rigorous approach would test for unit roots in the seasonal part of the process. Furthermore the series is assumed to contain a unit root. Thus, for $s = 4$: $m_t = (1 - L)(1 - L^4)y_t$
 - ACF and PACF: remaining seasonality in lag 4. The seasonal term is most likely to be in the form of an MA coefficient, since the ACF cuts to zero, whereas the PACF does not. Nevertheless, it is recommended to estimate several similar models and then select the best.
 - 3 models considered:

$m_t = a_0 + a_1 m_{t-1} + \varepsilon_t + \beta_4 \varepsilon_{t-4}$	Model 1: AR(1) with Seasonal MA
$m_t = a_0 + (1 + a_1 L)(1 + a_4 L^4)m_{t-1} + \varepsilon_t$	Model 2: Multiplicative Autoregressive
$m_t = a_0 + (1 + \beta_1 L)(1 + \beta_4 L^4)\varepsilon_t$	Model 3: Multiplicative Moving Average

- * All model coefficients are significant
- * Model 1 has better diagnostics: Smallest SSR, AIC and BIC + Q-statistics for lags 4, 8 and 12 not significant.
- * Model 2 has significant residual correlation at $Q(8)$ and $Q(12)$, suggesting that the SAR term does not adequately capture seasonality.
- * Model 3 adequately captures the seasonal pattern, with no significant residual correlation at $Q(4)$ and $Q(8)$.
- Supplements to Box-jenkins method with seasonal data:
 1. In the identification stage, check periodicity and seasonally difference the series if necessary¹. Then check the ACF of the resultant series and perform unit root tests. If necessary first-difference the data. Redo the procedure until the transformed series becomes stationary.

¹Consider unit root tests for seasonality.

2. Use the ACF and PACF to identify potential models. Try to estimate models with low-order nonseasonal ARMA coefficients. Consider both additive and multiplicative seasonality. Allow the appropriate form of seasonality to be determined by the various diagnostic statistics. Compare many possible choices in choosing a model.

Chapter 6

Structural Breaks and Parameter Instability

6.1 Introduction

If **structural breaks** are not taken into account, then:

- ACF may decay linearly (non-stationary).
- Unit root tests may point to non-stationarity.

Structural breaks are related to model instability due to **parameter change**.

6.2 Known break point

Suppose we know the possible break happened in T_1 . Consider the joint model:

$$Y_t = \begin{cases} M_1(W_t) + u_{1t}, & t = 1, \dots, T_1 \\ M_2(W_t) + u_{2t}, & t = T_1 + 1, \dots, T \end{cases} \quad (6.1)$$

where $M_1(\cdot)$ and $M_2(\cdot)$ denote parametric models involving the same regressors W_t (constant, trend, lags, exogenous variables, etc).

6.2.1 Break in the mean

Consider $M_1(W_t)$ and $M_2(W_t)$ involve a finite number of unknown parameters β_1 and β_2 , respectively

$$M_1(W_t) = M(W_t; \beta_1), \quad (6.2)$$

$$M_2(W_t) = M(W_t; \beta_2). \quad (6.3)$$

Testing for the presence of a known model break in the mean at T_1 is equivalent to testing the null hypothesis

$$H_0 : \beta_1 = \beta_2 \quad (6.4)$$

For now, we will only consider linear models of the form

$$Y_t = \begin{cases} Z_t\beta_0 + W_t'\beta_1 + u_{1t}, & t = 1, \dots, T_1 \\ Z_t\beta_0 + W_t'\beta_2 + u_{2t}, & t = T_1 + 1, \dots, T \end{cases} \quad (6.5)$$

or equivalently

$$Y_t = Z_t' \beta_0 + (W_t' \beta_1 + u_{1t}) \mathbf{1}_{\{t \leq T_1\}} + (W_t' \beta_2 + u_{2t}) \mathbf{1}_{\{t > T_1\}} \quad (6.6)$$

$$= X_t' \beta + v_t \quad (6.7)$$

where

- β_1 and β_2 are $(q \times 1)$ parameters to be tested for break before and after T_1 , respectively.
- W_t ($q \times 1$) are regressors associated with **unstable parameters**.
- β_0 ($r \times 1$) are parameters not tested for break, assumed stable a priori.
- Z_t ($r \times 1$) are regressors associated with **stable parameters**.

For estimation, we employ the usual OLS or MLE, since equation (6.7) is a linear regression with:

- $X_t = (Z_t', W_t' \mathbf{1}_{\{t \leq T_1\}}, W_t' \mathbf{1}_{\{t > T_1\}})'$
- $\beta := (\beta_0', \beta_1', \beta_2')'$ is a $(k \times 1)$ vector with $k = r + 2q$
- $v_t := u_{1t} \mathbf{1}_{\{t \leq T_1\}} + u_{2t} \mathbf{1}_{\{t > T_1\}}$

Estimation of equation (6.7) leads to

$$\hat{\beta} = (X'X)^{-1}(X'Y) \quad (6.8)$$

and the variance $\mathbb{V}(\hat{\beta})$ is given by

$$\hat{V} = T(X'X)^{-1} \hat{\Omega} (X'X)^{-1} \quad (6.9)$$

where $\hat{\Omega}$ is the estimator of

$$\Omega = \lim_{T \rightarrow \infty} \mathbb{V} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T X_t v_t \right) \quad (6.10)$$

Autocorrelation in the residuals v_t is an issue for inference. One way to fix this (Section 4.4) is to **ensure the residuals v_t are white noise** (e.g., ARMA modeling) and use (6.10). Another way is to **use robust errors against heteroskedasticity and autocovariance**, known as the **Newey–West estimator**, given by

$$\hat{\Omega} = \sum_{j=-m}^m \left(1 - \frac{|j|}{m+1} \right) \Gamma_j \quad (6.11)$$

where

$$\Gamma_j := \frac{1}{T} \sum_{t=j+1}^T X_t X_{t-j}' \hat{v}_t \hat{v}_{t-j}, \quad j \geq 0,$$

and $\Gamma_j = \Gamma_{-j}'$ for $j < 0$. The residuals are given by

$$\hat{v}_t = Y_t - X_t' \hat{\beta}.$$

Recall that under conditions that guarantee the validity of asymptotic OLS theorems, the CLT applies:

$$\hat{V}^{-1/2}(\hat{\beta} - \beta) \xrightarrow{d} N(0, I_k) \quad (6.12)$$

To test

$$H_0 : \beta_1 - \beta_2 = 0,$$

consider the difference $\hat{\beta}_1 - \hat{\beta}_2$. Let $\hat{V}_1$ and $\hat{V}_2$ denote the $(q \times q)$ covariance matrices associated with $\hat{\beta}_1$ and $\hat{\beta}_2$, respectively, and let $\hat{V}_{12}$ denote their estimated cross-covariance. Then, under H_0 ,

$$P_T := \left(\hat{V}_1 + \hat{V}_2 - \hat{V}_{12} - \hat{V}_{12}' \right)^{-1/2} (\hat{\beta}_1 - \hat{\beta}_2) \xrightarrow{d} N(0, I_q). \quad (6.13)$$

Equivalently, the Wald statistic is

$$W_T = (\hat{\beta}_1 - \hat{\beta}_2)' \left(\hat{V}_1 + \hat{V}_2 - \hat{V}_{12} - \hat{V}_{12}' \right)^{-1} (\hat{\beta}_1 - \hat{\beta}_2) \xrightarrow{d} \chi_q^2. \quad (6.14)$$

If the two estimators are independent¹, then $\hat{V}_{12} = 0$, and the statistic reduces to

$$W_T = (\hat{\beta}_1 - \hat{\beta}_2)' (\hat{V}_1 + \hat{V}_2)^{-1} (\hat{\beta}_1 - \hat{\beta}_2) \xrightarrow{d} \chi_q^2. \quad (6.15)$$

For a known candidate break date T_1 , the statistic is more precisely written as $W_T(T_1)$, since the estimates $\hat{\beta}_1$ and $\hat{\beta}_2$ depend on the partition of the sample at T_1 . Hence, H_0 of no structural break at T_1 is rejected whenever

$$W_T(T_1) > c_\alpha, \quad \mathbb{P}(\chi_q^2 > c_\alpha) = \alpha.$$

Chow test

Alternatively, equation (6.5) can be estimated allowing the coefficients β_1 and β_2 to differ across the two regimes while keeping β_0 fixed across regimes:

$$Y_t = \begin{cases} Z_t' \beta_0 + W_t' \beta_1 + u_{1t}, & t = 1, \dots, T_1, \\ Z_t' \beta_0 + W_t' \beta_2 + u_{2t}, & t = T_1 + 1, \dots, T. \end{cases}$$

Let SSR_{UR} denote the sum of squared residuals from this unrestricted model, and let SSR_R denote the sum of squared residuals from the restricted model obtained under $H_0 : \beta_1 = \beta_2$. Assuming $\beta_0 \in \mathbb{R}^r$ and $\beta_1, \beta_2 \in \mathbb{R}^q$, the unrestricted model contains $r + 2q$ parameters, while H_0 imposes q restrictions. The Chow test statistic is therefore

$$F = \frac{(SSR_R - SSR_{UR})/q}{SSR_{UR}/(T - r - 2q)} \sim F(q, T - r - 2q). \quad (6.16)$$

Thus, H_0 is rejected for sufficiently large values of F^2 .

Dummy variable test

We can also use a dummy variable to test for a break in one or more coefficients. For example, to test for a break in the coefficients of an AR(1) model at a known date T_1 , estimate

$$Y_t = c + \tilde{c}D_t + \phi Y_{t-1} + \tilde{\phi}D_t Y_{t-1} + \varepsilon_t, \quad (6.17)$$

where $D_t = \mathbf{1}(t > T_1)$. The coefficients before the break are c and ϕ , while those after the break are $c + \tilde{c}$ and $\phi + \tilde{\phi}$. Individual breaks can be tested using the *t*-**statistics** for $H_0 : \tilde{c} = 0$ and $H_0 : \tilde{\phi} = 0$, while a joint break can be tested using the *F*-**statistic** for $H_0 : \tilde{c} = \tilde{\phi} = 0$.

¹For a break at T_1 , $\hat{\beta}_1 - \beta_1 = (X_1' X_1)^{-1} X_1' v_1$ and $\hat{\beta}_2 - \beta_2 = (X_2' X_2)^{-1} X_2' v_2$. Hence, $\text{Cov}(\hat{\beta}_1, \hat{\beta}_2 | X) = (X_1' X_1)^{-1} X_1' \text{Cov}(v_1, v_2 | X) X_2' (X_2' X_2)^{-1}$. Thus we can assume $\hat{V}_{12} = 0$ when the errors are white noise.

²The unrestricted model estimates $r + 2q$ parameters: r common coefficients in β_0 and q in each of β_1 and β_2 . Hence, the residual degrees of freedom are $T - r - 2q$. Under $H_0 : \beta_1 = \beta_2$, q independent restrictions are imposed, yielding the $F(q, T - r - 2q)$ distribution.

6.2.2 Break in the variance

Suppose the model is the one in equation (6.7):

$$Y_t = X_t' \beta + u_t,$$

where X_t may contain constants, deterministic trends, and autoregressive terms. We consider the simple case in which u_t is serially uncorrelated, but its variance may exhibit a break at a known point T_1 :

$$\mathbb{V}(u_t) = \begin{cases} \sigma_1^2, & t = 1, \dots, T_1, \\ \sigma_2^2, & t = T_1 + 1, \dots, T. \end{cases} \quad (6.18)$$

Therefore, we would like to test

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad \text{against} \quad H_1 : \sigma_1^2 \neq \sigma_2^2. \quad (6.19)$$

To construct an exact F -test under normality, estimate the same regression specification separately over the two subsamples:

$$\begin{aligned} Y_t &= X_t' \beta_1 + u_{1t}, & t = 1, \dots, T_1, \\ Y_t &= X_t' \beta_2 + u_{2t}, & t = T_1 + 1, \dots, T, \end{aligned}$$

and obtain the corresponding OLS residuals

$$\begin{aligned} \hat{u}_{1t} &= Y_t - X_t' \hat{\beta}_1, & t = 1, \dots, T_1, \\ \hat{u}_{2t} &= Y_t - X_t' \hat{\beta}_2, & t = T_1 + 1, \dots, T. \end{aligned}$$

Although β_1 and β_2 are estimated separately, they are nuisance parameters in this test: the hypothesis being tested concerns only the equality of the error variances.³ The unbiased estimators of the regime-specific error variances are

$$s_1^2 = \frac{1}{T_1 - k} \sum_{t=1}^{T_1} \hat{u}_{1t}^2, \quad (6.20)$$

$$s_2^2 = \frac{1}{T - T_1 - k} \sum_{t=T_1+1}^T \hat{u}_{2t}^2. \quad (6.21)$$

Normality

If, in addition, the errors are normally distributed and independent across time,

$$\begin{aligned} u_{1t} &\sim N(0, \sigma_1^2), \\ u_{2t} &\sim N(0, \sigma_2^2), \end{aligned}$$

³Separate estimation is required for the exact finite-sample F distribution. To see this, let X_j and u_j denote the regressor matrix and error vector in regime j . Separate OLS estimation gives $\hat{u}_j = M_{X_j} u_j$, where $M_{X_j} = I - X_j(X_j' X_j)^{-1} X_j'$ is symmetric and idempotent with rank $T_j - k$. Hence, under normality, $\hat{u}_j' \hat{u}_j / \sigma_j^2 = u_j' M_{X_j} u_j / \sigma_j^2 \sim \chi_{T_j - k}^2$. Moreover, if the errors are independent across time, the quadratic forms from the two non-overlapping subsamples are independent. In contrast, if a single $\hat{\beta}$ is estimated from the full sample, then $\hat{u} = M_X u$, and the sum of squared full-sample residuals restricted to regime j is $u' M_X D_j M_X u$, where D_j selects the observations belonging to that regime. In general, $M_X D_j M_X$ is not idempotent and therefore this quadratic form does not have a $\chi_{T_j - k}^2$ distribution. Consequently, the ratio of the two regime-specific residual variances computed from a single full-sample regression does not have the exact F distribution used below.

then

$$(T_1 - k) \frac{s_1^2}{\sigma_1^2} \sim \chi_{T_1-k}^2,$$

$$(T - T_1 - k) \frac{s_2^2}{\sigma_2^2} \sim \chi_{T-T_1-k}^2,$$

and the two random variables are independent. Therefore,

$$\frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} \sim F_{T_1-k, T-T_1-k}.$$

Under $H_0 : \sigma_1^2 = \sigma_2^2$, it follows that

$$W := \frac{s_1^2}{s_2^2} \sim F_{T_1-k, T-T_1-k}. \quad (6.22)$$

Since the alternative hypothesis is two-sided, H_0 is rejected at significance level α whenever

$$W < F_{T_1-k, T-T_1-k} \left(\frac{\alpha}{2} \right) \quad \text{or} \quad W > F_{T_1-k, T-T_1-k} \left(1 - \frac{\alpha}{2} \right),$$

where $F_{\nu_1, \nu_2}(p)$ denotes the p -quantile of an F distribution with ν_1 and ν_2 degrees of freedom.

Non-Normality

Non-Normality requires an asymptotic argument for testing. Let $s^2 := (s_1^2, s_2^2)'$ and $\sigma^2 := (\sigma_1^2, \sigma_2^2)'$, and

$$K := \mathbb{V}(s^2) = \mathbb{E}[(s^2 - \sigma^2)(s^2 - \sigma^2)']. \quad (6.23)$$

Then, by the **central limit theorem** for i.i.d. vectors, we have

$$K^{-1/2}(s^2 - \sigma^2) \xrightarrow{d} N(0, I_2) \quad (6.24)$$

Then, we obtain

$$\frac{(s_1^2 - \sigma_1^2) - (\sigma_1^2 - \sigma_2^2)}{\sqrt{K_1 + K_2}} \xrightarrow{d} N(0, 1)$$

where $K_1 := \frac{\kappa_1}{T_1}$, $K_2 := \frac{\kappa_2}{T-T_1}$, $\kappa_1 := \mathbb{V}(\sqrt{T_1}s_1^2)$ and $\kappa_2 := \mathbb{V}(\sqrt{T-T_1}s_2^2)$.

Estimating the variance of the sample variance

A consistent estimator for κ_1 and κ_2 is the Newey–West estimator ⁴:

$$\hat{\kappa}_1 := \sum_{j=-m}^m \left(1 - \frac{|j|}{m+1} \right) a_j; \quad \hat{\kappa}_2 := \sum_{j=-m}^m \left(1 - \frac{|j|}{m+1} \right) b_j, \quad (6.25)$$

where

$$a_j = \frac{1}{T_1} \sum_{t=j+1}^{T_1} (\hat{u}_t^2 - s_1^2)(\hat{u}_{t-j}^2 - s_1^2).$$

$$b_j = \frac{1}{T-T_1} \sum_{t=j+T_1+1}^T (\hat{u}_t^2 - s_2^2)(\hat{u}_{t-j}^2 - s_2^2).$$

⁴We assumed that only that the errors were uncorrelated, not that the squared errors were uncorrelated!

Under $H_0 : \sigma_1^2 - \sigma_2^2 = 0$, the **test statistic for a variance break** is given by

$$W_T \equiv \frac{s_1^2 - s_2^2}{\sqrt{\hat{K}_1 + \hat{K}_2}} \xrightarrow{d} N(0, 1)$$

where we can consider two-sided or one-sided tests ⁵.

6.3 Unknown Break Point

Endogenous Breaks correspond to breaks at a data not prespecified by the researcher. By performing a **Chow test** or a **Wald test** for all potential break dates, the break date that results in the largest value of the test statistic provides a consistent estimate of the actual break date.

To ensure an adequate number of observations in each of the two subsamples, it is necessary to have a **“trimming”** such that the break could not occur before the first t_1 observations or after the last $t_2 = T - t_1$ observations.

$$T_1 \in \mathcal{T} = \{t_1, \dots, t_2\}, \quad 1 < t_1 < t_2 < T$$

In this case, we could calculate the Wald statistic $W_T(t)$ or F-statistic $F_T(t)$ for each $t \in \mathcal{T}$ and take the maximum as our test statistic:

$$W_T^* \equiv \max_{t \in \mathcal{T}} W_T(t)$$

$$F_T^* \equiv \max_{t \in \mathcal{T}} F_T(t)$$

This is known as the **Supremum test**. Searching for the most likely break date means that the F-statistic or Wald statistics for the null hypothesis of no break is inflated. Therefore, it has a non-standard asymptotic distribution⁶ which cannot be obtained from a traditional F-table. These distributions are computed via simulation/bootstrapping. Under H_0 :

$$W_T^* \xrightarrow{d} \mathcal{W}$$

The critical values of $\mathcal{W}$ were tabulated in the above paper. In particular, these critical values $c = c(\alpha, k, \pi_1, \pi_2)$ depend on:

- α : the significance level (0.1, 0.005, 0.001)
- k : the number of parameters being tested
- $\pi_1 = t_1/T$ and $\pi_2 = t_2/T$, only through the value

$$\lambda = \frac{\pi_2(1 - \pi_1)}{\pi_1(1 - \pi_2)}$$

Thus, we reject H_0 whenever

$$W_T^* > c(\alpha, k, \pi_1, \pi_2).$$

⁵Consider the following:

- in favor of $H_1 : \sigma_1^2 \neq \sigma_2^2$ if $|W_T| > c_{\alpha/2}$
- in favor of $H_1 : \sigma_1^2 > \sigma_2^2$ if $W_T < -c_\alpha$
- in favor of $H_1 : \sigma_1^2 < \sigma_2^2$ if $W_T > c_\alpha$

⁶See Andrews [1993].

6.4 Parameter instability and Diagnostics

There are many processes that evolve over time; without a sudden specific break point in the data (parameter gradually changes). Some diagnostics to check this type of parameters instability are:

- Estimate model in expanding window and plot the parameters along with its ± 2 standard deviation band. If the magnitude of the coefficient suddenly begins to change, you should suspect of a structural change at that point.

6.5 More on endogenous breaks

See Enders [1995] section 7.12.

Chapter 7

ARFIMA models

7.1 Introduction

An integrated process of order d can be represented by

$$(1 - L)^d Y_t = \Psi(L) \varepsilon_t, \quad (7.1)$$

with $\sum_{j=0}^{\infty} |\psi_j| < \infty$ and $\varepsilon_t \sim wn(0, \sigma_\varepsilon^2)$. Multiplying both sides of (7.1) by $(1 - L)^{-d}$ yields¹:

$$Y_t = (1 - L)^{-d} \Psi(L) \varepsilon_t. \quad (7.2)$$

For z a scalar, define the function

$$f(z) = (1 - z)^{-d},$$

with derivatives given by:

$$\begin{aligned} \frac{\partial f}{\partial z} &= d(1 - z)^{-d-1}, \\ \frac{\partial^2 f}{\partial z^2} &= (d+1)d(1 - z)^{-d-2}, \\ \frac{\partial^3 f}{\partial z^3} &= (d+2)(d+1)d(1 - z)^{-d-3}, \\ &\vdots \\ \frac{\partial^j f}{\partial z^j} &= (d+j-1)(d+j-2) \cdots (d+1)d(1 - z)^{-d-j}. \end{aligned}$$

A power series expansion for $f(z)$ around $z = 0$ is thus given by:

$$\begin{aligned} (1 - z)^{-d} &= f(0) + \left. \frac{\partial f}{\partial z} \right|_{z=0} z + \frac{1}{2!} \left. \frac{\partial^2 f}{\partial z^2} \right|_{z=0} z^2 + \frac{1}{3!} \left. \frac{\partial^3 f}{\partial z^3} \right|_{z=0} z^3 + \cdots \\ &= 1 + dz + \frac{1}{2!} (d+1)d z^2 + \frac{1}{3!} (d+2)(d+1)d z^3 + \cdots \end{aligned}$$

This implies that the operator $(1 - L)^{-d}$ can be represented by the filter:

$$\begin{aligned} (1 - L)^{-d} &= 1 + dL + \frac{1}{2!} (d+1)dL^2 + \frac{1}{3!} (d+2)(d+1)dL^3 + \cdots \\ &= \sum_{j=0}^{\infty} h_j L^j. \end{aligned} \quad (7.3)$$

¹The inverse operator $(1 - L)^{-d}$ admits a square-summable linear-process representation provided that $d < \frac{1}{2}$, which guarantees that Y_t is covariance stationary.

where $h_0 = 1$ and

$$h_j = \frac{1}{j!}(d+j-1)(d+j-2)\cdots(d+1)d = \frac{\Gamma(j+d)}{\Gamma(d)\Gamma(j+1)}. \quad (7.4)$$

where $\Gamma(j+1) = j!$ denotes the gamma function and $\Gamma(j+d) = (d+j-1)(d+j-2)\cdots(d+1)d\Gamma(d)$. Moreover, h_j can be approximated as $j \rightarrow \infty$ by²:

$$\tilde{h}_j \approx (j+1)^{d-1}. \quad (7.5)$$

Theorem 7.1. *The sequence $\{h_j\}_{j=0}^\infty$ with h_j given by (7.4) is square summable provided that $d < \frac{1}{2}$, that is $\sum_{j=0}^\infty h_j^2 < \infty$ for $d < \frac{1}{2}$. Consequently, for $d < \frac{1}{2}$ and $\Psi(L) = 1$ the corresponding process Y_t given in (7.1) is **covariance stationary**.*

Proof. Section 7.6.2. □

Equivalently to (7.3), the operator $(1-L)^d$ can be represented by the filter

$$\begin{aligned} (1-L)^d &= 1 - dL + \frac{1}{2!}d(d-1)L^2 - \frac{1}{3!}d(d-1)(d-2)L^3 + \cdots \\ &= \sum_{j=0}^{\infty} \pi_j L^j, \end{aligned} \quad (7.6)$$

where $\pi_0 = 1$ and

$$\pi_j = \frac{1}{j!}(-d+j-1)(-d+j-2)\cdots(-d+1)(-d) = \frac{\Gamma(j-d)}{\Gamma(-d)\Gamma(j+1)}. \quad (7.7)$$

Moreover, π_j can be approximated as $j \rightarrow \infty$ by

$$\tilde{\pi}_j \approx (j+1)^{-d-1}. \quad (7.8)$$

Theorem 7.2. *The sequence $\{\pi_j\}_{j=0}^\infty$ with π_j given by (7.7) is square summable provided that $d > -\frac{1}{2}$, that is, $\sum_{j=0}^\infty \pi_j^2 < \infty$ for $d > -\frac{1}{2}$. Consequently, for $d > -\frac{1}{2}$ and $\Psi(L) = 1$ the corresponding process Y_t given in (7.1) is **invertible**.*

Proof. Proceeds similarly to the one in Section 7.6.2. □

If $d > \frac{1}{2}$, the standard practice to achieve stationarity is to difference the series. For example, if $d = 0.7$, then

$$(1-L)^{-0.3}(1-L)Y_t = \Psi(L)\varepsilon_t,$$

where ΔY_t is fractionally integrated with parameter $d = -0.3 < \frac{1}{2}$. The following table provides a summary of the behavior of the fractionally integrated process:

d	Behavior
$-\frac{1}{2} < d < 0$	Stationary, antipersistent, and mean reverting
$d = 0$	Short memory
$0 < d < \frac{1}{2}$	Stationary, persistent, and mean reverting
$\frac{1}{2} \leq d < 1$	Nonstationary but mean reverting
$d = 1$	Unit root; shocks have permanent effects

Mean reversion refers to the property that the effect of a shock ε_t eventually vanishes in the limit. Thus, from (7.5), $\lim_{j \rightarrow \infty} h_j = 0$ for $\frac{1}{2} \leq d < 1$, even though the process is not covariance stationary because its variance is infinite. In contrast, for $d = 1$, $h_j = 1$ for all j , so shocks have permanent effects and the process is not mean reverting.

²Proof in Section 7.6.1.

7.2 ARFIMA Models

7.2.1 ARFIMA(p, d, q)

An ARFIMA(p, d, q) process is defined by

$$\Phi(L)(1-L)^d Y_t = \Theta(L)\varepsilon_t, \quad (7.9)$$

where $\{\varepsilon_t\}$ is white noise with finite variance σ_ε^2 . Equivalently, using $\Psi(L) = \Phi(L)^{-1}\Theta(L)$,

$$Y_t = (1-L)^{-d}\Psi(L)\varepsilon_t. \quad (7.10)$$

The parameter d governs the long-run dependence of the process, while $\Phi(L)$ and $\Theta(L)$ characterize its short-run dynamics. It follows from (7.10) that the ARFIMA(p, d, q) can be written as the following MA(∞):

$$Y_t = \left(\sum_{j=0}^{\infty} h_j L^j \right) \left(\sum_{j=0}^{\infty} \psi_j L^j \right) \varepsilon_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}, \quad (7.11)$$

where

$$c_j = \sum_{k=0}^j h_{j-k} \psi_k \quad (7.12)$$

Theorem 7.3. *Consider the ARFIMA(p, d, q) process. If $d < \frac{1}{2}$ and all roots of $\Phi(z) = 0$ lie outside the unit circle, then the coefficients $\{c_j\}_{j=0}^{\infty}$ in (7.11) are square summable, that is,*

$$\sum_{j=0}^{\infty} c_j^2 < \infty.$$

Consequently, Y_t is **covariance stationary**.

Similarly, applying $(1-L)^d$ and the inverse of the MA polynomial to (7.9) gives

$$\varepsilon_t = \Theta(L)^{-1}\Phi(L)(1-L)^d Y_t. \quad (7.13)$$

Theorem 7.4. *Consider the ARFIMA(p, d, q) process. If $d > -\frac{1}{2}$ and all roots of $\Theta(z) = 0$ lie outside the unit circle, then the inverse filter in (7.13) is square summable. Consequently, Y_t is **invertible**.*

Both Theorems 7.3 and 7.4 can be proved using the Cauchy–Schwarz inequality, by showing that the convolution of an absolutely summable sequence with a square-summable sequence is square summable. Therefore, if all roots of both $\Phi(z) = 0$ and $\Theta(z) = 0$ lie outside the unit circle, an ARFIMA(p, d, q) process is both covariance stationary and invertible for

$$-\frac{1}{2} < d < \frac{1}{2}. \quad (7.14)$$

Further notice that, for covariance stationarity, the coefficients of an ARMA process satisfy absolute summability, whereas those of an ARFIMA process satisfy square summability.

7.2.2 ACF - ARFIMA(p, d, q)

From the $MA(\infty)$ representation in (7.11), the autocovariance function at lag k is

$$\begin{aligned}\gamma_k &= \text{Cov}(Y_t, Y_{t-k}) = \sum_{j=0}^{\infty} \sum_{\ell=0}^{\infty} c_j c_{\ell} \text{Cov}(\varepsilon_{t-j}, \varepsilon_{t-k-\ell}) \\ &= \sigma_{\varepsilon}^2 \sum_{\ell=0}^{\infty} c_{\ell} c_{\ell+k},\end{aligned}\tag{7.15}$$

where the last equality follows ε_t white noise. Therefore, the autocorrelation function is

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\sum_{\ell=0}^{\infty} c_{\ell} c_{\ell+k}}{\sum_{\ell=0}^{\infty} c_{\ell}^2}.\tag{7.16}$$

It can be shown that the autocorrelation function satisfies

$$\rho_k \sim C k^{2d-1}, \quad k \rightarrow \infty,\tag{7.17}$$

for some constant C . This implies that, for $0 < d < \frac{1}{2}$, the autocorrelations decay hyperbolically to zero and are not absolutely summable, i.e., $\sum_{k=0}^{\infty} |\rho_k| = \infty$, characterizing a **long-memory** process. In contrast, for $-\frac{1}{2} < d < 0$, the autocorrelations are asymptotically negative, decay hyperbolically to zero at a faster rate, and are absolutely summable, i.e., $\sum_{k=0}^{\infty} |\rho_k| < \infty$, characterizing an **antipersistent** process. In contrast, for a stationary $ARMA(p, q)$ process, the autocorrelations decay exponentially to zero, characterizing a **short-memory** process.

For example, consider the $ARFIMA(0, d, 0)$ process: $Y_t = (1-L)^{-d} \varepsilon_t = \sum_{j=0}^{\infty} h_j \varepsilon_{t-j}$, where $h_j \sim j^{d-1}/\Gamma(d)$, so the impulse-response coefficients decay **hyperbolically**. In contrast, for the stationary $AR(1)$ process $Y_t = (1 - \phi L)^{-1} \varepsilon_t$, with $|\phi| < 1$, the impulse-response coefficients are ϕ^j and decay **exponentially**. Furthermore, for the nonstationary random walk the ACF decays **linearly**, approaching 1 as $t \rightarrow \infty$ ³. Figure 7.1 displays the ACF's of a simulated $ARFIMA(0, 0.4, 0)$ process and a random walk. Notice that the ACF of the random walk remains close to one, while that of the $ARFIMA(0, 0.4, 0)$ process gradually decays toward zero, reflecting its stationary but highly persistent behavior.

In a finite sample, the long-memory process can be approximated arbitrarily well with a suitably large-order ARMA representation. Thus, we can interpret the **fractional-difference model** as a way to **capture parsimoniously long-run multipliers that decay very slowly**. In such cases, a low-order $ARMA(p, q)$ may explain only the short-lag ACF and PACF. Additionally, treating the series as $I(0)$ may suggest nonstationarity, while treating it as $I(1)$ may lead to overdifferencing.

Another feature is that the $ARFIMA(p, d, q)$ allows modelling the short-term behavior of the series as an $ARMA(p, q)$ process while retaining the hyperbolic behavior of the ACF. Accordingly, the effects of the $\Phi(L)$ and $\Theta(L)$ parameters decay exponentially, while the effect of the fractional difference operator decays hyperbolically.

The **PACF** of an ARFIMA process also exhibits a slow decay rather than the sharp cutoff associated with finite-order AR processes. In particular, for an $ARFIMA(0, d, 0)$ process,

$$\alpha_k \sim \frac{d}{k}, \quad k \rightarrow \infty,$$

so the PACF decays hyperbolically to zero.

³See Section 4.1.

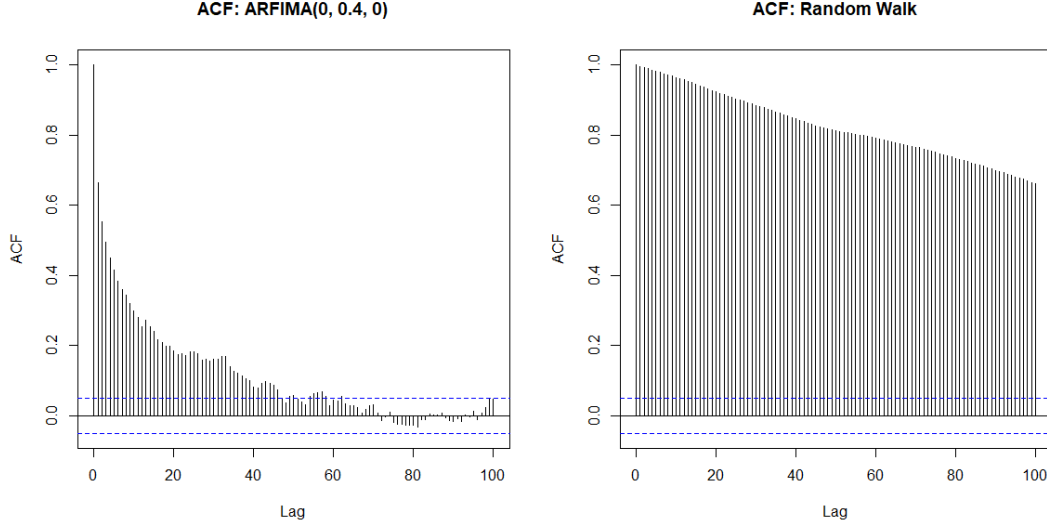


Figure 7.1: ACF of a simulated ARFIMA(0, 0.4, 0) process and a random walk for 1500 observations.

7.3 Estimation

Estimation of an ARFIMA(p, d, q) model can proceed either semiparametrically in the frequency domain, such as through the Geweke–Porter–Hudak (GPH) estimator, or parametrically by maximum likelihood estimation (MLE). We consider only the latter approach, referring Morettin and Toloi [2018] for an understanding on GPH.

7.3.1 Maximum Likelihood Estimation

Consider the ARFIMA(p, d, q) model

$$\Phi(L)(1 - L)^d Y_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma_\varepsilon^2). \quad (7.18)$$

For a given parameter vector $\boldsymbol{\theta} = (d, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q)'$, the innovations can be recovered as

$$\varepsilon_t(\boldsymbol{\theta}) = \Theta(L)^{-1}\Phi(L)(1 - L)^d Y_t. \quad (7.19)$$

Under Gaussianity, the conditional log-likelihood is

$$\ell(\boldsymbol{\theta}, \sigma_\varepsilon^2) = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2} \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2. \quad (7.20)$$

For fixed $\boldsymbol{\theta}$, maximizing with respect to σ_ε^2 yields

$$\hat{\sigma}_\varepsilon^2(\boldsymbol{\theta}) = \frac{1}{T} \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2. \quad (7.21)$$

Hence, the remaining parameters can be estimated numerically by minimizing the sum of squared innovations,

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2, \quad (7.22)$$

subject to the appropriate stationarity and invertibility restrictions. However, estimating the ARFIMA(p, d, q) requires pre-specifying the orders p and q . The GPH estimator may first be used to estimate d and test for the presence of fractional integration, i.e. $H_0 : d = 0$. Next the ACF and PACF of the fractionally differenced series $(1 - L)^d Y_t$ is used to obtain candidate values for p and q . Each candidate ARFIMA model can then be estimated by MLE, with the final specification selected using IC and diagnostic checks.

Another alternative is to estimate several candidate ARFIMA(p, d, q) models by MLE and select the one that minimizes the AIC/BIC, while satisfying the diagnostic checks.

7.3.2 Example: Conditional MLE for the ARFIMA(1, d , 1) Model

Consider the ARFIMA(1, d , 1) model:

$$(1 - \phi L)(1 - L)^d Y_t = (1 + \theta L)\varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma_\varepsilon^2). \quad (7.23)$$

where $\boldsymbol{\theta} = (d, \phi, \theta)'$. Define

$$X_t(d) = (1 - L)^d Y_t = \sum_{j=0}^{\infty} \pi_j(d) Y_{t-j}, \quad (7.24)$$

where $\pi_0(d) = 1$ and the fractional-differencing coefficients satisfy

$$\pi_j(d) = \frac{j-1-d}{j} \pi_{j-1}(d), \quad j = 1, 2, \dots \quad (7.25)$$

Using $X_t(d)$, the residuals of (7.23) can be written as

$$\varepsilon_t(\boldsymbol{\theta}) = X_t(d) - \phi X_{t-1}(d) - \theta \varepsilon_{t-1}(\boldsymbol{\theta}). \quad (7.26)$$

Under Gaussianity, the conditional log-likelihood is

$$\ell(\boldsymbol{\theta}, \sigma_\varepsilon^2) = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2} \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2. \quad (7.27)$$

For fixed $\boldsymbol{\theta}$, differentiation with respect to σ_ε^2 gives

$$\frac{\partial \ell}{\partial \sigma_\varepsilon^2} = -\frac{T}{2\sigma_\varepsilon^2} + \frac{1}{2(\sigma_\varepsilon^2)^2} \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2 = 0 \implies \hat{\sigma}_\varepsilon^2(\boldsymbol{\theta}) = \frac{1}{T} \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2. \quad (7.28)$$

Substituting (7.28) into (7.27) yields the concentrated conditional log-likelihood

$$\ell_c(\boldsymbol{\theta}) = -\frac{T}{2} \left[\log(2\pi) + 1 + \log \left(\frac{1}{T} \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2 \right) \right]. \quad (7.29)$$

Since the logarithm is strictly increasing, maximizing $\ell_c(\boldsymbol{\theta})$ is equivalent to minimizing

$$Q(\boldsymbol{\theta}) = \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta})^2. \quad (7.30)$$

Therefore,

$$\hat{\boldsymbol{\theta}} = (\hat{d}, \hat{\phi}, \hat{\theta})' = \arg \min_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}). \quad (7.31)$$

which is conducted using a nonlinear optimization procedure. For now, we consider Newton–Raphson. The first derivative with respect to d , is given by

$$\dot{\pi}_j(d) = -\frac{1}{j}\pi_{j-1}(d) + \frac{j-1-d}{j}\dot{\pi}_{j-1}(d). \quad (7.32)$$

It follows from (7.24) that

$$\frac{\partial X_t(d)}{\partial d} = \sum_{j=0}^{\infty} \dot{\pi}_j(d) Y_{t-j}. \quad (7.33)$$

The first derivatives of the innovations are obtained directly by differentiating (7.26):

$$\frac{\partial \varepsilon_t}{\partial d} = \frac{\partial X_t}{\partial d} - \phi \frac{\partial X_{t-1}}{\partial d} - \theta \frac{\partial \varepsilon_{t-1}}{\partial d}, \quad (7.34)$$

$$\frac{\partial \varepsilon_t}{\partial \phi} = -X_{t-1} - \theta \frac{\partial \varepsilon_{t-1}}{\partial \phi}, \quad (7.35)$$

$$\frac{\partial \varepsilon_t}{\partial \theta} = -\varepsilon_{t-1} - \theta \frac{\partial \varepsilon_{t-1}}{\partial \theta}. \quad (7.36)$$

Hence, defining the innovation gradient as

$$\mathbf{g}_t(\boldsymbol{\theta}) := \nabla_{\boldsymbol{\theta}} \varepsilon_t = \begin{pmatrix} \partial \varepsilon_t / \partial d \\ \partial \varepsilon_t / \partial \phi \\ \partial \varepsilon_t / \partial \theta \end{pmatrix}, \quad (7.37)$$

the gradient of the objective function $Q(\boldsymbol{\theta})$ is

$$\nabla Q(\boldsymbol{\theta}) = 2 \sum_{t=1}^T \varepsilon_t(\boldsymbol{\theta}) \mathbf{g}_t(\boldsymbol{\theta}). \quad (7.38)$$

For the exact Newton–Raphson step, second derivatives are also required. Let

$$\mathbf{H}_{\varepsilon,t}(\boldsymbol{\theta}) := \nabla_{\boldsymbol{\theta}}^2 \varepsilon_t(\boldsymbol{\theta}).$$

Differentiating (7.38) gives

$$\mathbf{H}_Q(\boldsymbol{\theta}) = 2 \sum_{t=1}^T [\mathbf{g}_t(\boldsymbol{\theta}) \mathbf{g}_t(\boldsymbol{\theta})' + \varepsilon_t(\boldsymbol{\theta}) \mathbf{H}_{\varepsilon,t}(\boldsymbol{\theta})]. \quad (7.39)$$

Starting from an initial value $\boldsymbol{\theta}^{(0)} = (d^{(0)}, \phi^{(0)}, \theta^{(0)})'$, the Newton–Raphson iteration is then

$$\boldsymbol{\theta}^{(m+1)} = \boldsymbol{\theta}^{(m)} - \mathbf{H}_Q(\boldsymbol{\theta}^{(m)})^{-1} \nabla Q(\boldsymbol{\theta}^{(m)}), \quad m = 0, 1, 2, \dots \quad (7.40)$$

At each iteration, the procedure therefore consists of computing $\{\pi_j(d^{(m)})\}$, constructing $\{X_t(d^{(m)})\}$, recursively obtaining $\{\varepsilon_t(\boldsymbol{\theta}^{(m)})\}$ and their derivatives, evaluating the gradient and Hessian, and updating the parameter vector according to (7.40). The iterations proceed until convergence. Finally, the innovation variance is estimated at the converged parameter estimates by

$$\hat{\sigma}_\varepsilon^2 = \frac{1}{T} \sum_{t=1}^T \varepsilon_t(\hat{d}, \hat{\phi}, \hat{\theta})^2. \quad (7.41)$$

In practice, direct computation of the exact Hessian in (7.39) can be cumbersome. Consider Quasi-Newton methods (e.g. BFGS).

Stationarity and Invertibility Restrictions

For the $ARFIMA(1, d, 1)$ model, covariance stationarity and invertibility require

$$-\frac{1}{2} < d < \frac{1}{2}, \quad |\phi| < 1, \quad |\theta| < 1.$$

These restrictions can be imposed through a reparameterization. Let $\boldsymbol{\lambda} = (\lambda_d, \lambda_\phi, \lambda_\theta)' \in \mathbb{R}^3$ and define

$$d = \frac{1}{2} \tanh(\lambda_d), \quad \phi = \tanh(\lambda_\phi), \quad \theta = \tanh(\lambda_\theta). \quad (7.42)$$

Since $-1 < \tanh(x) < 1$, the stationarity and invertibility restrictions are automatically satisfied for every $\boldsymbol{\lambda} \in \mathbb{R}^3$. The constrained optimization problem can therefore be written as the unconstrained problem

$$\hat{\boldsymbol{\lambda}} = \arg \min_{\boldsymbol{\lambda} \in \mathbb{R}^3} Q \left(\frac{1}{2} \tanh(\lambda_d), \tanh(\lambda_\phi), \tanh(\lambda_\theta) \right). \quad (7.43)$$

After convergence, the parameter estimates are recovered through

$$\hat{d} = \frac{1}{2} \tanh(\hat{\lambda}_d), \quad \hat{\phi} = \tanh(\hat{\lambda}_\phi), \quad \hat{\theta} = \tanh(\hat{\lambda}_\theta).$$

7.4 Forecasting

An $ARFIMA(p, d, q)$ process can be written as

$$\Phi(L)(1-L)^d Y_t = \Theta(L)\varepsilon_t. \quad (7.44)$$

Under invertibility, the process admits an $AR(\infty)$ representation. Define

$$A(L) := \Theta(L)^{-1} \Phi(L)(1-L)^d = 1 - \sum_{j=1}^{\infty} a_j L^j. \quad (7.45)$$

Then, from $A(L)Y_t = \varepsilon_t$, we write

$$Y_t = \sum_{j=1}^{\infty} a_j Y_{t-j} + \varepsilon_t. \quad (7.46)$$

Then, truncating the filter at some lag M leads to:

$$Y_t \approx \sum_{j=1}^M a_j Y_{t-j} + \varepsilon_t. \quad (7.47)$$

and the corresponding truncated forecasts are given by

$$\hat{Y}_{T+h|T}^{(M)} = \sum_{j=1}^M a_j \hat{Y}_{T+h-j|T}^{(M)}, \quad (7.48)$$

where $\hat{Y}_{T+h-j|T}^{(M)} = Y_{T+h-j}$ whenever $h-j \leq 0$. Alternatively, under covariance stationarity the $ARFIMA(p, d, q)$ process admits the $MA(\infty)$ representation

$$Y_t = C(L)\varepsilon_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}, \quad c_0 = 1, \quad (7.49)$$

where

$$C(L) = (1 - L)^{-d} \frac{\Theta(L)}{\Phi(L)}. \quad (7.50)$$

At time T , write

$$Y_{T+h} = \sum_{j=0}^{h-1} c_j \varepsilon_{T+h-j} + \sum_{j=h}^{\infty} c_j \varepsilon_{T+h-j}. \quad (7.51)$$

The optimal linear h -step-ahead forecast is

$$\hat{Y}_{T+h|T} = \sum_{j=h}^{\infty} c_j \varepsilon_{T+h-j}. \quad (7.52)$$

Subtracting (7.52) from (7.51) gives the forecast error

$$u_{T+h|T} := Y_{T+h} - \hat{Y}_{T+h|T} = \sum_{j=0}^{h-1} c_j \varepsilon_{T+h-j}. \quad (7.53)$$

Since the innovations are uncorrelated with variance σ_ε^2 , the forecast-error variance is

$$\mathbb{V}(u_{T+h|T}) = \mathbb{V} \left(\sum_{j=0}^{h-1} c_j \varepsilon_{T+h-j} \right) = \sigma_\varepsilon^2 \sum_{j=0}^{h-1} c_j^2. \quad (7.54)$$

Therefore, we can forecast ARFIMA(p, d, q) using either (7.48) or (7.52).

7.5 Testing for fractional difference

Consider

$$H_0 : d = 0 \text{ vs } H_1 : d \neq 0.$$

This hypothesis can be tested using the asymptotic GPH estimator or the asymptotic distribution of the MLE.

7.6 Appendix: Proofs and Derivations

7.6.1 Derivation of equation (7.6.1)

Equation (7.4) for h_j may be written as:

$$\begin{aligned}
 h_j &= \frac{1}{j!} (d+j-1)(d+j-2)(d+j-3) \cdots (d+1)(d) \\
 &= \left[\frac{d+j-1}{j} \right] \left[\frac{d+j-2}{j-1} \right] \left[\frac{d+j-3}{j-2} \right] \cdots \left[\frac{d+1}{2} \right] \left[\frac{d}{1} \right] \\
 &= \left[\frac{j+d-1}{j} \right] \left[\frac{j-1+d-1}{j-1} \right] \left[\frac{j-2+d-1}{j-2} \right] \cdots \\
 &= \left[1 + \frac{d-1}{j} \right] \left[1 + \frac{d-1}{j-1} \right] \left[1 + \frac{d-1}{j-2} \right] \cdots \left[1 + \frac{d-1}{j-(j-1)} \right]. \tag{7.55}
 \end{aligned}$$

For large j , we have the approximation⁴

$$\left[1 + \frac{d-1}{j} \right] \approx \left[1 + \frac{1}{j} \right]^{d-1} = \left[\frac{j+1}{j} \right]^{d-1}, \tag{7.56}$$

which allows us to write (7.55) as:

$$\begin{aligned}
 h_j &\approx \left[\frac{j+1}{j} \right]^{d-1} \left[\frac{j}{j-1} \right]^{d-1} \left[\frac{j-1}{j-2} \right]^{d-1} \cdots \left[\frac{3}{2} \right]^{d-1} \left[\frac{2}{1} \right]^{d-1} \\
 &= (j+1)^{d-1}. \tag{7.57}
 \end{aligned}$$

7.6.2 Proof of Theorem (7.1)

Consider (7.4) and (7.5). From the Gamma-ratio asymptotic relation,

$$\frac{\Gamma(j+d)}{\Gamma(j+1)} \sim j^{d-1},$$

where $\sim$ denotes asymptotic equivalence, that is, the ratio of the two sides converges to one as $j \rightarrow \infty$. Then, we obtain

$$\begin{aligned}
 \lim_{j \rightarrow \infty} \frac{h_j}{\tilde{h}_j} &= \frac{1}{\Gamma(d)} \lim_{j \rightarrow \infty} \frac{\Gamma(j+d)}{\Gamma(j+1)(j+1)^{d-1}} \\
 &= \frac{1}{\Gamma(d)} \lim_{j \rightarrow \infty} \frac{j^{d-1}}{(j+1)^{d-1}} = \frac{1}{\Gamma(d)}.
 \end{aligned}$$

Consequently,

$$\lim_{j \rightarrow \infty} \frac{h_j^2}{\tilde{h}_j^2} = \frac{1}{\Gamma(d)^2},$$

which is finite and strictly positive. Hence, by the limit comparison test⁵,

$$\sum_{j=0}^{\infty} h_j^2 < \infty \iff \sum_{j=0}^{\infty} \tilde{h}_j^2 < \infty.$$

⁴This follows from the first-order Taylor expansion $(1+x)^a = 1+ax+O(x^2)$ around $x_0=0$, with $x=1/j$ and $a=d-1$. For large j , we have $x \rightarrow 0$, which is the point around which the approximation is performed.

⁵The limit comparison test states that, for two sequences of positive terms $\{a_j\}$ and $\{b_j\}$, if $\lim_{j \rightarrow \infty} a_j/b_j = c$ for some $0 < c < \infty$, then $\sum_{j=0}^{\infty} a_j$ and $\sum_{j=0}^{\infty} b_j$ either both converge or both diverge.

Assume initially that $d < 1$. Then $2(d-1) < 0$ and the function $f(x) = x^{2(d-1)}$ is decreasing on $(0, \infty)$. From squaring and summing over $\tilde{h}_j$ we get⁶:

$$\begin{aligned} \sum_{j=0}^{N-1} (j+1)^{2(d-1)} &= \sum_{j=1}^N j^{2(d-1)} < 1 + \int_1^N x^{2(d-1)} dx \\ &= 1 + \left[\frac{1}{2d-1} \right] x^{2d-1} \Big|_1^N \\ &= 1 + \left[\frac{1}{2d-1} \right] [N^{2d-1} - 1], \end{aligned}$$

This converges as $N \rightarrow \infty$ provided that $2d-1 < 0$, or equivalently, $d < \frac{1}{2}$.

⁶Since $d < 1$, the function $x^{2(d-1)}$ is decreasing on $(0, \infty)$, so $j^{2(d-1)} < x^{2(d-1)}$ for $x \in [j-1, j)$ and $j \geq 2$. Since $[j-1, j]$ has unit length, integrating both sides gives $j^{2(d-1)} < \int_{j-1}^j x^{2(d-1)} dx$. Summing from $j=2$ to N yields $\sum_{j=2}^N j^{2(d-1)} < \int_1^N x^{2(d-1)} dx$, and adding the $j=1$ term, which equals 1, gives the inequality.

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